Machine Learning in Materials Discovery

Metals, Alloys, Catalysts, Magnetic Materials

Critical raw material-free multiprincipal alloy design for a net-zero future

Swati Singh Mingwen Bai, Allan Matthews, Saurav Goel & Shrikrishna N. Joshi

Country of Origin:

- India and the United Kingdom.

Institutes/Research Labs Involved:

- Indian Institute of Technology Guwahati (India), University of Petroleum and Energy Studies, Dehradun (India), University of Leeds (UK), Henry Royce Institute, The University of Manchester (UK), London South Bank University (UK)

Funding Received: The Department of Science and Technology (DST), India.

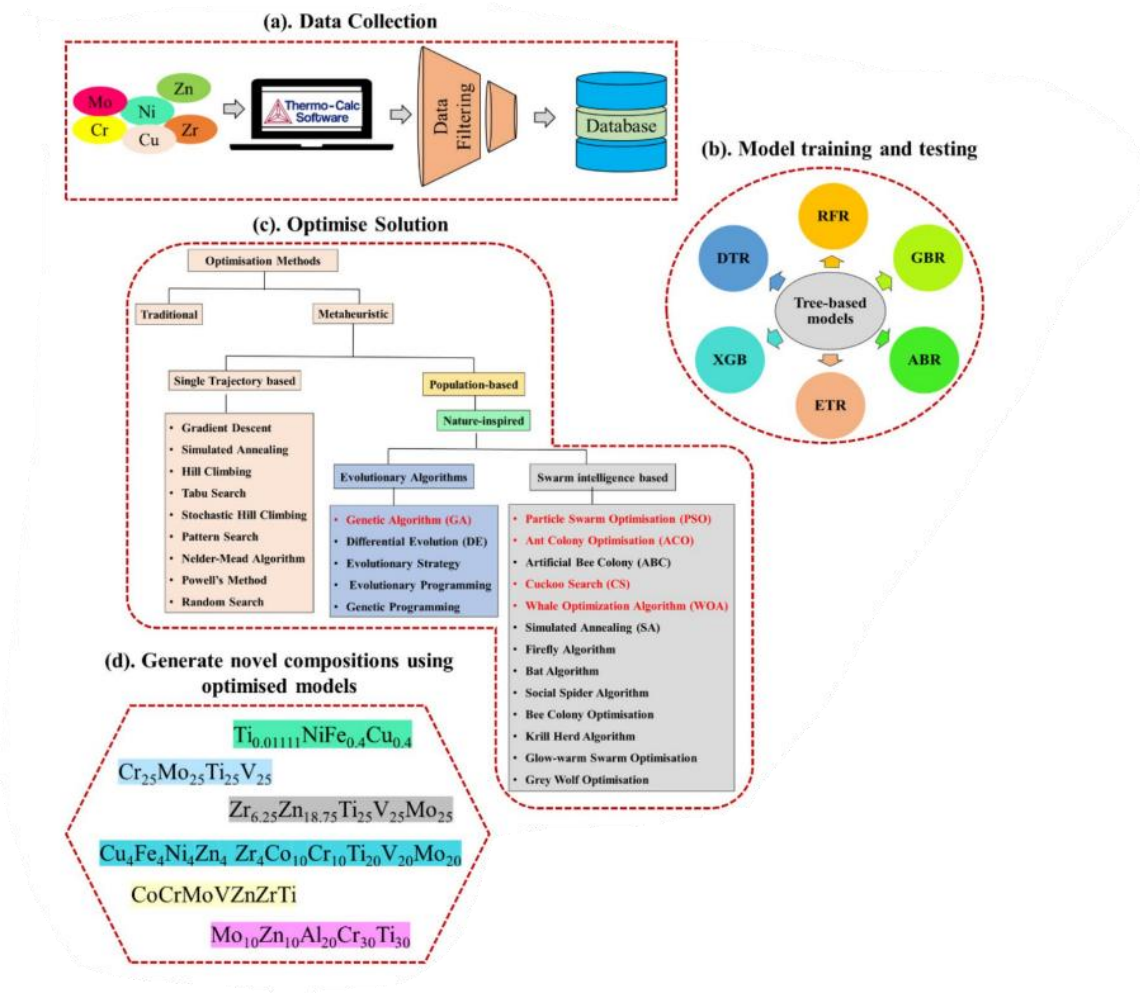
- UK Research and Innovation (UKRI) via EPSRC Grant No. EP/T024607/1.
- The Royal Society through an International Exchange Cost Share award (IEC\NSFC\223536)

Journal:

- *Scientific Reports* (published by Nature Portfolio).

Links:

- https://www.nature.com/articles/s41598-025-87784-0
- https://www.nature.com/articles/s41598-023-31461-7

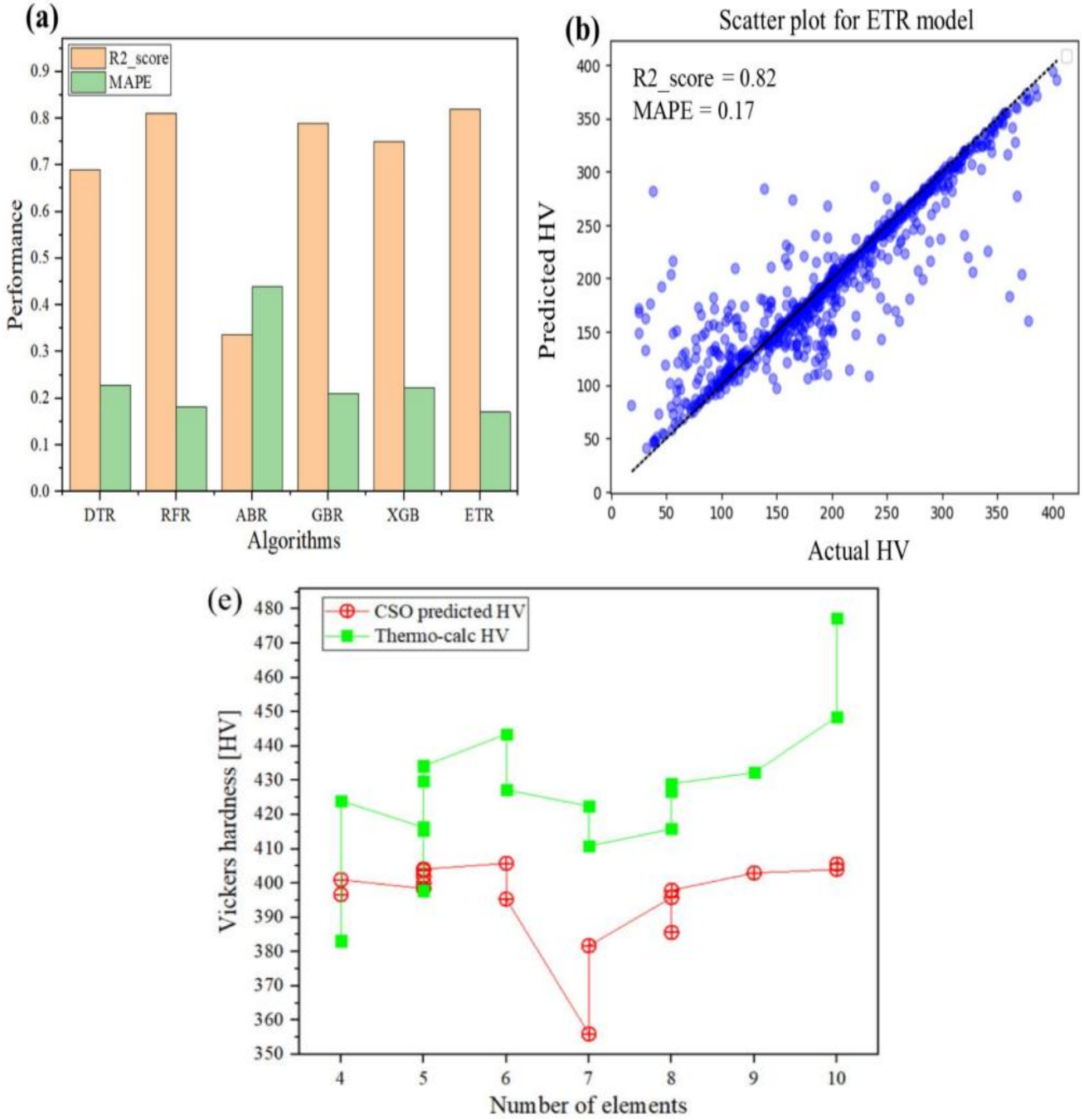


Material Focus:

- HEAs comprised of 3d transition metals (Co, Cr, Fe, Ni, Ti, Ta, Mo, W) refractory elements (Nb, Mo, Ta, V, Hf, Zr). These are referred to as critical raw materials (CRMs)
- Wide applications in aerospace, nuclear and energy sectors due to superior performance in high temperature applications

Methodology:

- Pipeline to predict novel multicomponent alloys having reduced CRM percentages while having similar desirable properties
- Computational database (TCHEA7) of 3608 unary and binary alloys entries used for data
- Machine learning models used to map Vickers hardness of candidates (validation metric) and act as an extrapolation base, which is proposed by metaheuristic optimization algorithms



Key Concept Explained - Optimization:

- Mathematical process of finding the best possible input parameters within a defined search space to maximize or minimize a specific objective function.
- Traditional gradient based methods fail when the search space is non-differentiable and highly rugged
- Metaheuristic methods are stochastic frameworks that treat the objective function as a black box, driven by a balance between two opposing search dynamics: exploration and exploitation.
- Exploration searches the global search domain to uncover new chemical solution basins in the HEA compositional space, while exploitation refines local neighbourhoods of known solutions to dial in the exact atomic percentages.

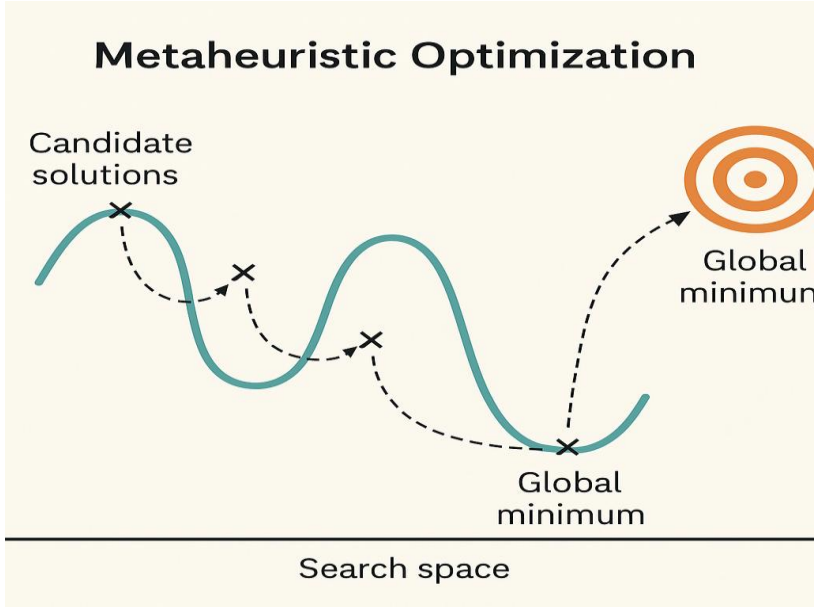
Technology:

- The Extra Tree Regressor (ETR) model shows the best performance as a surrogate evaluator model in the optimization loop, while Cuckoo Search Optimization (CSO) proposed reduced RE novel HEA compositions which were closest to the ThermoCalc predictions

Validation:

- A predicted FCC-phase alloy Al6.25Cu18.75Fe25Co25Ni25 was synthesized, which showed excellent agreement with the model predicted hardness and ThermoCalc prediction

No. of elements	Composition suggested from CSO	HV predicted by CSO	Thermo-calc HV	Percentage Error (%)
4	$Mo_{25}Sn_{25}Ti_{25}V_{25}$	397	383	4
4	$Cr_{25}Mo_{25}Ti_{25}V_{25}$	401	424	5
4	$Ti_{0.01111}NiFe_{0.4}Cu_{0.4}$	**468**	**488**	**4**
5	$Co_{12.5}Cr_{12.5}Mo_{25}Ti_{25}V_{25}$	398	416	4
5	$Mo_{10}Zn_{10}Al_{20}Cr_{30}Ti_{30}$	400	398	0.5
5	$Co_{11.11111}Cr_{11.11111}Mo_{22.22222}V_{22.22222}Ti_{33.33333}$	404	430	6
5	$Zn_{6.25}Cr_{18.75}Mo_{25}Ti_{25}V_{25}$ (similar to novel one)	402	415	3
5	$Zr_{6.25}Zn_{18.75}Ti_{25}V_{25}Mo_{25}$ (similar to novel one	404	434	7
6	$Zn_5Cr_{10}Zr_{10}Mo_{25}Ti_{25}V_{25}$	406	443	8
6	$Co_{16.66667}Cr_{16.66667}Mo_{16.66667}Ti_{16.66667}V_{16.66667}Zr_{16.66667}$	395	427	7
7	CoCrMoVZnZrTi	356	422	16
7	$Cu_5Zr_5Co_{12}Cr_{12}Mo_{13}V_{13}Ti_{40}$	382	411	7
8	$Co_{10}Mn_{10}V_{10}Zn_{10}Zr_{10}Cr_{12.5}Ti_{17.5}Mo_{20}$	396	416	5
8	$Co_{12.5}Cu_{12.5}Cr_{12.5}Mo_{12.5}Ti_{12.5}V_{12.5}Zn_{12.5}Zr_{12.5}$	385	427	10
8	$Ni_2Zn_2Zr_{11}Cr_{12.5}Co_{12.5}Mo_{20}Ti_{20}V_{20}$	398	429	7
9	$Cu_5Ni_5Zn_5\ Zr_5Co_8Cr_8Mo_{10}V_{10}Ti_{44}$	403	432	7
10	$Co_{3.57143}Cr_{3.57143}Cu_{3.57143}Fe_{3.57143}Ni_{3.57143}$ $Zn_{3.57143}Zr_{3.57143}Mo_{25}Ti_{25}V_{25}$ OR $(CoCrCuFeNiZnZr)_{0.1428572}MoTiV$	404	448	10
10	$Cu_4Fe_4Ni_4Zn_4\ Zr_4Co_{10}Cr_{10}Ti_{20}V_{20}Mo_{20}$	405	477	15



DiffCrysGen: A Score-Based Diffusion Model for Design of Diverse Inorganic Crystalline Materials

Sourav Mal, Subhankar Mishra, Prasenjit Sen

Country of Origin:

- India

Institutes/Research Labs Involved:

- Harish-Chandra Research Institute (HRI), National Institute of Science Education and Research (NISER), Odisha, Homi Bhabha National Institute (HBNI), Mumbai

Industry Collaboration:

- There are no direct industry collaborations listed. The research is aimed at designing sustainable alternatives to conventional permanent magnets.

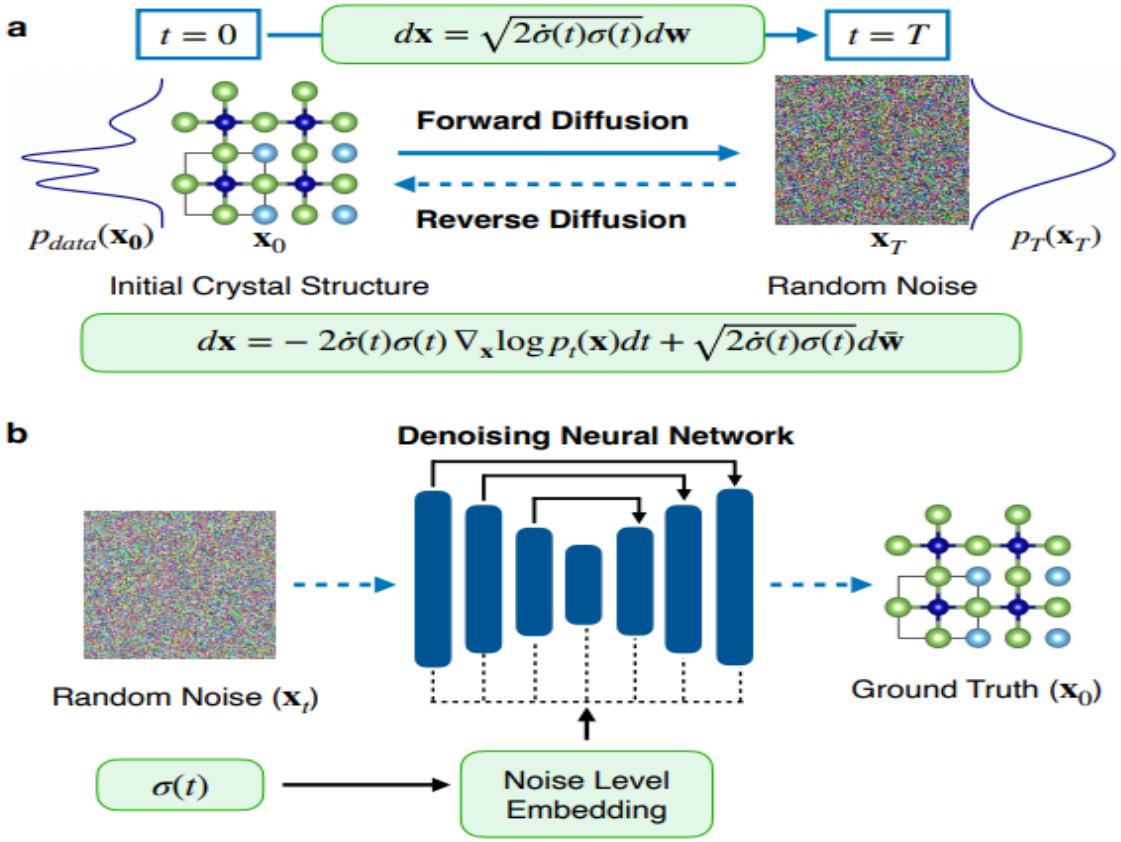
Funding Received:

- Department of Atomic Energy (DAE), Government of India

Journal: Preprint on ArXiv, not published yet in peer-reviewed journal

Links:

- https://arxiv.org/abs/2505.07442v1
- https://arxiv.org/abs/2510.12329



Material Focus:

- Rare-earth-free magnetic materials. The generated datasets span transition-metal alloys, oxides, halides, and chalcogenides.
- Applications in energy sector and spintronics

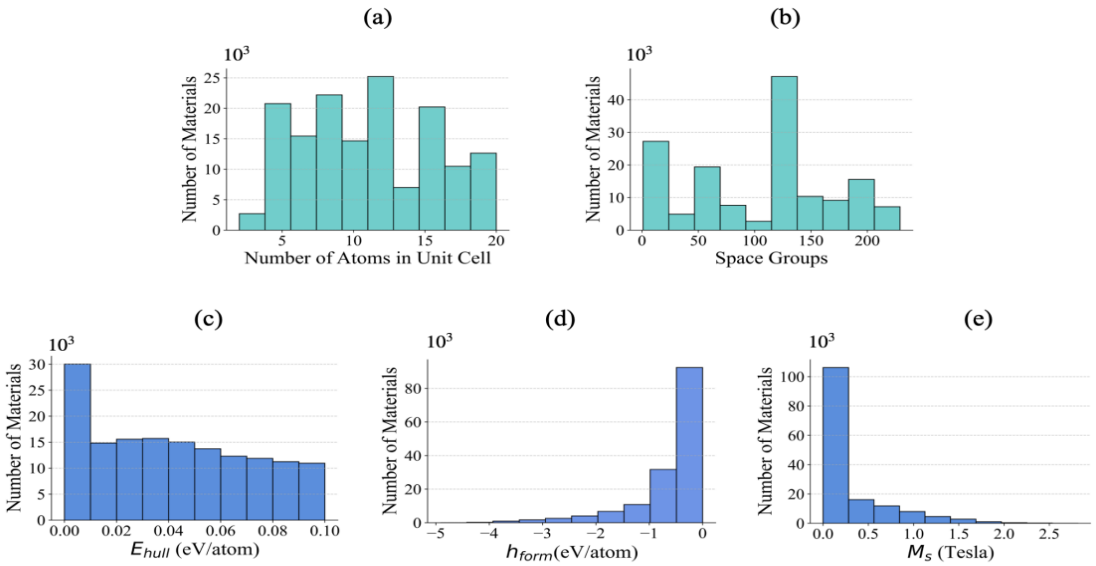
Methodology:

- Diffusion based generative model to discover new stable and property aligned compositions of sustainable magnets
- Removes complexity added by physics-based priors applied to atom types, coordinates and lattices, which mandate customized masked and symmetric diffusion
- Invertible Real-Space Crystallographic Representation (IRCR) used. Five matrices encoded vertically into a single 2D array, each capturing a distinct aspect
- Dataset Alex-1, curated from Alexandria DFT database, filtered down to 155,534 magnetic candidates

Validation:

- DFT structure optimization and formation energy minimization
- Phonon dispersion calculations for dynamic stability

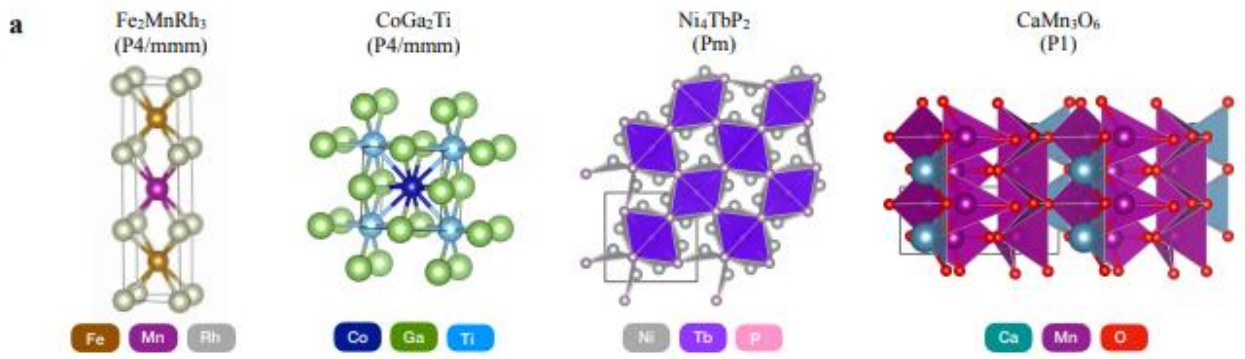
Component	Shape	Description
E (Element Matrix)	94×3	One-hot encoding of chemical elements (up to $Z = 94$).
L (Lattice Matrix)	2×3	Lattice constants (a, b, c) and angles (α, β, γ).
C (Coordinate Matrix)	$n_{\text{sites}} \times 3$	Fractional atomic coordinates in the unit cell.
O (Occupancy Matrix)	$n_{\text{sites}} \times 3$	One-hot encoding of site occupancy.
P (Property Matrix)	6×3	Elemental properties, including atomic number (Z), electronegativity (e), period number (p), group number (g), valence electron number (n_v), and atomic fraction (s).

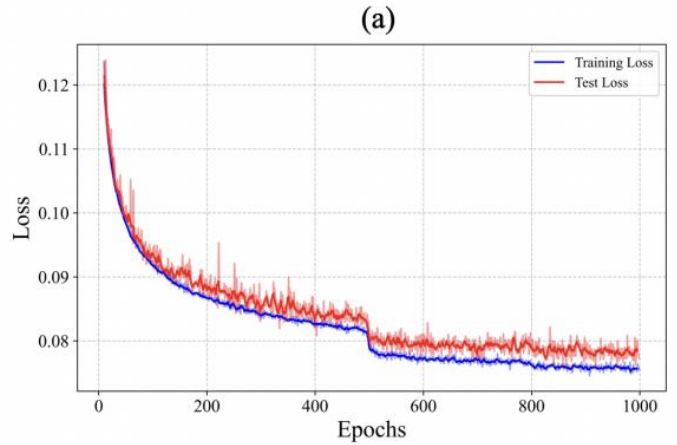


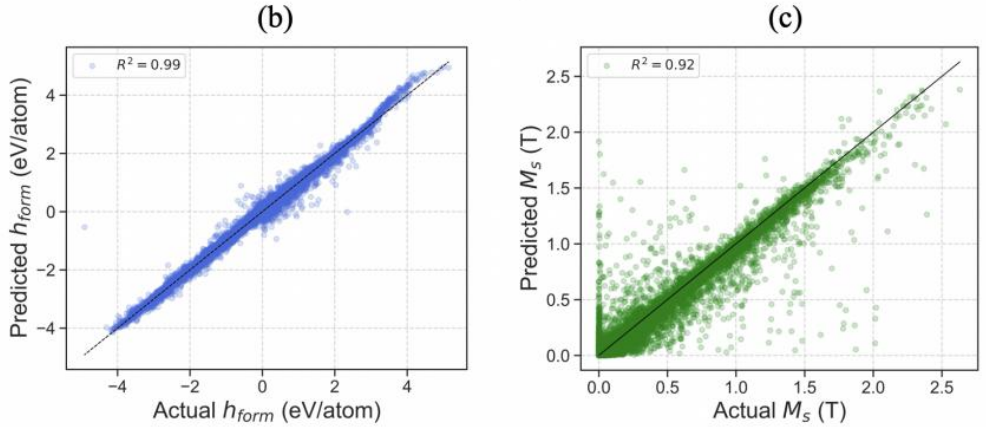
Key Concept Explained - Diffusion based Generative Model:

- A diffusion model learns the probability distribution of a dataset by learning to invert a known corruption process
- DiffCrysGen specifically implements the CT-S-SDE formulation (Song et al): original data is not rescaled, it is buried under noise of growing variance, preserving the absolute physical scale of quantities
- The denoiser algebraically recovers the data through: $\nabla_{x_t} log\, p_t\,(x_t) = [D_\theta(x_t; \sigma(t)) - x_t]/\sigma^2(t)$
- Every component of the crystal is diffused together in one unified process by isotropic Gaussian noise within one shared latent space.
- Periodic boundary conditions are handled entirely by the representation rather than by the diffusion process
- The denoising network is a noise-conditional one-dimensional UNet with 1.32 million trainable parameters, built from an encoder and decoder whose fundamental building block is a ResNet block

NOTE: No experimental verification, the only contact with experiment is a cross-check of the 28 final candidates against ICSD, and while a different cubic polymorph of Fe_2NiSi has been synthesized previously, the tetragonal high-anisotropy phase claimed here has not







Exploration of crystal chemical space using text-guided generative artificial intelligence

Hyunsoo Park, Anthony Onwuli, Aron Walsh

Country of Origin:

- United Kingdom.

Institutes/Research Labs Involved:

- Department of Materials, Imperial College London, London, UK

Funding Received:

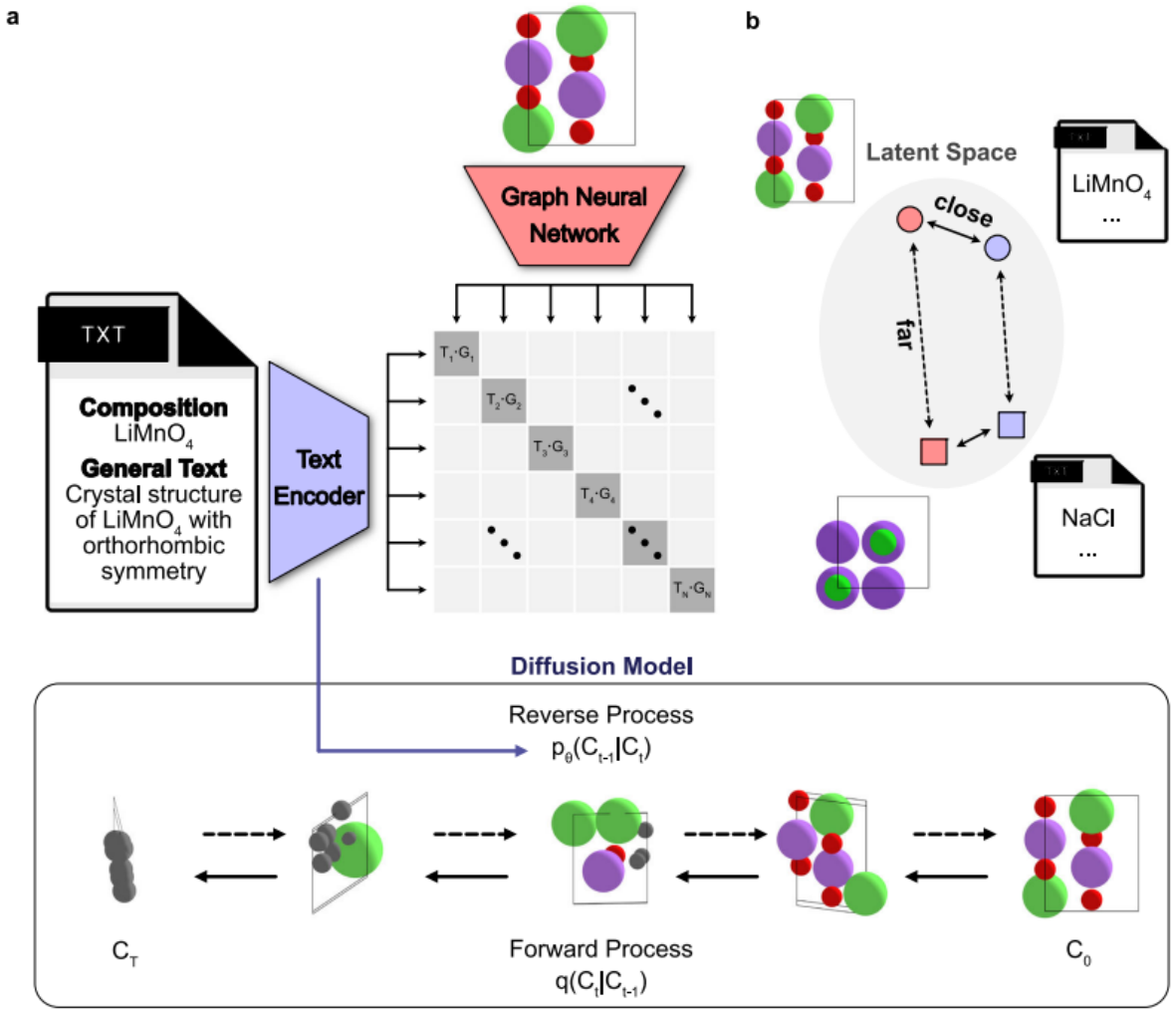
- EPSRC project EP/X037754/1, and the AI for Chemistry (AIchemy) hub under EPSRC grants EP/Y028775/1 and EP/Y028759/1

Journal:

- Nature Communications (2025) 16:4379

Links:

- https://www.nature.com/articles/s41467-025-59636-y



Material Focus:

- Inorganic crystalline compounds, with the demonstration systems being TiO_2 polymorphs, Ti-Zn-O ternary space, and the Li-P-S-Cl quaternary space

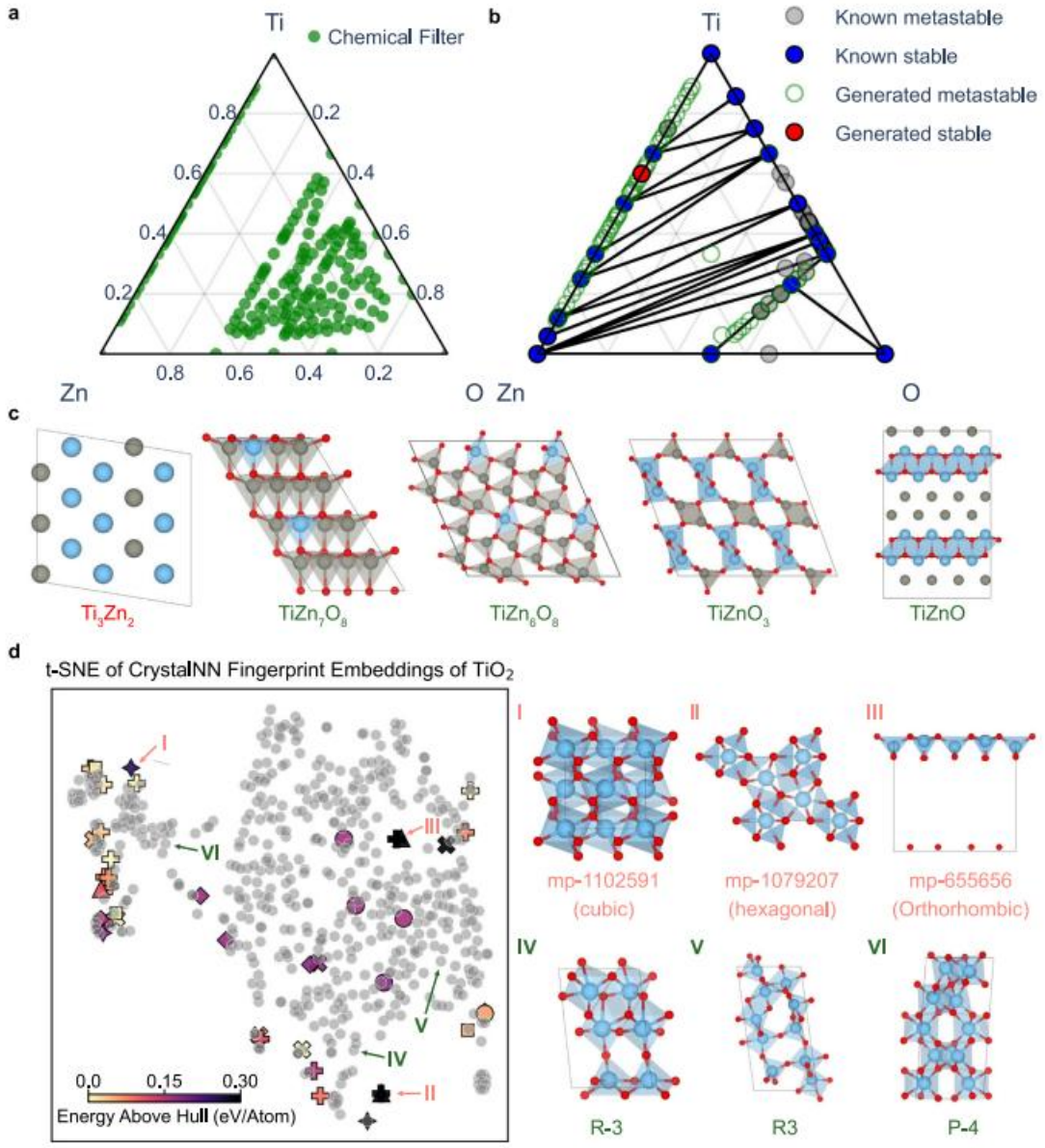
Methodology:

- Chemeleon generates the chemical composition and the crystal structure from a written prompt, rather than taking a fixed formula as input
- The model has two components:

  - Crystal CLIP, a transformer text encoder pre-trained by cross-modal contrastive learning

  - Text embeddings are aligned with graph embeddings from equivariant GNNs

  - Classifier-free guidance denoising diffusion model conditioned on those text embeddings
- A crystal is represented as C = (A, X, L) and each component gets its own diffusion process
- The denoising network is an E(3)-equivariant GNN built on the DiffCSP and EGNN frameworks, with a Fourier transform applied to the fractional coordinates to give periodic E(3)-invariant outputs
- Training used 32,525 Materials Project structures with 40 or fewer atoms per primitive cell

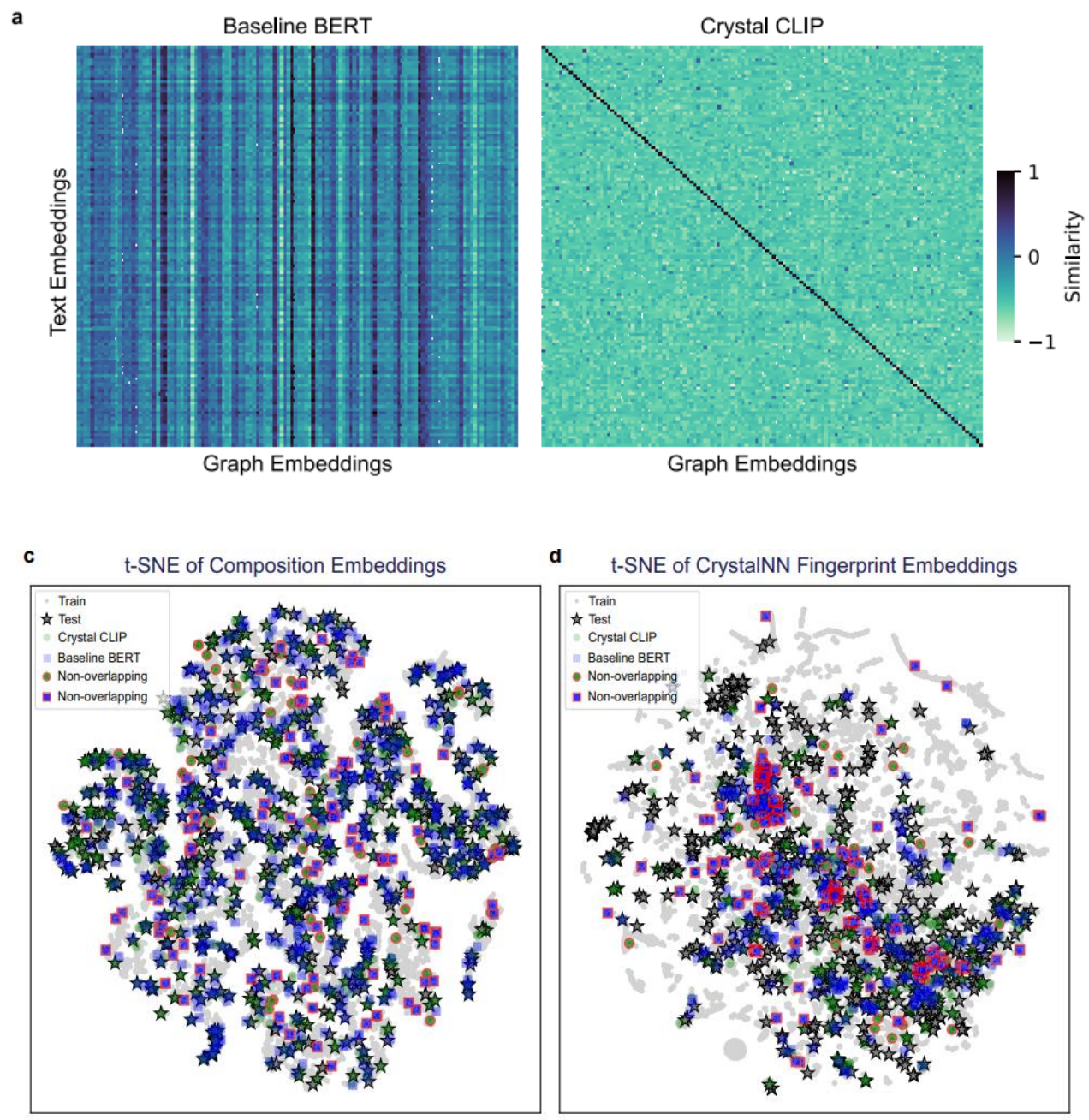


Key Concept Explained - Contrastive Language Grounding:

- t-SNE of the element embeddings from Crystal CLIP shows clean clustering into transition metals, halogens and noble gases, which does not emerge in the baseline
- Generation begins from pure noise, atom types start in the D3PM absorbing state as dummy atoms of atomic number zero, the lattice from a Gaussian, and the fractional coordinates from a wrapped normal over the unit cube. Text conditioning enters through FiLM layers, which biases the denoising trajectory, creating small stoichiometric mismatches
- The samples then pass through two-tiers: MACE-MP for preliminary geometry optimisation, followed by DFT relaxation and static calculations in atomate2/VASP

Validation:

- For TiO_2, 549 novel polymorphs were sampled and 539 converged under MACE-MP
- In Ti-Zn-O the model added one new structure below the convex hull, the ordered intermetallic Ti_3Zn_2 with space group I4/mmm
- Phonon dispersions confirm Li_5PS_4Cl, Li_6PS_4Cl, Li_4PS_4Cl and Li_3PS_3Cl as dynamically stable



Guiding Generative Models to Uncover Diverse and Novel Crystals via Reinforcement Learning

Hyunsoo Park, Anthony Onwuli, Aron Walsh

Country of Origin:

- United Kingdom.

Institutes/Research Labs Involved:

- Department of Materials, Imperial College London, London, UK

Funding Received:

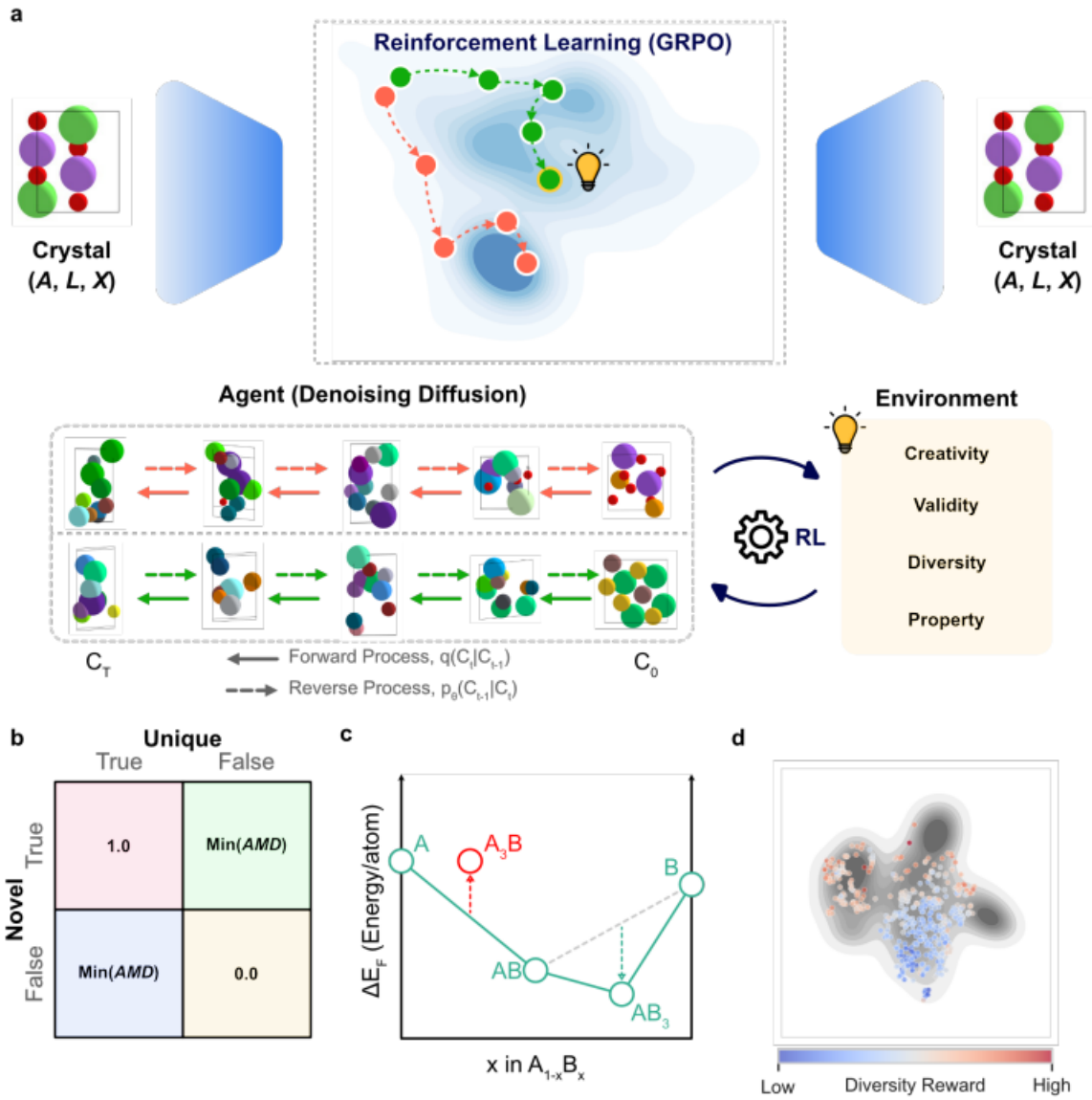
- EPSRC project EP/X037754/1, and the AI for Chemistry (AIchemy) hub under EPSRC grants EP/Y028775/1 and EP/Y028759/1

Journal:

- Preprint on arXiv

Links:

- https://arxiv.org/abs/2511.07158

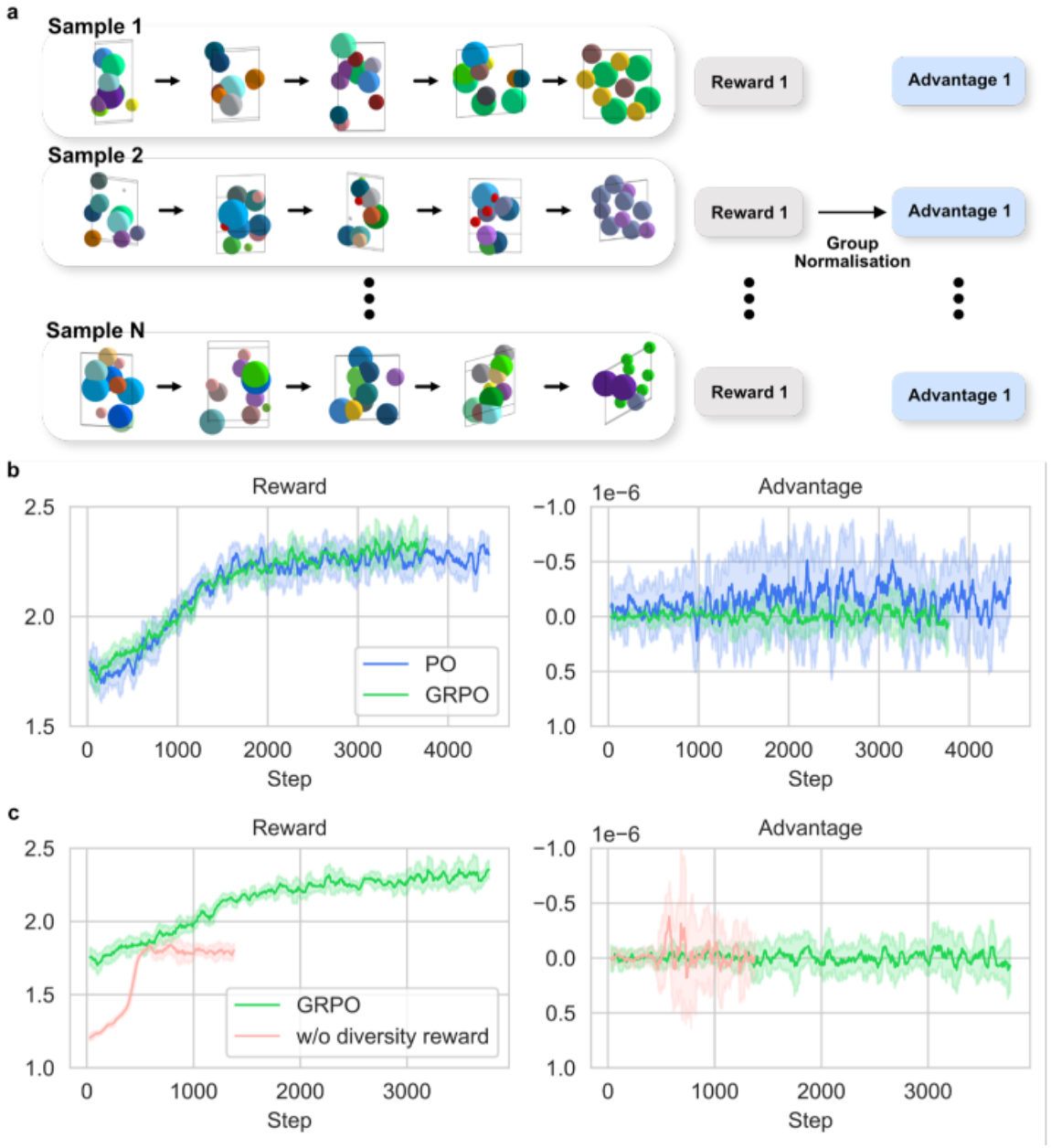


Material Focus:

- Crystalline multi-component intermetallics, mixed-alkali halides, alkali and lanthanide chalcogenides and lanthanide pnictides
- Applications across semiconductors, battery materials and catalysts

Methodology:

- Reinforcement learning is used to correct the misalignment between training data reproduction and search space discovery
- A Transformer VAE first encodes the crystal (atom types **A**, lattice matrix **L**, fractional coordinates **X**) into a continuous 8-dimensional latent space
- The denoising model starts from random noise in this latent space and cleans it up until a sensible latent vector remains, which the decoder turns back into a full crystal structure.
- Group Relative Policy Optimisation (GRPO) is the learning rule. A batch of 64 structures is generated under the same conditions and compared against each other, rewarding the ones that beat the batch average.
- A multi-objective reward combines creativity, stability and diversity
- Trained on MP-20 and the larger Alex-MP-20 datasets, with 43,295 DFT bandgap values used for the property-guided task

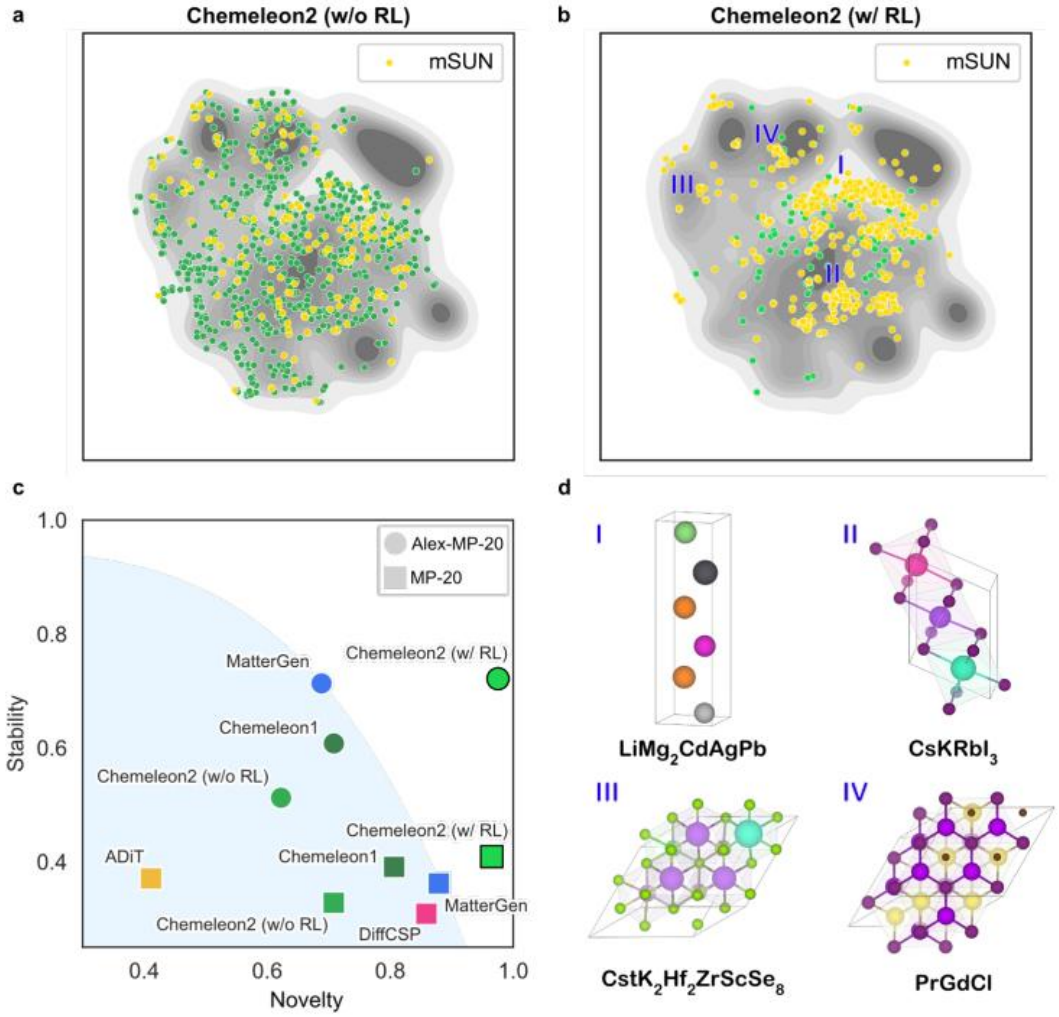


Key Concept Explained - Reward-Driven Exploration:

- Compounds chemically close to known materials are usually stable, while genuinely novel chemistries are more likely to be synthetically inaccessible.
- Reinforcement learning optimises expected reward instead of data likelihood, so the model can be pushed into low-probability regions of chemical space while stability is still enforced as an explicit reward term
- The creativity reward uses the Average Minimum Distance descriptor, giving a continuous isometry-invariant measure of how far a structure sits from the batch and from the reference dataset
- Removing the diversity reward causes severe mode collapse, where the policy repeatedly samples small geometric perturbations of a single favourable composition
- A KL penalty against the pre-trained denoiser and an entropy bonus keep the policy from drifting towards physically implausible structures

Validation:

- On Alex-MP-20 mSUN (metastable, unique and novel fraction) rises from 15.9 % to 61.3
- Novelty rises to 97.5 % and metastability from 51.2 % to 72.1 %
- On the smaller MP-20 dataset mSUN improves from 7.6 % to 26.0 %
- Compositional statistics shifts, with transition-metal content rising from 26.2 % to 35.6 % and non-metal content falling from 27.2 % to 16.3 %



Machine learning-enabled high-entropy alloy discovery

Ziyuan Rao, PoYen Tung, Ruiwen Xie, Ye Wei

Country of Origin:

- Germany, United Kingdom, Netherlands, China and Sweden

Institutes/Research Labs Involved:

- Max-Planck-Institut für Eisenforschung, TU Darmstadt, University of Cambridge, TU Delft, Southeast University, KTH Royal Institute of Technology

Funding Received:

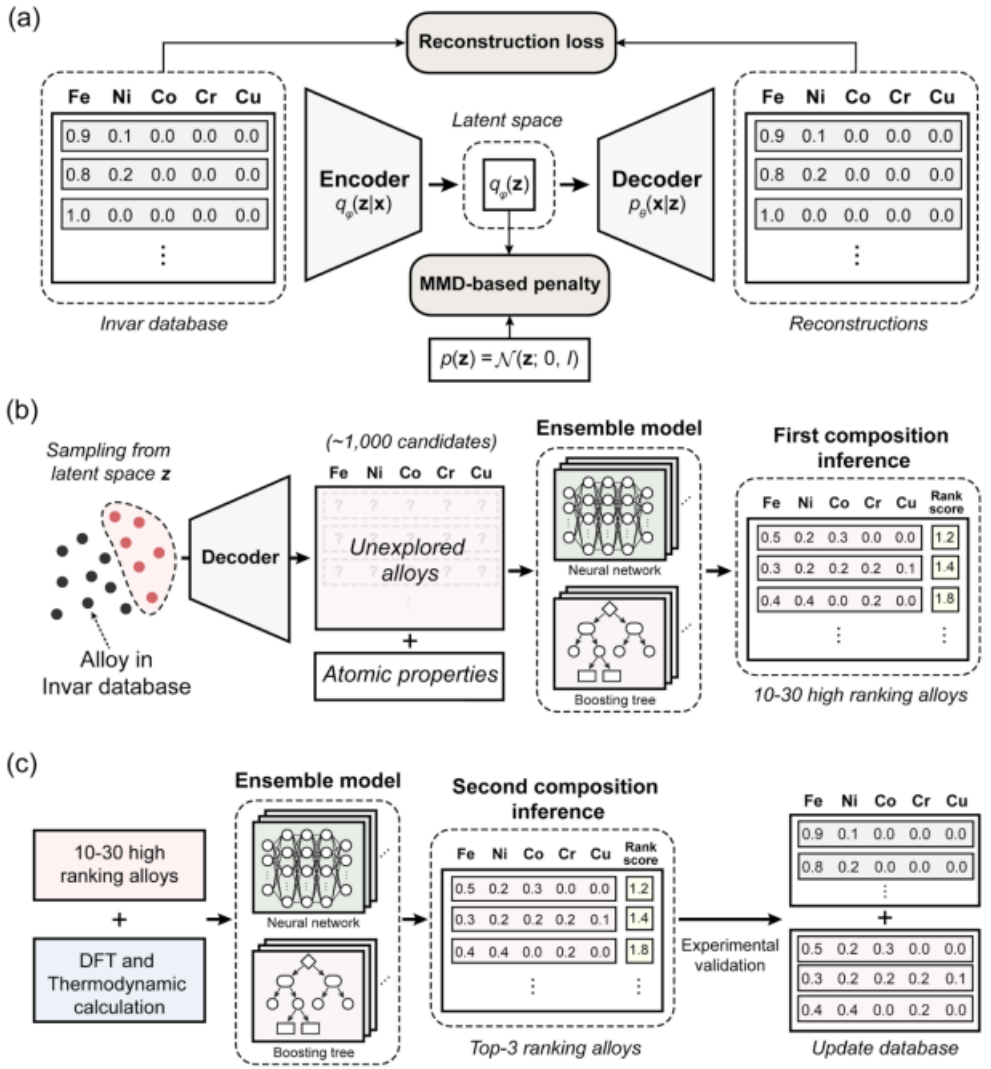
- Deutsche Forschungsgemeinschaft (DFG), Project-ID 405553726, TRR 270
- European Research Council Consolidator Grant, ERC-CoG-SHINE-771602

Journal:

- Preprint on arXiv

Links:

- https://arxiv.org/abs/2202.13753



Material Focus:

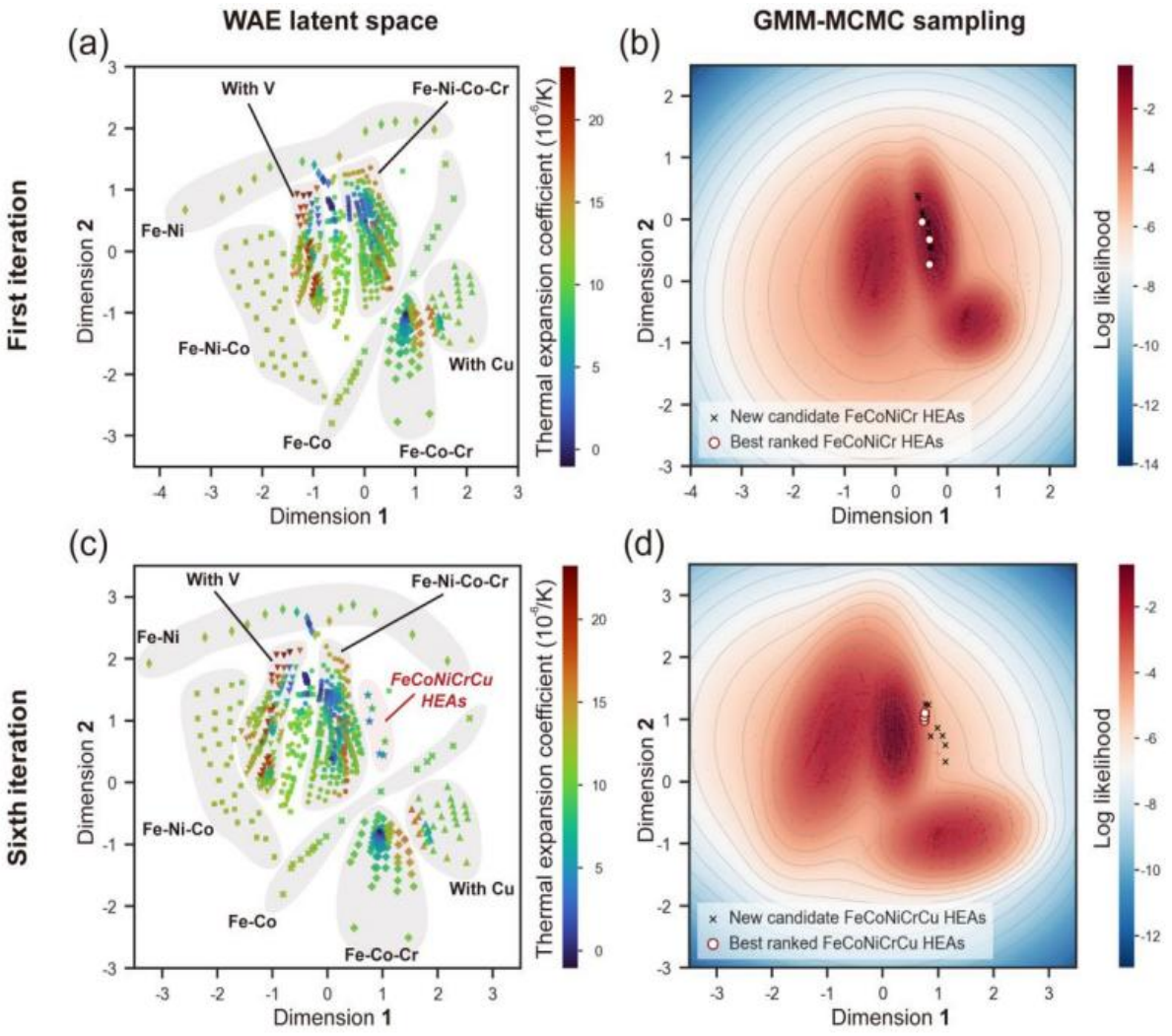
- High-entropy Invar alloys based on FeNiCoCr and FeNiCoCrCu, targeting very low thermal expansion coefficient
- Essential for cryogenic transport, satellite components and precision instrumentation

Methodology:

- HEA composition space contains at least 10^{50} possible variants
- Through active learning, the model proposes a small number of compositions, which upon computational validation are appended to the training data and retraining takes place iteratively
- A Wasserstein Autoencoder encodes every composition down, producing a physically meaningful latent space where low TEC alloys cluster together.
- A GMM and a binary classifier is fitted onto the map, and Markov Chain Monte Carlo draws fresh candidates, which the decoder converts into compositions
- Filtration takes place through VEC, Curie temperature and magnetisation from DFT and thermodynamic databases, using TERM
- Top 3 compositions are cast and measured

Validation:

- TECs were measured experimentally and the results were fed back into the training database
- Six AL iterations were completed, casting 17 alloys in total



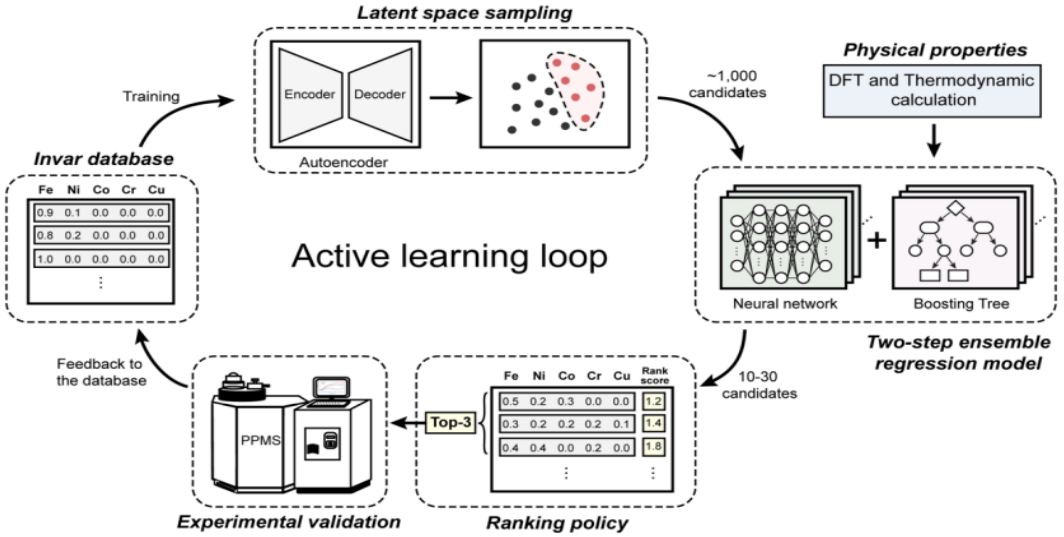
Key Concept Explained - Active Learning:

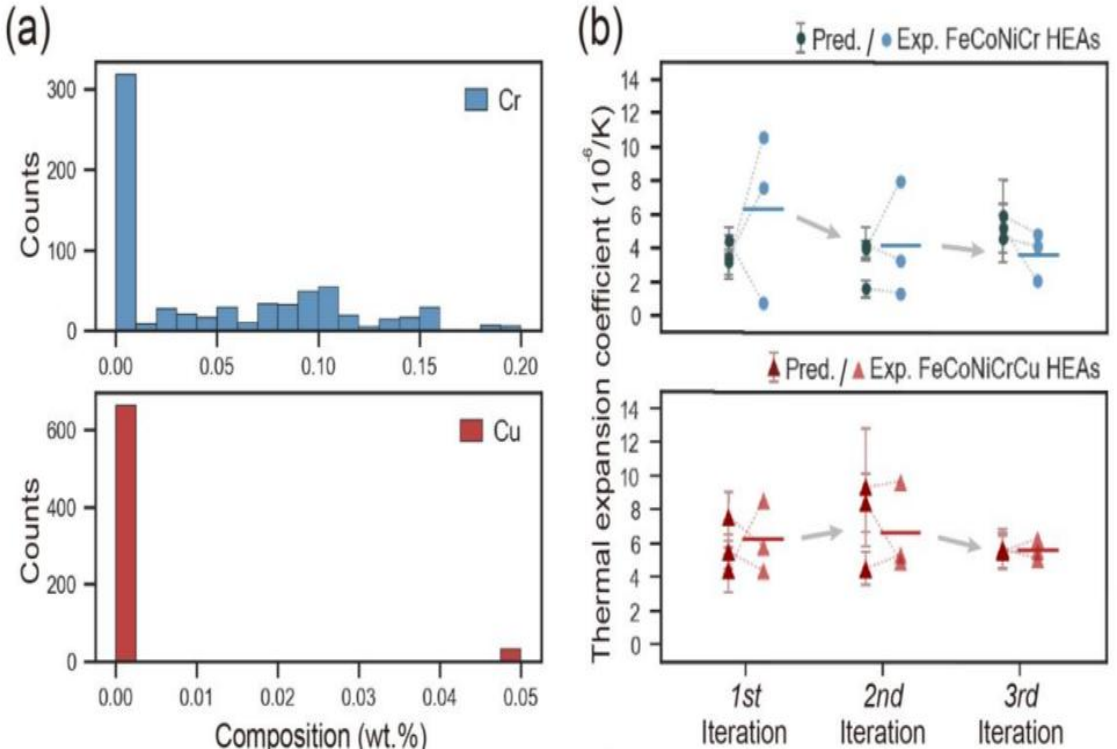
- Active learning is a closed-loop pipeline that continuously updates training dataset and iteratively retrains the model to steer its performance accordingly
- With 699 data points,the generative stage restricts sampling to a realistic 2D map, the physics-informed second stage narrows 1,000 candidates to 10-30.
- New measurements redraw the latent map: later rounds sample from a different density regions

Technology:

- Mean TEC falls from 6.49 to 5.61 to 3.65×10^{-6} K^{-1} across the first three iterations
- Two alloys, A3 (1.41×10^{-6} K^{-1}) and A9 (2.02×10^{-6} K^{-1}), reach thermal expansion around 2×10^{-6} K^{-1} at 300 K, comparable to the classic FeNi binary Invar alloy but with four principal elements
- Seven new Invar alloys with TEC below 5×10^{-6} K^{-1} were found from only 17 cast compositions.
- The new alloys occupy an otherwise empty corner of the design space, combining low thermal expansion with high configurational entropy

NOTE: The framework optimises a single property from composition alone and deliberately ignores processing and microstructure, which is defensible for the Invar effect but would not transfer directly to properties that are microstructure-controlled.





Citrine Informatics

Company Profile:

- Enterprise SaaS company applying generative artificial intelligence to materials and chemicals development.
- Aggregation and analysis of large quantities of scientific data for rapid design and manufacture of novel materials with tailored properties
- The platform is delivered through Citrine DataManager, Citrine VirtualLab, Citrine Catalyst and Citrine Professional Services
- Case studies span eleven product classes including Metals, Catalysts, Glass, Polymers and Formulations.

Country of Origin:

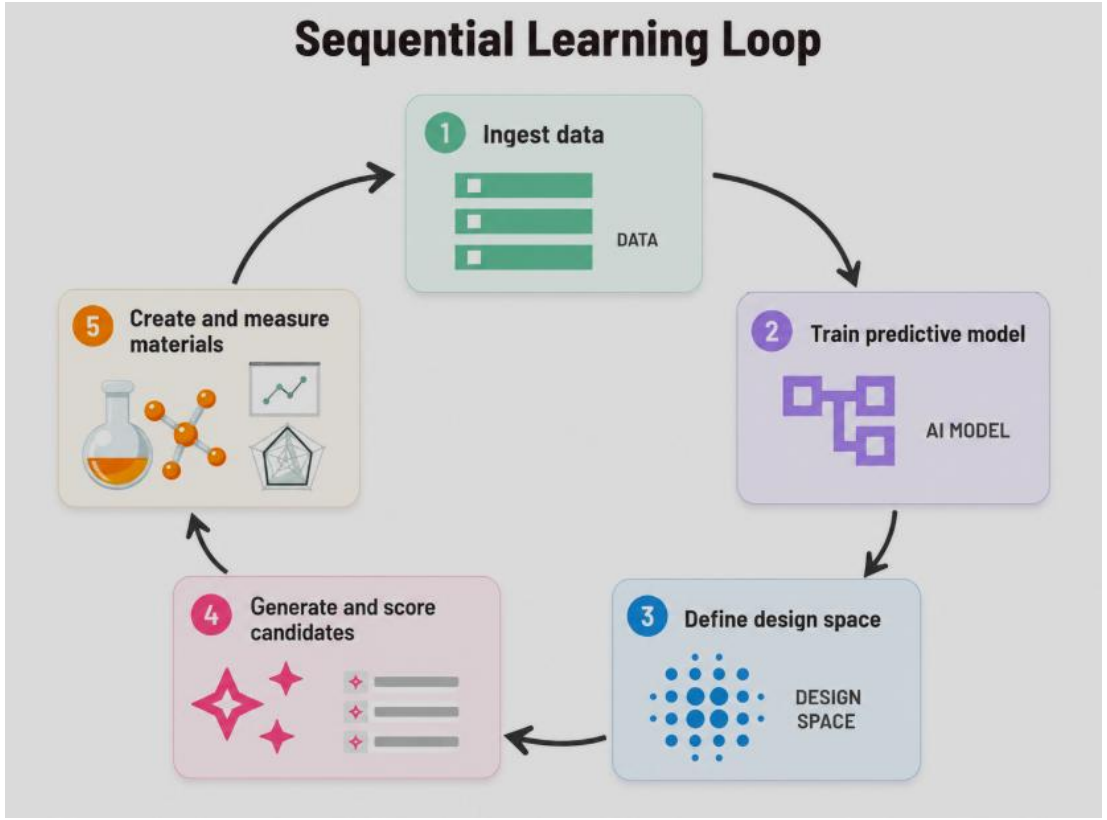
- United States

Institutes/Research Labs Involved (across the six projects):

- HRL Laboratories (jointly owned by Boeing and General Motors), NASA Marshall Space Flight Center, SLAC National Accelerator Laboratory and the Stanford Synchrotron Radiation Lightsource

Industry Collaboration:

- HRL Laboratories used the Citrine Platform to find a 3D-printable aluminium alloy strong enough for aerospace applications, and BASF collaborated with Citrine from June 2018 to accelerate development of new environmental catalyst technologies.



Material Focus:

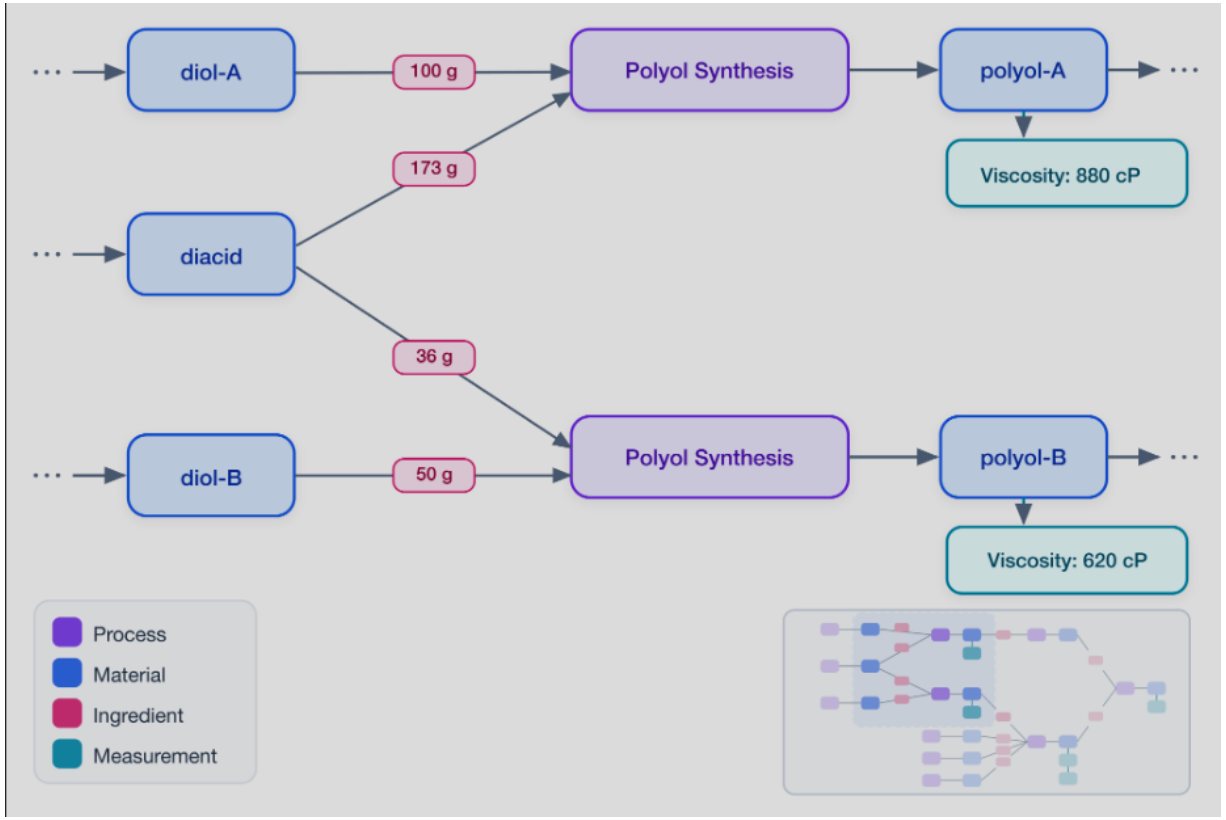
- Aluminium alloy powders for additive manufacturing, nickel-based superalloys, shape memory alloys, IN718 printed components, cold-spray-deposited alloys, colloidal platinum nanoparticles, and environmental catalysts for greenhouse gas capture
- Nickel-based superalloys typically comprise over half the weight of advanced aircraft engines, so high-temperature alloys sit at the centre of the portfolio
- Applications concentrate in aerospace, defence ground vehicles, turbomachinery, emissions control and heterogeneous catalysis, where qualification cost rather than raw material cost dominates development time

Methodology (Platform-Level Approach):

- Sequential learning: a model trained on existing data drives the next batch of experiments; those experiments are run, the results return to the database, and the model is retrained

Key Concept: GEMD

- Alloy data is hard to model because a single measured property depends on a full chain of processing
- GEMD solves this by representing material history as a connected graph. Each node is either a:
  - **Material: Batch or sample**
  - **Process: Processing steps like** melting, heat treatment, printing
  - **Measurement:** Observed property
  - **Ingredient: A** material consumed by a process
- Because the schema is generic (Material / Process / Measurement / Ingredient), it works identically across metals, catalysts, glasses and formulations.



Applied Projects I: Alloy and Part Design:

HRL Laboratories: 3D-printable high-strength aluminium alloy

- Target: aluminium alloy nanoparticle combinations that nucleate a microstructure resistant to hot cracking during 3D printing
- The GEMD-featurized model, combined with physics constraints, reduced 11.5 million candidate powder/nanoparticle combinations to 100 testable candidates in seconds
- Result: AL 7A77, the first high-strength aluminium powder feedstock registered with the Aluminum Association; NASA Marshall Space Flight Center is the first commercial customer

ADAPT / TARDEC: additively manufactured MRAP door hinge

- Adding armour plate exceeded the original hinge's design strength, causing field failures with up to two-year replacement lead times
- Citrine, Elementum 3D and University of Pittsburgh used sequential learning to qualify a completely new material and printer combination with almost no prior data for that pairing
- Result: 84% prediction accuracy on the first print, hinge reduced from six parts to one, maximum joint stress cut from 900 MPa to below 90 MPa, weight reduced 38%, print-to-delivery time cut from up to two years to days

Applied Projects II — Catalysts and Process Qualification:

BASF: AI for environmental catalyst technology (announced June 2018)

- Goal: faster development of environmental catalysts
- BASF supplied experimental data to build proprietary models on the Citrine Platform
- Result: thousands of new materials screened rapidly, with BASF reporting improving predictions across iterations

SLAC: closed-loop nanoparticle synthesis for catalysis

- Goal: colloidal platinum particles of controlled size and shape, optimising three objectives simultaneously: minimise size variation, hit a target radius, maintain yield
- ML models chose each next experiment's inputs in a flow reactor with autonomous pumps and furnaces, retrained after every run, starting from zero historical data
- Result: over 50 unique syntheses in 24 hours uninterrupted; all constraints met 21 recipes after the seed grid

Validation:

- Demonstrated across aluminium alloys, nickel-based superalloys and shape memory alloys
- Every applied project is validated by physical build and test, not by held-out data alone

Links:

- https://citrine.io/resources/case-studies/
- https://tomkat.stanford.edu/innovation-transfer/citrine-informatics
- https://citrine.io/first-to-market-high-strength-3d-printable-aluminum-alloy/
- https://citrine.io/wp-content/uploads/2021/04/Case-Study-HRL-AM-powder-development.pdf
- https://citrine.io/optimization-with-ai-in-additive-manufacturing/
- https://citrine.io/wp-content/uploads/2019/07/citrine_informatics-tardec_case_study.pdf
- https://citrine.io/ai-guided-closed-loop-synthesis-of-nanoparticles-for-catalysis/
- https://citrine.io/wp-content/uploads/2021/11/Case-Study-AI-Guided-Closed-loop-design-of-nanoparticles.pdf
- https://www.basf.com/us/en/media/news-releases/2018/06/P-US-18-075
- https://citrine.io/addressing-challenges-in-high-temperature-alloy-design/
- https://citrine.io/machine-learning-for-alloy-composition-and-process-optimization/ and https://doi.org/10.1115/GT2018-75207
- https://citrine.io/rapid-qualification-of-additive-manufactured-parts/
- https://citrine.io/modeling-additive-manufacturing-performance/
- https://pubs.aip.org/aip/jcp/article/154/22/224201/313373
- https://citrineinformatics.github.io/gemd-docs
- https://citrine.io/product-class/metals/ and https://citrine.io/product-class/catalysts/

QuesTek Innovations LLC

Company Overview:

- Has used Integrated Computational Materials Engineering (ICME) as its core technology to solve alloy development problems for industry
- ICMD® - a cloud-based SaaS platform for simulation-led materials design. ICMD® 2.0 introduces an AI Assist feature for workflow streamlining and results review

Country of Origin:

- United States

Institutes/Research Labs Involved:

- US Air Force Research Laboratory and America Makes, US Navy, US Department of Energy, and commercial partners including Daido Steel, Aubert & Duval, Stoke Space, Pangea Propulsion, McCoy Global and CBMM

Industry Collaboration:

- Extensive. Daido Steel adopted ICMD® to shorten specialty steel development and improve production quality stability; Aubert & Duval collaborates on predictive alloy design; CBMM collaborated on the niobium capability; and QuesTek alloys are deployed commercially, including flight-qualified parts

References:

- https://questek.com/
- https://questek.com/products-client-services/icmd-materials-design-platform/
- https://questek.com/case-studies/
- https://questek.com/case-study-ferrium-n63-questeks-answer-to-one-of-aerospaces-biggest-challenges/
- https://questek.com/questeks-single-crystal-nickel-answers-the-department-of-energys-call/
- https://questek.com/case-study-questeks-ferrium-c64-additive-manufacturing-process-make-army-helicopters-tougher-lighter-and-safer/
- https://questek.com/questek-daido-steel-partners-in-innovation-through-materials-by-design/
- https://questek.com/burn-resistant-alloy-for-stoke-spaces-reusable-rocket-engines/
- https://questek.com/questek-announces-niobium-suite-in-icmd-platform-expanding-advanced-materials-capability-with-collaboration-from-cbmm/
- https://www.ansys.com/partner-ecosystem/technology-partners/questek-innovations

Ferritico AB

- **Company Overview:**
- Swedish machine learning software company applying AI to the global steel industry
- CEO: Claes Holmström, CSO: Joakim Odqvist, CTO: Peter Hedström
- Based in Stockholm, Sweden. It is a KTH Holding portfolio company

Country of Origin:

- Sweden

Institutes/Research Labs Involved:

- KTH Royal Institute of Technology

Industry Collaboration:

- Ferritico reports being in production use at a global steel company, and collaborates with Outokumpu, Boliden and Outotec through EIT consortium projects. A partnership with STILFOLD produced Ferrogami, announced February 2025

Material Focus:

- Steels, covering development of novel grades, manufacturing process optimisation and downstream implementation
- Ferrogami, developed with STILFOLD, is a steel designed for lower carbon emissions and resource efficiency while retaining foldability, durability and strength

Methodology:

- Steel properties and process data are collected, then physics-based and machine learning models are applied for property prediction and data correlation
- Customized models are delivered as modular services rather than as a single monolithic tool.
- The heat treatment modules target optimised heat treatment schemes for advanced steels
- Two consortium projects extend the model library into degradation properties: CorTools is a corrosion resistance prediction model and EndureIT is an embrittlement prediction model
- The company also builds customised AI digital twin simulation solutions

Validation:

- The software is in production use at a global steel company, a huge commercial validation
- Consortium projects provide multi-partner experimental grounding for the corrosion and embrittlement models

Key Concept Explained – Domain-Narrow Materials Informatics:

Technology:

- The platform enables prediction of mechanical and thermodynamic process properties of steels
- Heat treatment modules reduce the experimental testing required when optimising heat treatment processes
- Ferrogami, produced by combining Ferritico's AI-driven alloy design with STILFOLD's patented industrial origami technology, achieves an improved strength-to-weight ratio in folded components, targeting humanoid robotics applications with dynamic motion demands

NOTE: Ferritico publishes almost no quantitative performance data. No model accuracies, dataset sizes, benchmark comparisons or customer-verified time savings are disclosed publicly, and the customer is unnamed.

Links:

- https://ferritico.com/
- https://www.ferritico.com/company
- https://www.ferritico.com/machine-learning-software-services
- https://kthholding.se/en/project/ferritico/
- https://kthventures.se/en/portfolio/ferritico
- https://www.crunchbase.com/organization/ferritico

CuspAI

Company Profile:

- British AI-for-materials company headquartered in Cambridge, United Kingdom, incorporated in March 2024, and it builds generative foundation models for the design of new chemical compounds and engineered materials.
- The company was co-founded by Dr Chad Edwards and Professor Max Welling

Country of Origin:

- United Kingdom

Institutes/Research Labs Involved:

- Georgia Institute of Technology, University of Amsterdam

Industry Collaboration:

- CuspAI has reported partnerships across carbon capture with Meta, automotive materials with Hyundai Motor Group announced in November 2025, and water treatment and PFAS remediation with Kemira

Funding Received:

- CuspAI raised total funding of $130 million.

Journal:

- No standalone peer-reviewed company paper. The associated public research output is the OpenDAC dataset, produced jointly with Meta FAIR and Georgia Tech.

- **Material Focus:**
- Metal-organic frameworks and related porous materials, targeting carbon dioxide capture
- Materials for further applications including energy storage, catalysis, and gas and water purification

Core ML Innovation — Generative Foundation Models Trained on Purpose-Built Data:

- LLMs trained on broad structural and property data, adapted to material design tasks
- The company's flagship model is MOFGEN, facilitating property driven candidate generation and exploration of chemical space. The model is based on geometric deep learning and equivariant neural networks
- Working with Meta's FAIR team and Georgia Tech, the OpenDAC dataset was created, described as the world's largest open dataset on carbon capture materials and containing more than 100 million data points derived from DFT calculations

Validation:

- Validation has been primarily computational, with physics-based simulation used to test and rank structures that the generative model proposes before any physical synthesis is attempted.
- Commercial validation is provided through named industrial partnerships, in which partner companies supply application-specific requirements and evaluate candidate materials against their own performance criteria.

Links:

- https://cusp.ai/
- https://medium.com/@CuspAI/cuspai-secures-30m-to-combat-climate-change-with-ai-designed-materials-e334b4caf560
- https://www.nea.com/blog/algorithms-to-atoms-our-investment-in-cuspai
- https://open-dac.github.io/
- https://app.dealroom.co/companies/cuspai

Radical AI

Company Profile:

- Radical AI is a New York-based materials company founded in 2024 that is developing an autonomous, closed-loop platform for materials discovery
- Joseph, Jorge Colindres and Professor Gerbrand Ceder are the founding team
- The company's laboratory operates out of the Alexandria Center for Life Science in Manhattan, where robotic stations weigh and mix elemental feedstock, melt the mixture into an alloy, and then analyse its composition, structure, hardness, and resistance to oxidation and heat
- Radical AI positions itself as vertically integrated rather than software-only, and it emphasises intellectual property around processing and scaling rather than composition patents alone.

Country of Origin:

- United States

Institutes/Research Labs Involved:

- Lawrence Berkeley National Laboratory

Industry Collaboration:

- Strategic co-investors include NVIDIA and the energy company Eni.

Funding Received:

- Total reported funding of $65 million

Journal:

- No company publications have been identified.

Material Focus:

- Rare-earth-free permanent magnets for wind turbines and motors, critical minerals
- Alloys for hypersonic flight, reusable space systems, submarines, turbines and jet engines

Core ML Innovation: Property-Driven Inverse Design Closed by an Autonomous Laboratory:

- Radical AI's approach is described as a materials flywheel, in which a generative AI engine proposes candidate compositions, an atomistic modelling layer screens those candidates computationally, a robotic laboratory synthesises and characterises them
- Property driven inverse design
- A machine-learning interatomic potential underlies the atomistic screening step, providing fast approximate simulations of how atoms in a candidate alloy will arrange themselves and interact
- The AI engine is designed to ingest multimodal information rather than numerical data alone, drawing on published scientific papers, textbooks, patents, experimental data and microscopy images

Validation:

- The company reports testing tens of thousands of materials per week, and states that its laboratory can complete in a single day work that would take a team of human scientists several weeks using conventional methods.

Technology and Outcomes:

- Proprietary materials are reportedly under development in the hypersonics and nuclear fusion application areas, in addition to the company's stated work on rare-earth-free magnets and electric vehicle battery chemistries.
- The operational laboratory at the Alexandria Center for Life Science in Manhattan distinguishes Radical AI from several competitors whose robotic facilities were, as of recent reporting, still being commissioned or had not yet been built out to full scale.

NOTE: Radical AI has not published any discovered material, and no independently characterised result has been reported to date. Throughput figures such as testing tens of thousands of materials per week are company statements without a public definition of what qualifies as a completed test, and there is no third-party benchmark or peer-reviewed validation available. Every discovery claim is unevidenced

Links:

- https://www.radical-ai.com/
- https://workingcapitalfund.com/why-we-invested-in-radical-ai/
- https://www.fastcompany.com/91543082/in-this-manhattan-lab-ai-designs-materials-from-scratch
- https://cen.acs.org/physical-chemistry/computational-chemistry/Self-driving-labs-changing-chemists/104/web/2026/06
- https://www.technologyreview.com/2025/12/15/1129210/ai-materials-science-discovery-startups-investment/

The India Picture: Substantial AI & DiscoveredMaterials

Substantial AI:

Material Focus:

- Substantial AI is a New Delhi-based startup applying machine learning to glass and ceramics composition-property design
- Flagship: Python for Glass Genomics (PyGGi), was launched by M3RG group at IIT Delhi

Core ML Approach:

- PyGGi is structured as three linked modules:
  - PyGGi Bank is a composition-property database compiled from literature and the open-source SciGlass database, functioning as the training data layer.
  - PyGGi Seer is a property-prediction model, using either a neural network or a Gaussian process model, selectable by the user.
  - PyGGi Zen is the inverse-design layer, searching for novel glass compositions that meet a specified target property and constraint set.
- The project originated as sponsored academic research within IIT Delhi's Department of Civil Engineering and Department of Materials Science

Validation:

- Published documentation, including a peer-reviewed feature article in the American Ceramic Society Bulletin, describes the tool's methodology and use cases in glass composition optimisation, giving Substantial AI a more academically transparent public record than most of the Western startups covered in this review

Discovered Materials:

Material Focus:

- Discovered Materials targets thermal management and structural materials for semiconductor chips and data-centre infrastructure
- Its stated ambition extends across the entire semiconductor materials stack, including advanced packaging technologies such as 3D stacking, rather than remaining confined to a single material class.

Core ML Approach:

- The company is built by Advaith Sridhar and Akash Ramdas, both IIT Madras alumni.
- Agentic Approach: AI agents are deployed across the full discovery pipeline spanning candidate proposal, simulation, synthesis and physical laboratory validation
- The company has released called Material Discovery Bench, a research benchmark for evaluating how well AI models search for new materials in a real-world chip-relevant problem space

Funding Received:

- $9 million in seed funding

Links:

- https://www.startus-insights.com/innovators-guide/materials-informatics-companies/
- https://substantial.ai/
- https://pyggi.iitd.ac.in/about
- https://ceramics.org/wp-content/bulletin/2024/articles/F-Sachan_aug24.pdf
- https://yourstory.com/2026/08/deeptech-startup-discovered-materials-raises-9-million-in-seed-round

Materials Discovery: Thermodynamic Modelling

A Deep Dive into ThermoCalc, MatCalc, Pandat, etc in Materials Discovery

The CALPHAD Approach

- Construction and optimization of thermodynamic models of each phase in a material system
- Gibbs Free Energy = f(State Variables):
  - Temperature, Pressure, Composition
  - Reference, Ideal Mixing, Excess
- Compound Energy Formalism
- Redlich-Kister relation to model non-ideal GFE
- Interaction parameters (coefficients of Redlich-Kister polynomial) fitted to experimental and first-principles calculation data
- Lagrangian constrained, nonlinear least squares minimization for G(min) determines equilibrium phases and compositions

$$G^{ex} = x_A x_B \sum_{i=0}^{n} L_i (x_A - x_B)^i$$

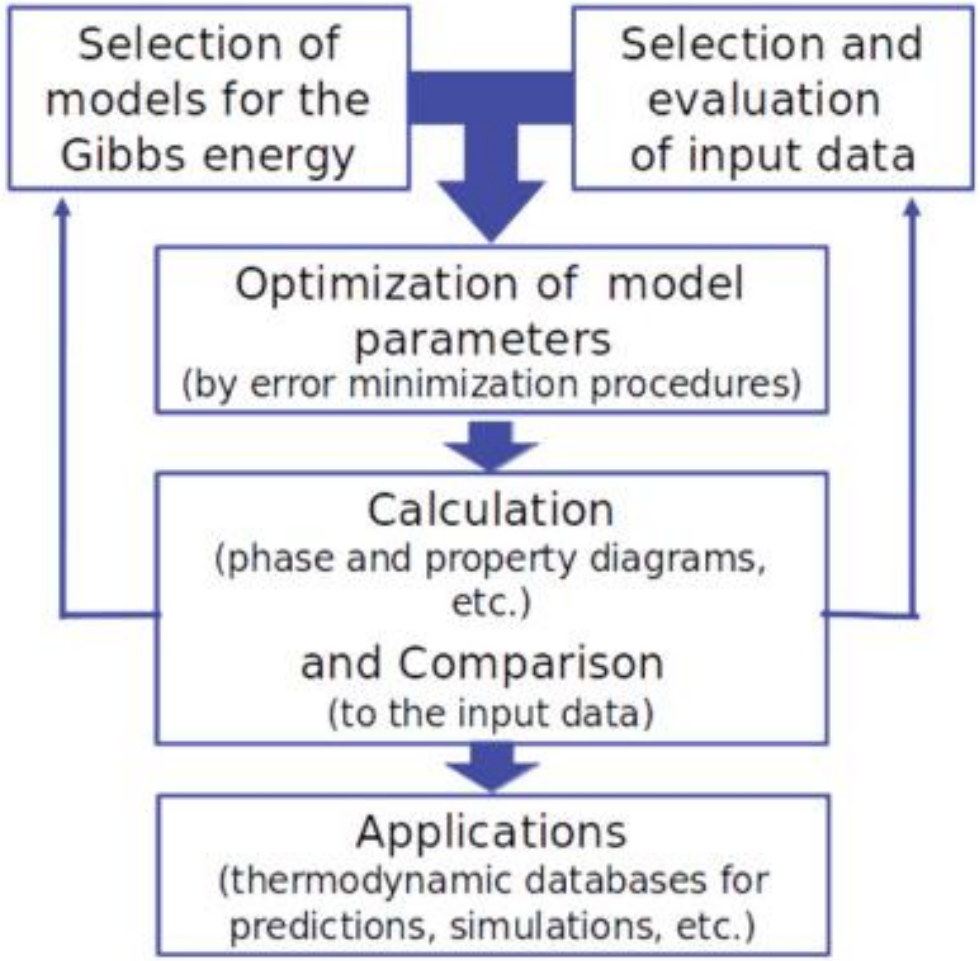


Figure 1. Flowchart representing the core of the CALPHAD method.

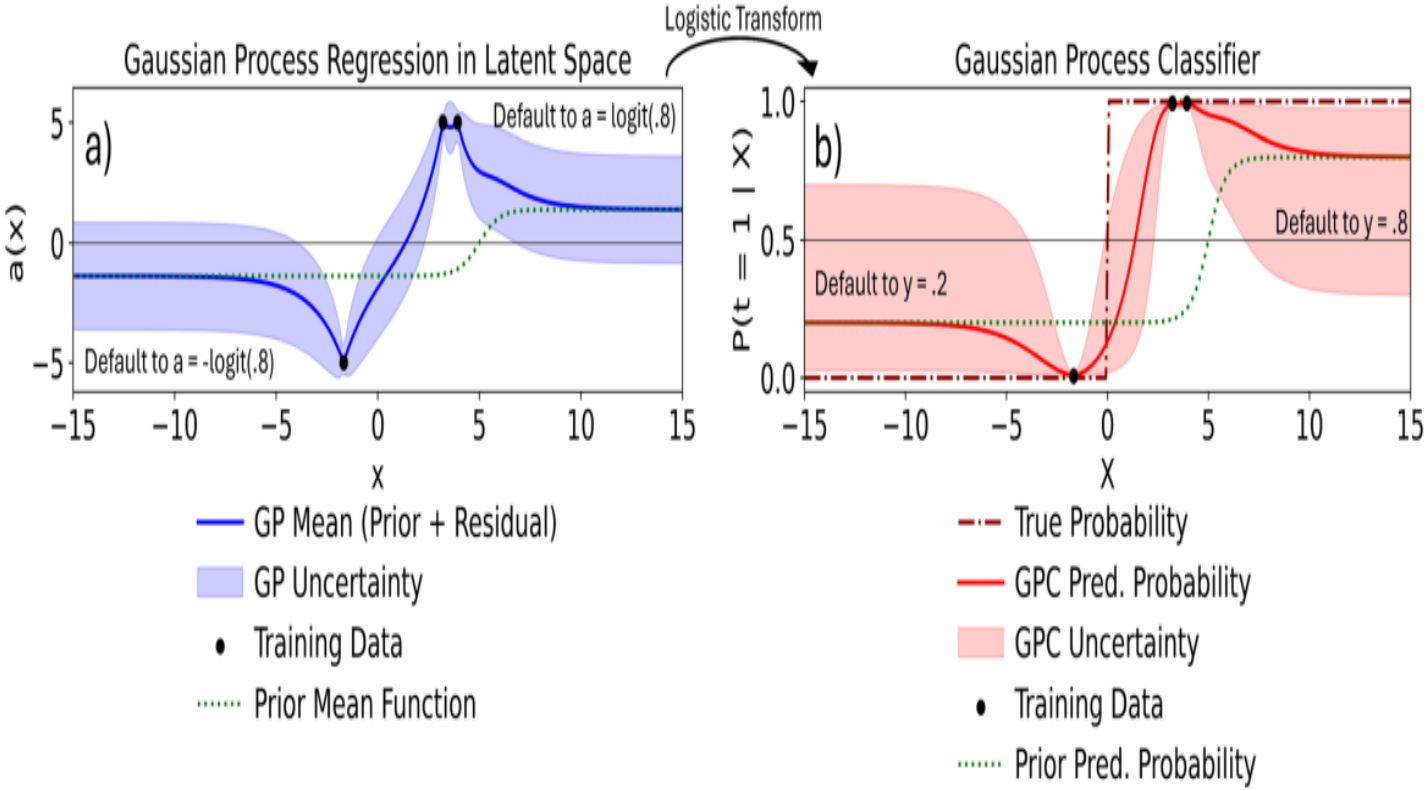
ThermoCalc Workflow

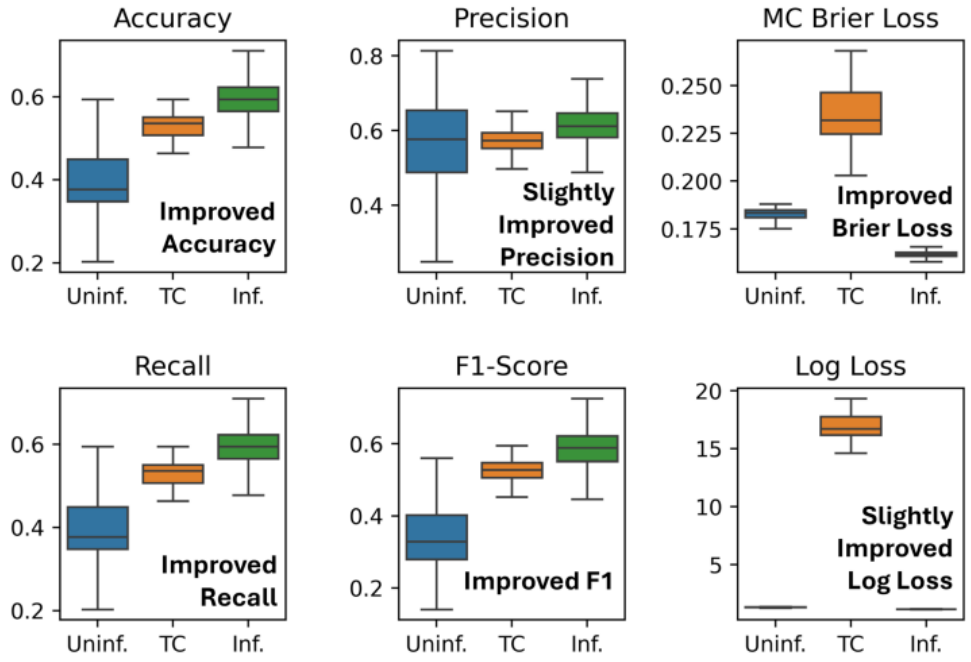


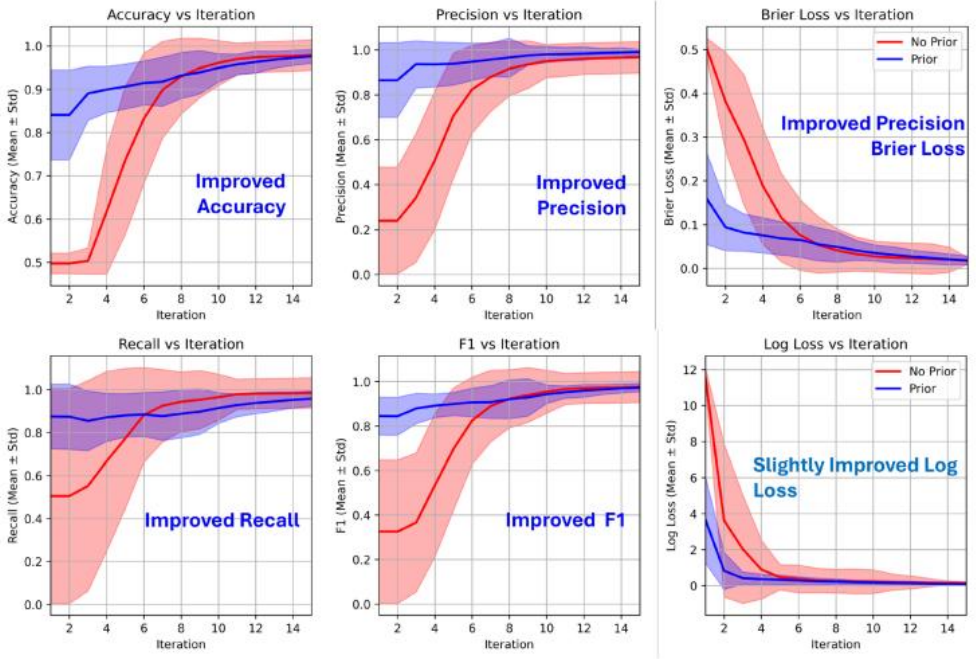
- **Data Capture:** Experimental data collection on phase equilibria, thermochemical properties like enthalpy of mixing, formation energies, empirical relations, ab initio calculations.
- **Critical Assessment and Model Selection:** Model selection. In the case of a binary system, there exists a number of thermodynamically distinct phases. For each of these, a GFE model is assigned by a human depending on its crystal structure, a polynomial in chemical composition and temperature.
- **Assessment and Optimization:** The free parameters of the model are fitted to the input data, a multi-objective optimization with constraints. In Thermo-Calc, optimization is completed using the built-in PARROT Module.
- **Storage:** After fitting, the Gibbs energy functions with their optimized free parameters are stored in a database.
- **Validation:** If the agreement with real multicomponent commercial alloys is not good for key data points, a re-optimization of one or more lower order systems is the way to rectify this.

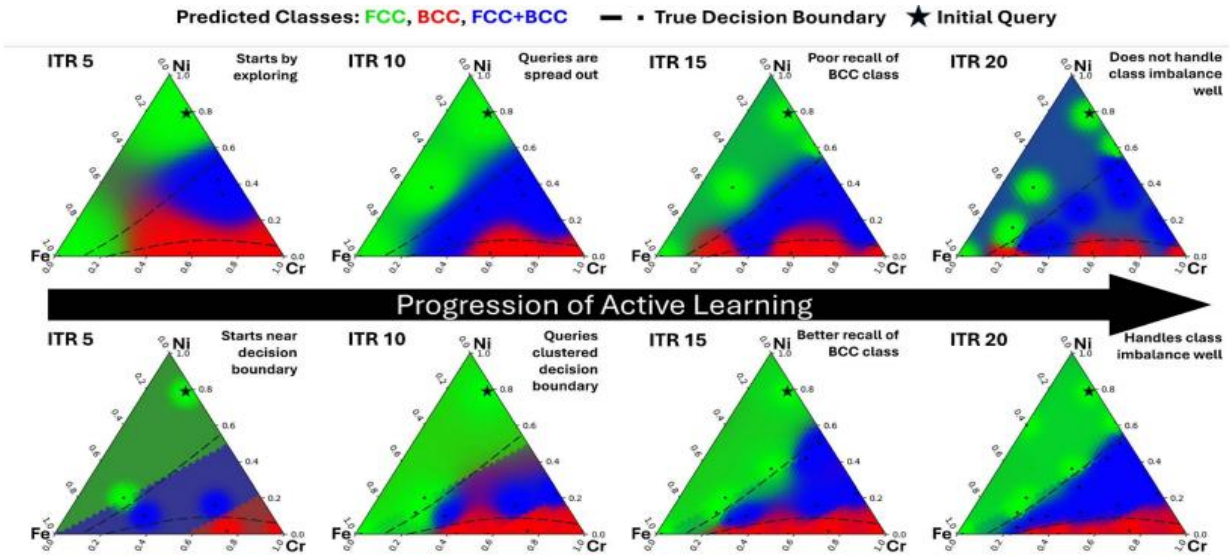
Case Study: 1 (USA)

- Physics informed Gaussian process classification of alloy design
- Modelling alloy design as a constraint problem (non negotiable aspects which must be satisfied) instead of optimization (aspects to be maximized/minimized)
- CALPHAD predictions used as a physics-based prior mean function for their Bayesian classification function
- TCHEA-6 ThermoCalc database utilized
- Link: https://pubs.rsc.org/dd/article/4/7/1884/889019/Physics-informed-Gaussian-process-classification

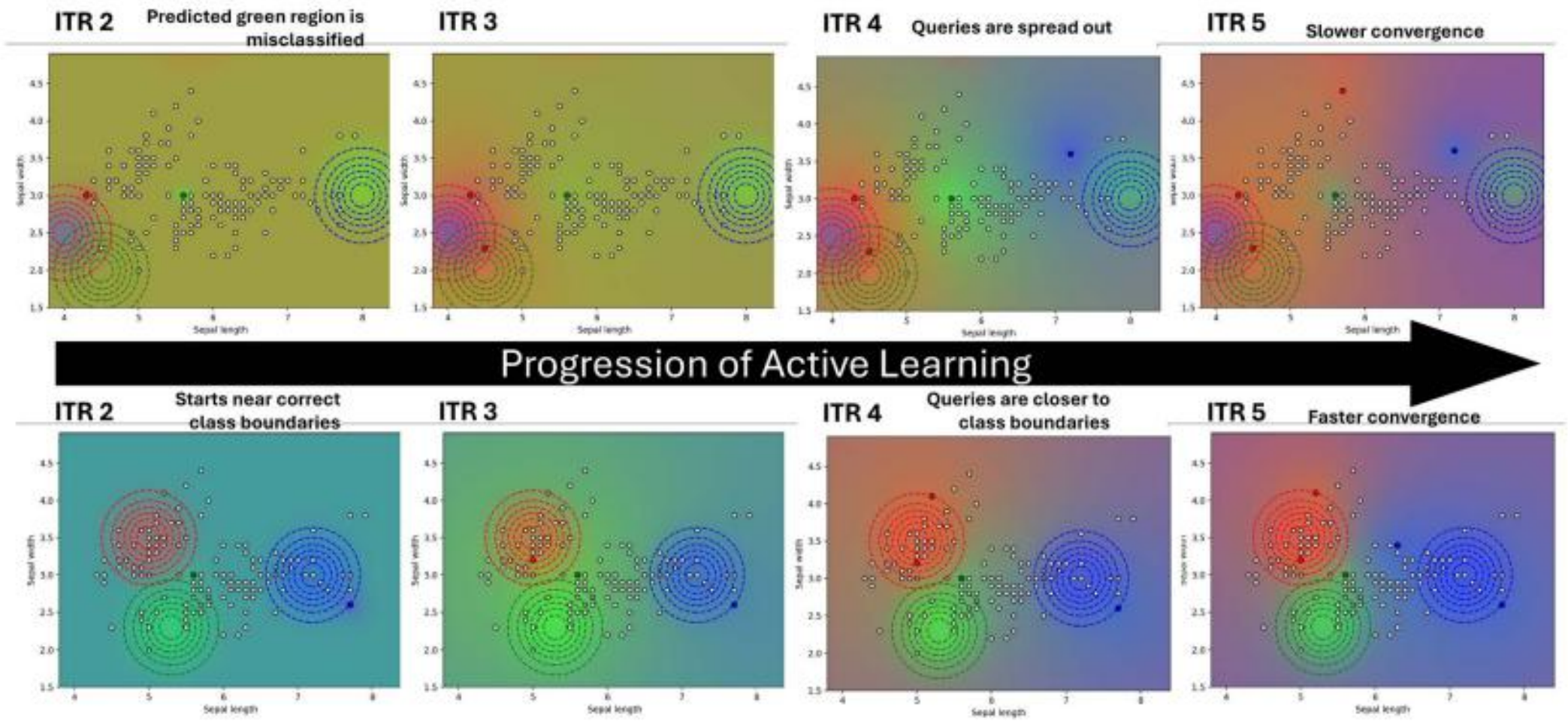








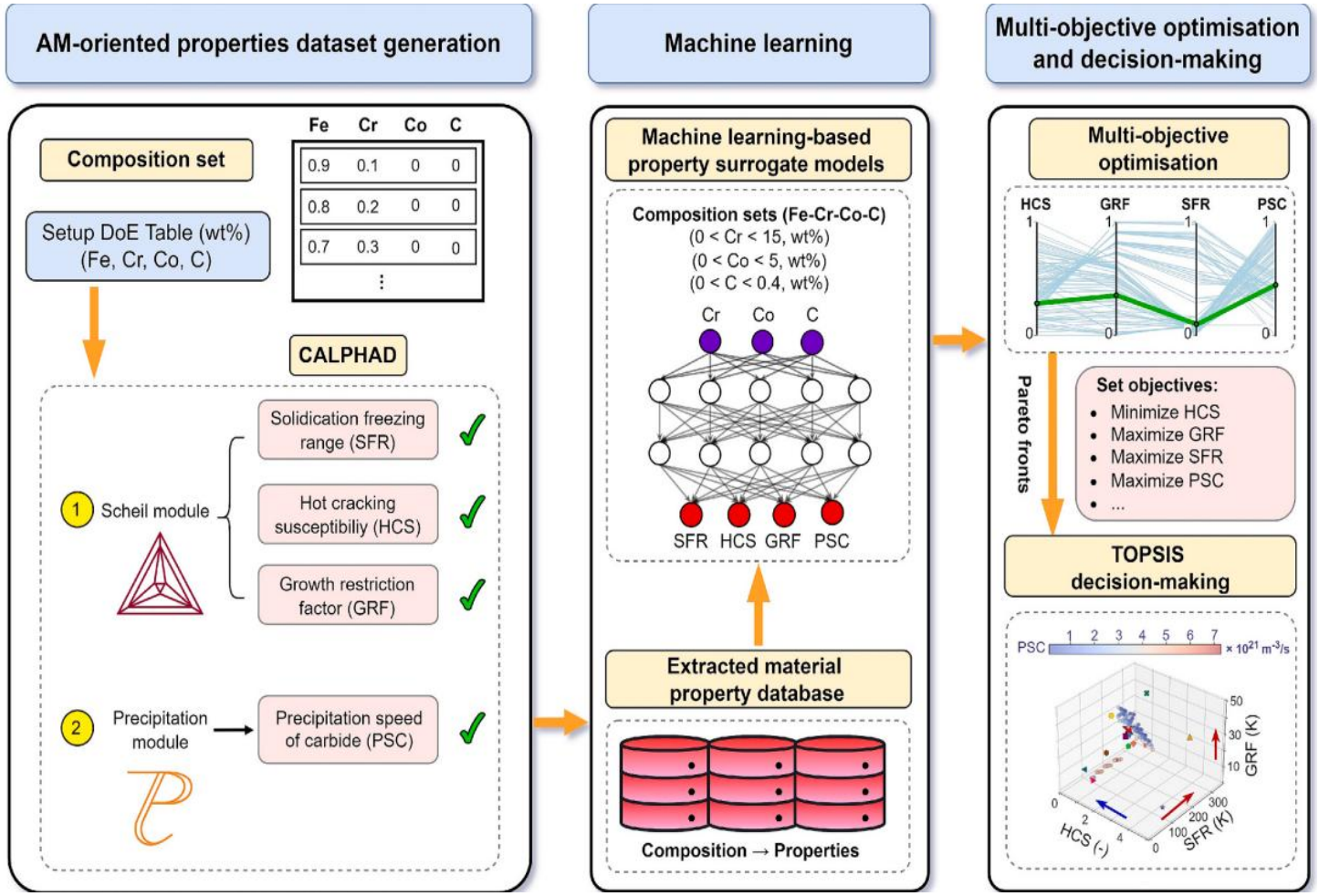
Model Performance Metrics

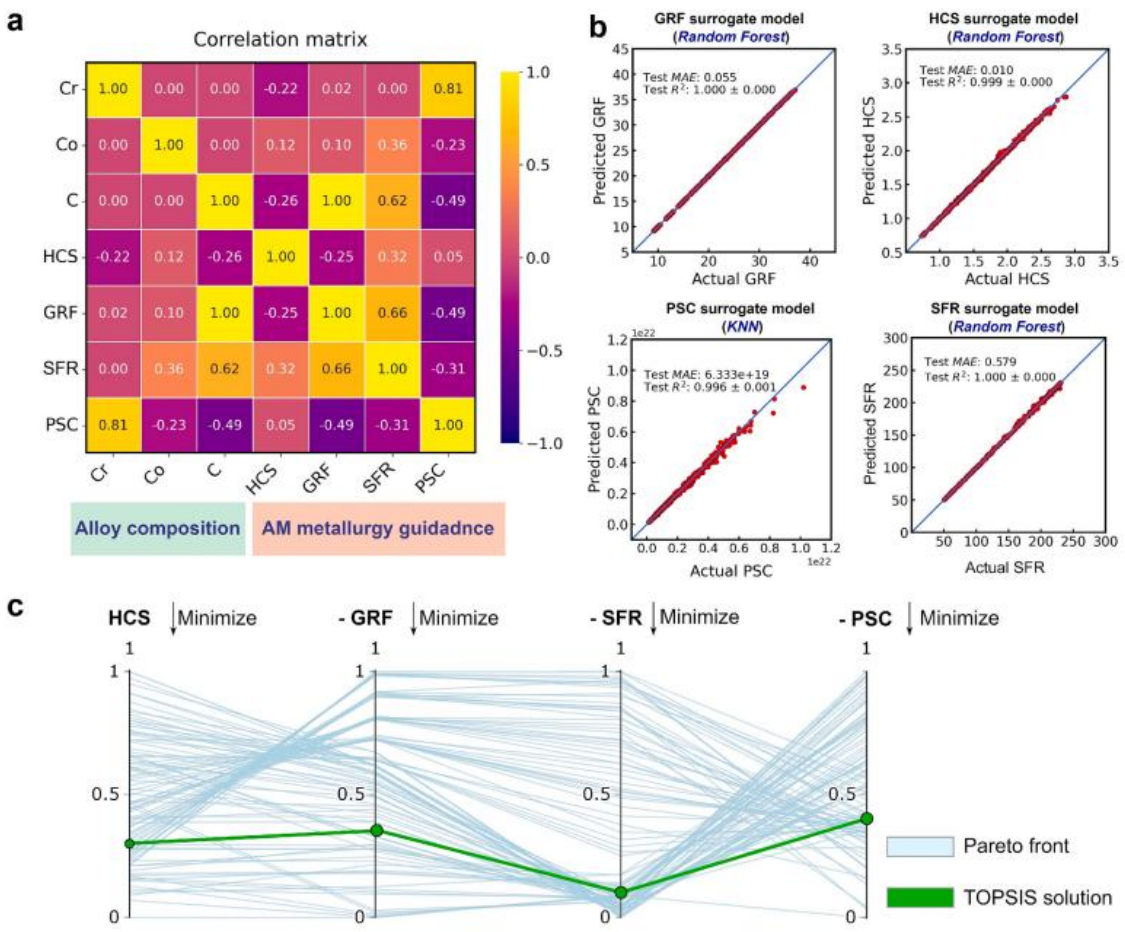


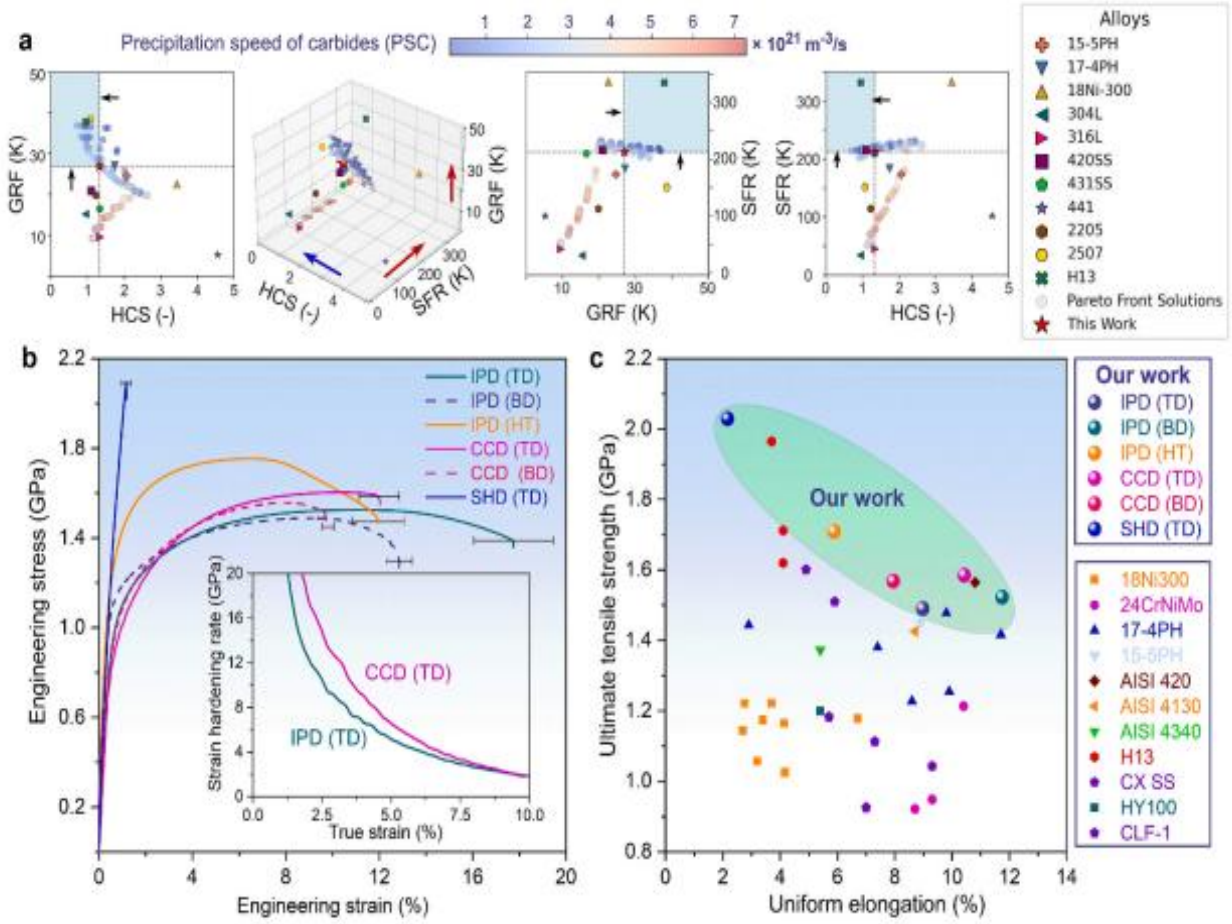
Active Learning Schemes compared using Informative vs Bad Priors

Case Study: 2 (Singapore)

- Novel AM metallurgy-guided ML alloy design framework
- Target properties: SFR, GRF, HCS, PSC
- Evaluated using high-throughput CALPHAD simulations and predicted via ML-based surrogate models.
- Optimized using multi-objective evolutionary algorithms (NSGA-3) coupled with decision-making strategies
- Validation: A newly designed Fe–Cr–Co–C martensitic stainless steel (MSS) was fabricated using laser-directed energy deposition
- Link: https://www.sciencedirect.com/science/article/pii/S136970212500272X?via%3Dihub



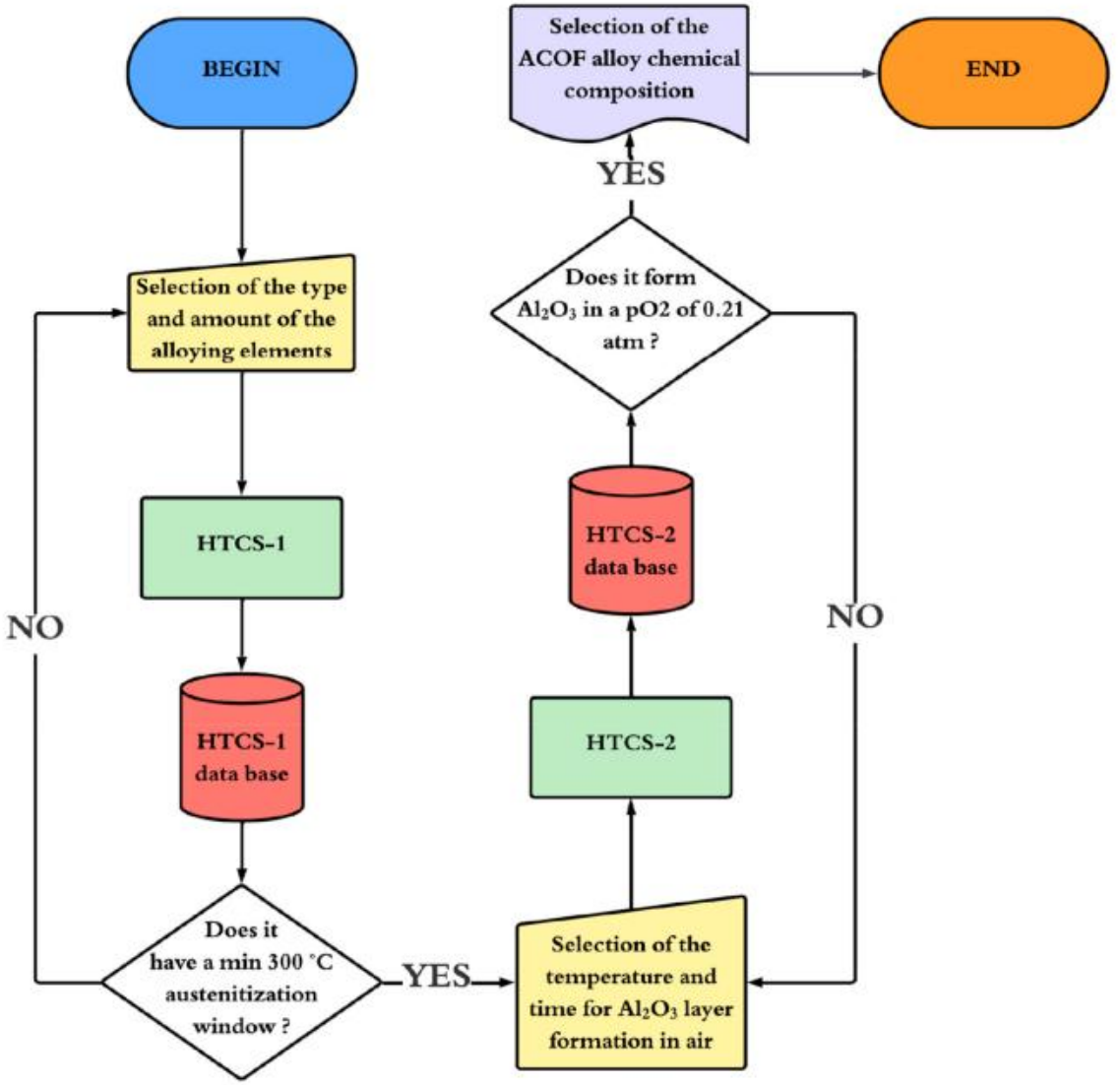


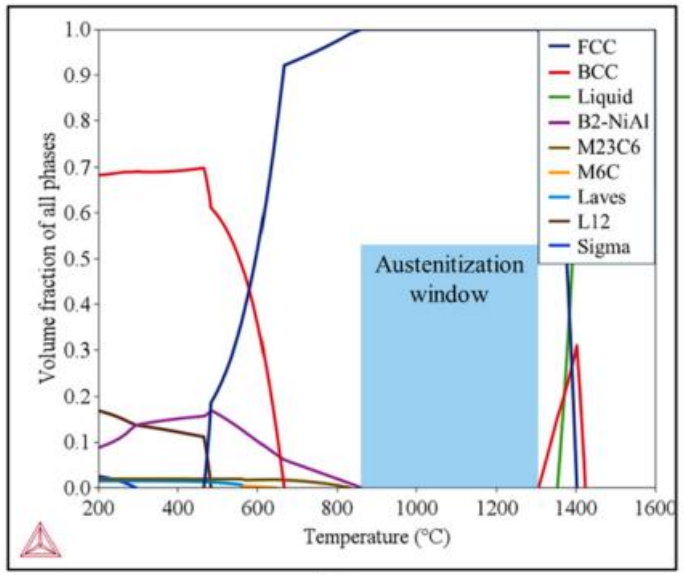


Spearman correlation matrix between the alloy composition and physical properties, model performance metrics, Pareto front solutions, highlighting the TOPSIS-selected solution in green

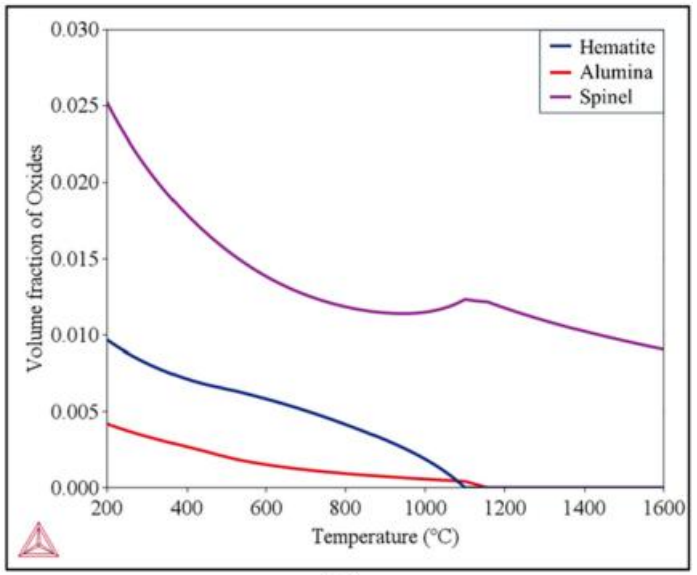
Case Study: 3 (Sweden)

- Development of an alumina-containing oxide-forming alloy for enhanced oxide formation, austenitization window and hydrogen trapping
- High Throughput Computational Screening: Elemental selection, automated CALPHAD calculations, data filtering, visualization
- CALPHAD prediction was that the selected alloy should have a broad austenitic stability range of approximately $850 - 1300^{\circ}C$ and form mixed Fe–Al-based oxides during oxidation.
- Validation; The cast alloy reduced total hydrogen uptake by >50% through trapping at BCC_B2–NiAl/austenite interfaces
- Link: https://www.sciencedirect.com/science/article/pii/S2238785426014079?fr=RR-2&ref=pdf_download&rr=a3fa0c98adb1d40b

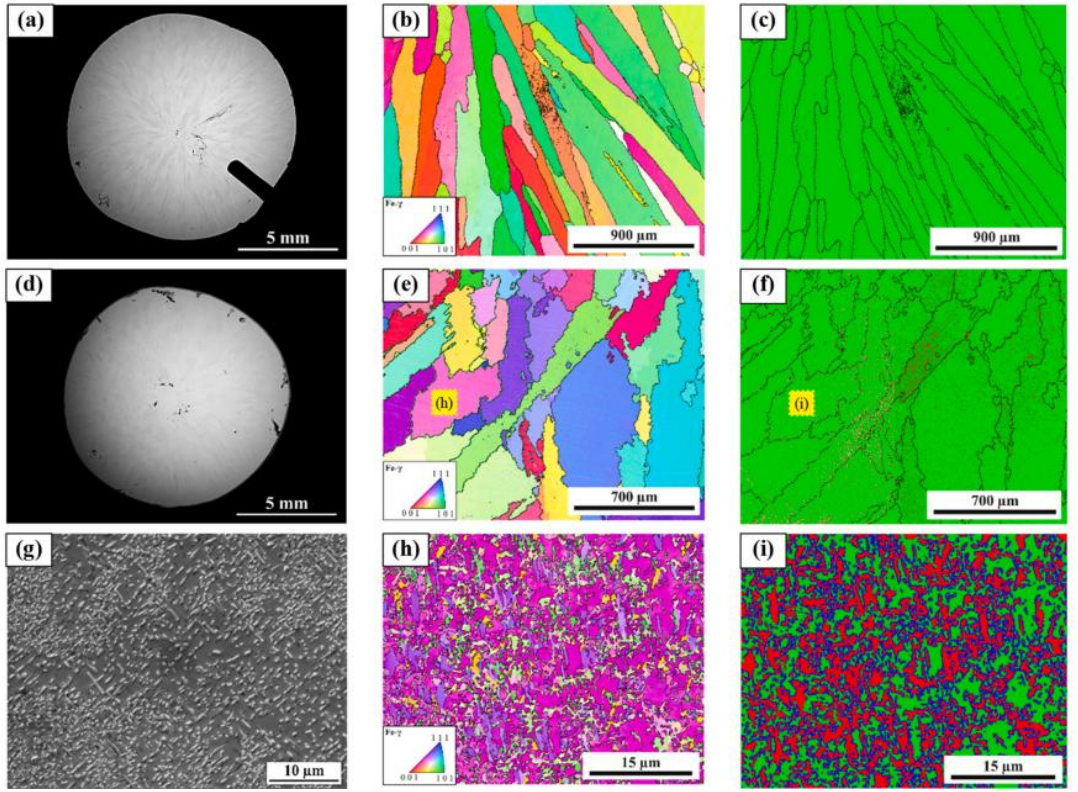


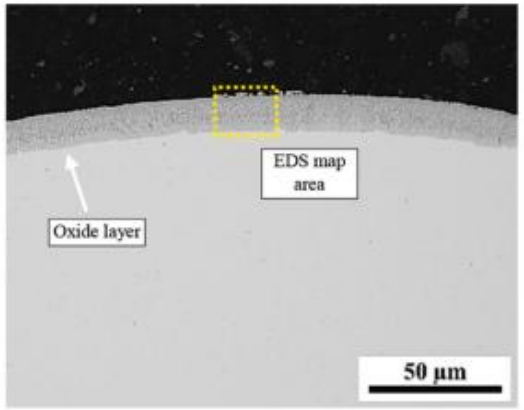


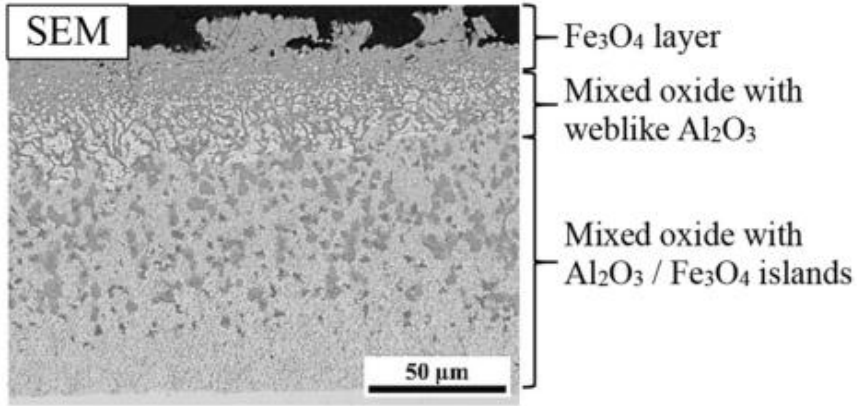
(a)



(b)



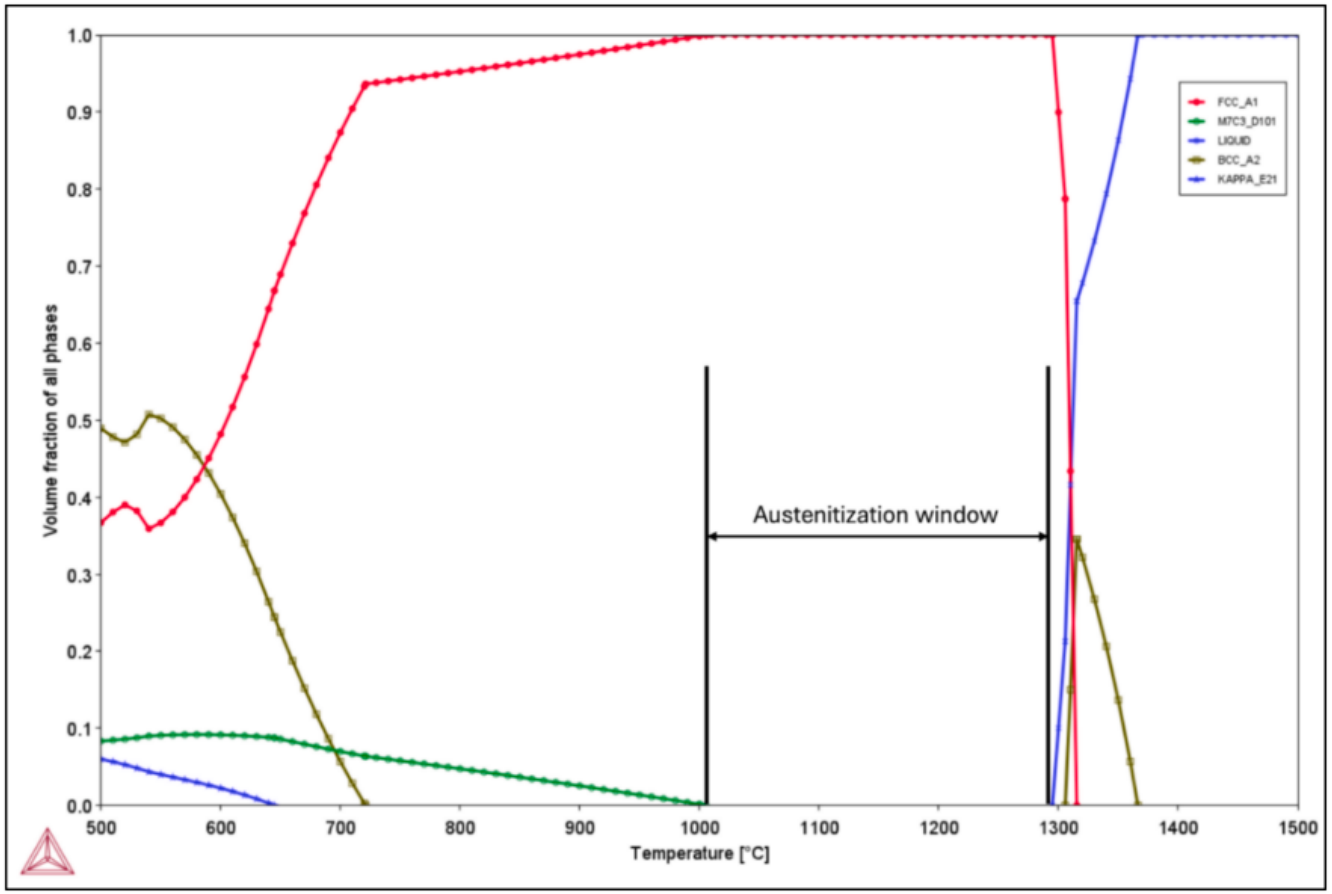


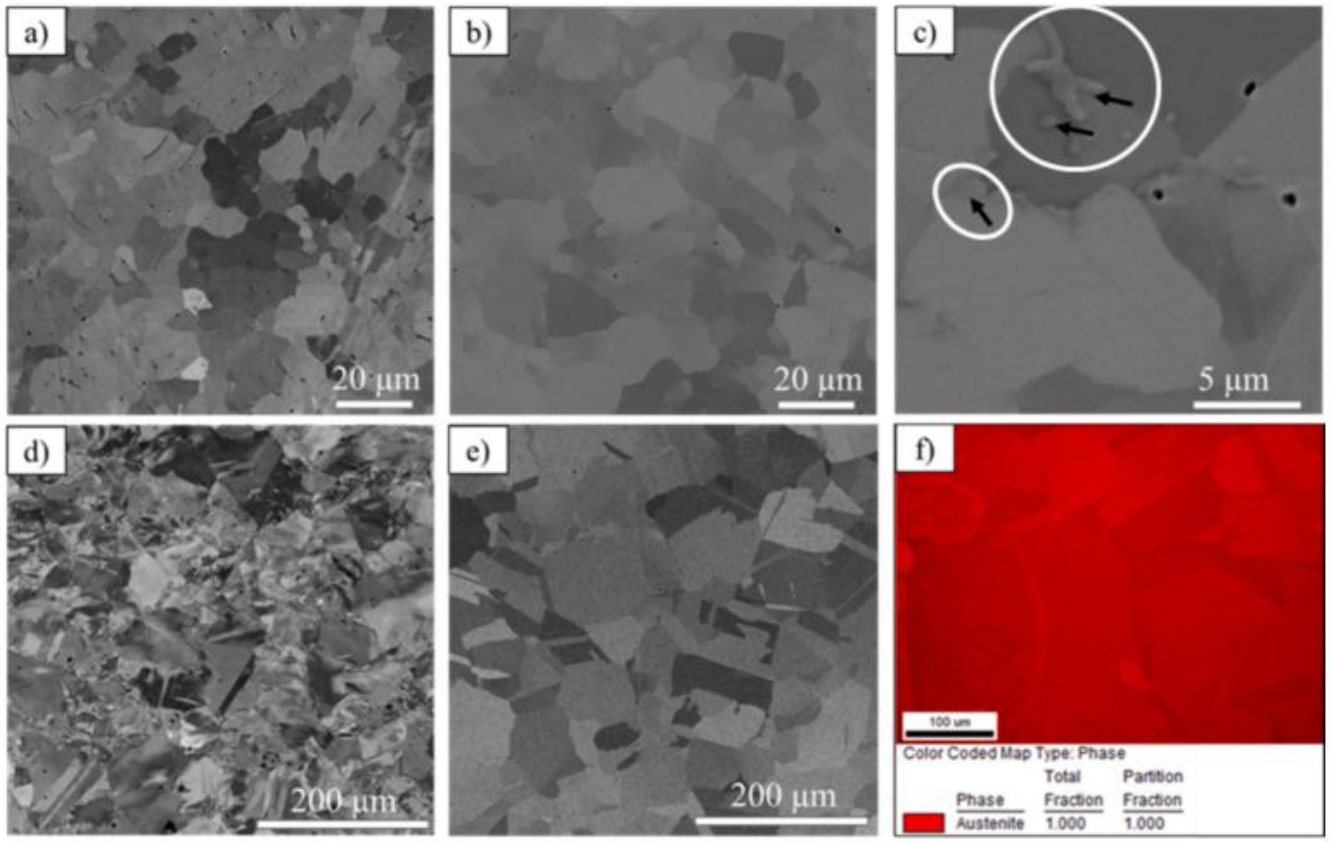


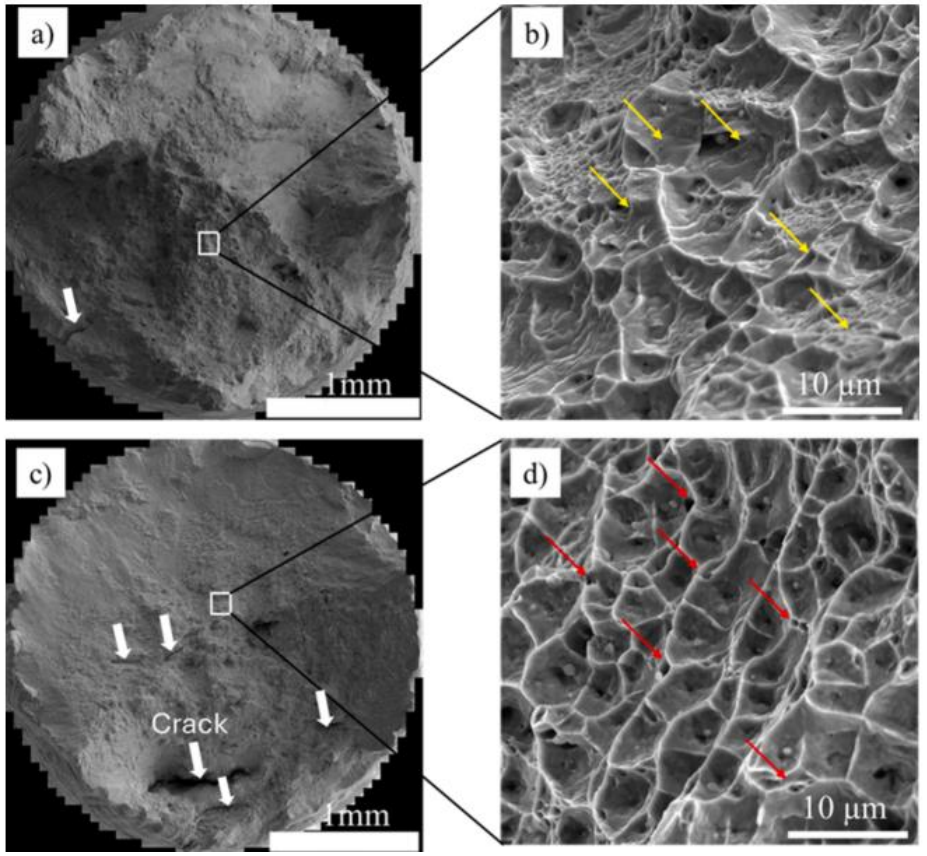
Phase Diagram of ACOF alloy, Microstructural Analysis

Case Study: 4 (Sweden, UK)

- Development of high-Mn steel with increased potential against hydrogen embrittlement through high throughput screening through ThermoCalc
- Alloy compositions optimized to increased stacking fault energy (calculated using Olson and Cohen model) and widen austenite phase
- “Islands” of FCC stability determined in a 7-dimensional compositional space to select final alloy composition
- ThermoCalc also used as a final validation layer
- The designed alloy exhibited an elevated SFE. YS and UTS aligned with literature values. Superior hydrogen resistance with a low elongation loss was demonstrated
- Link: https://www.sciencedirect.com/science/article/pii/S1350630724004849?via%3Dihub



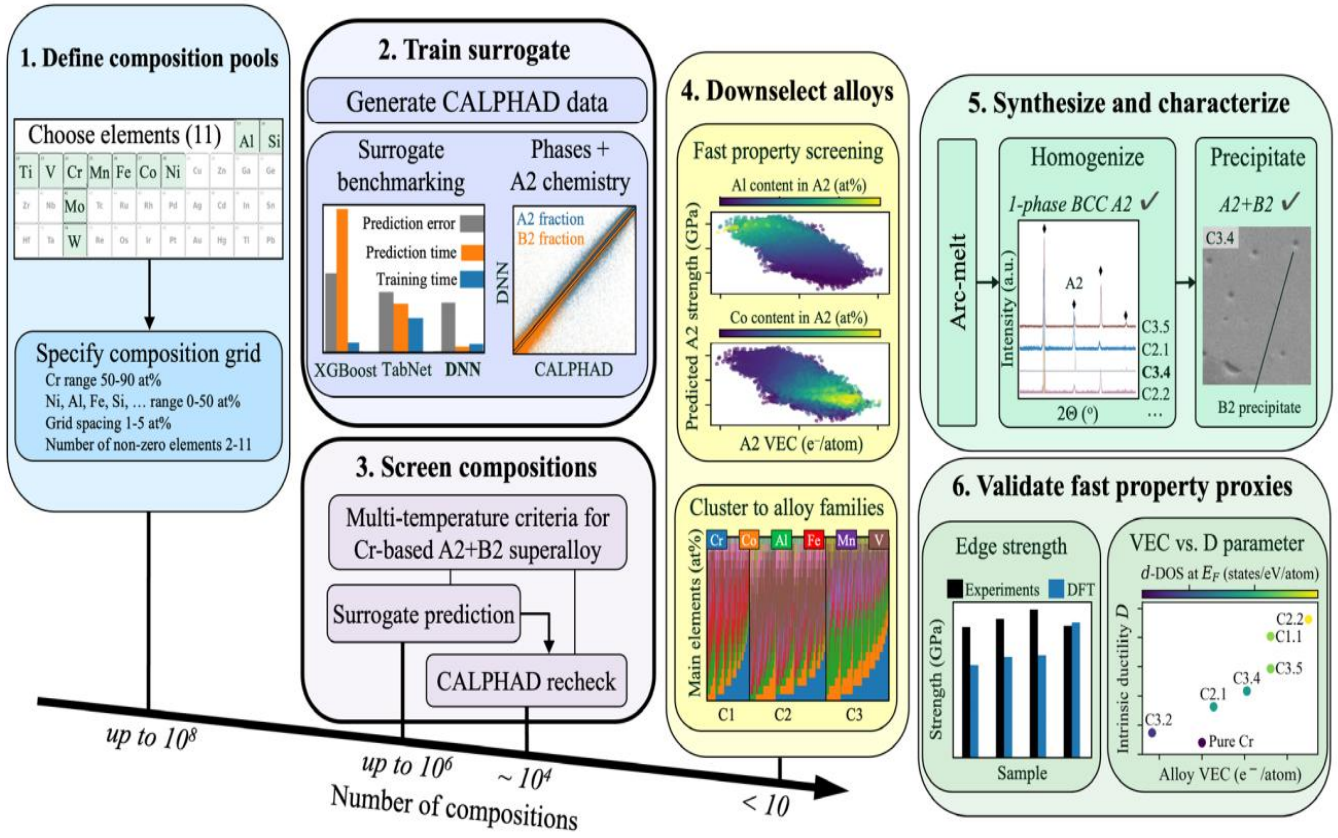


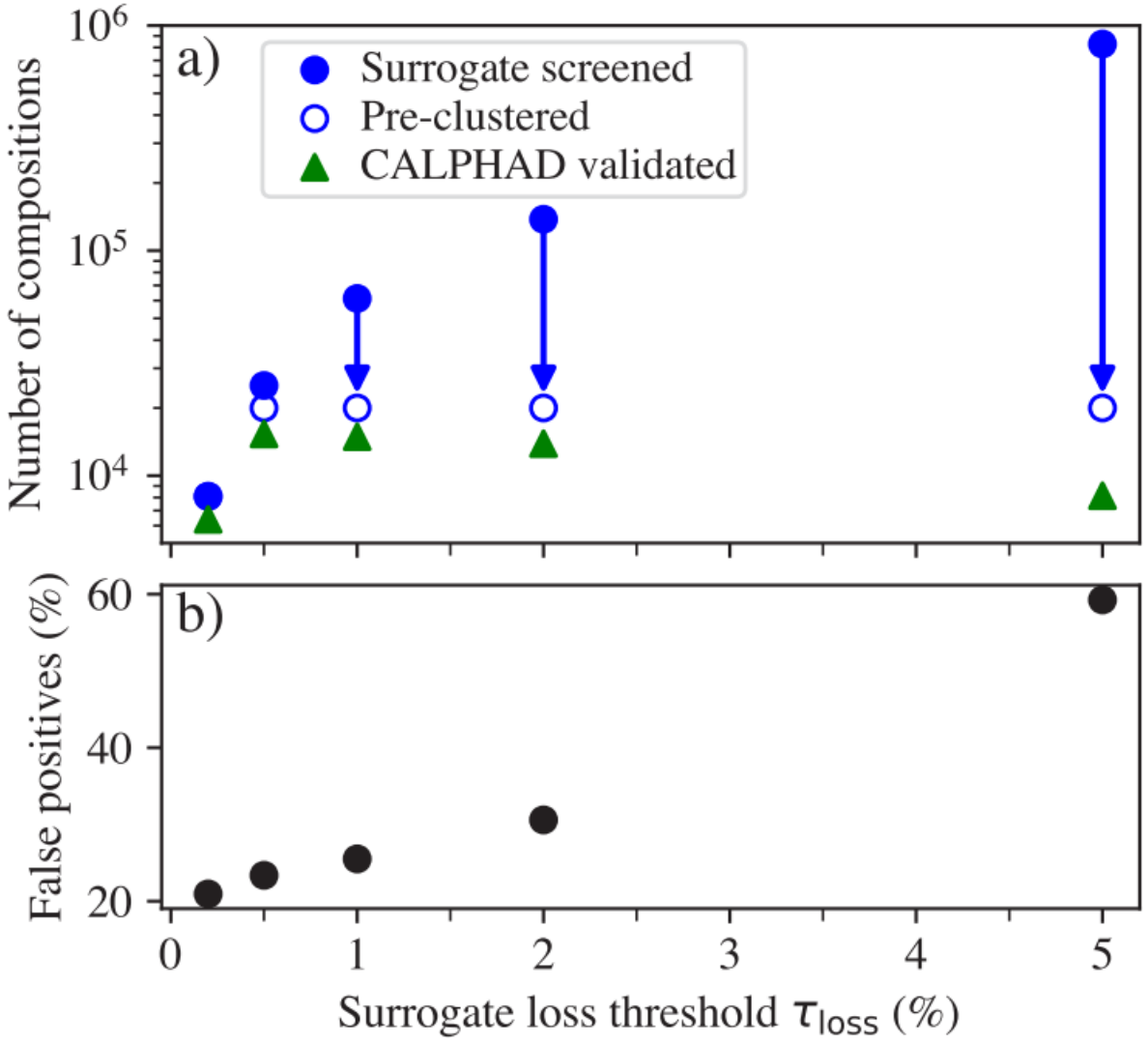


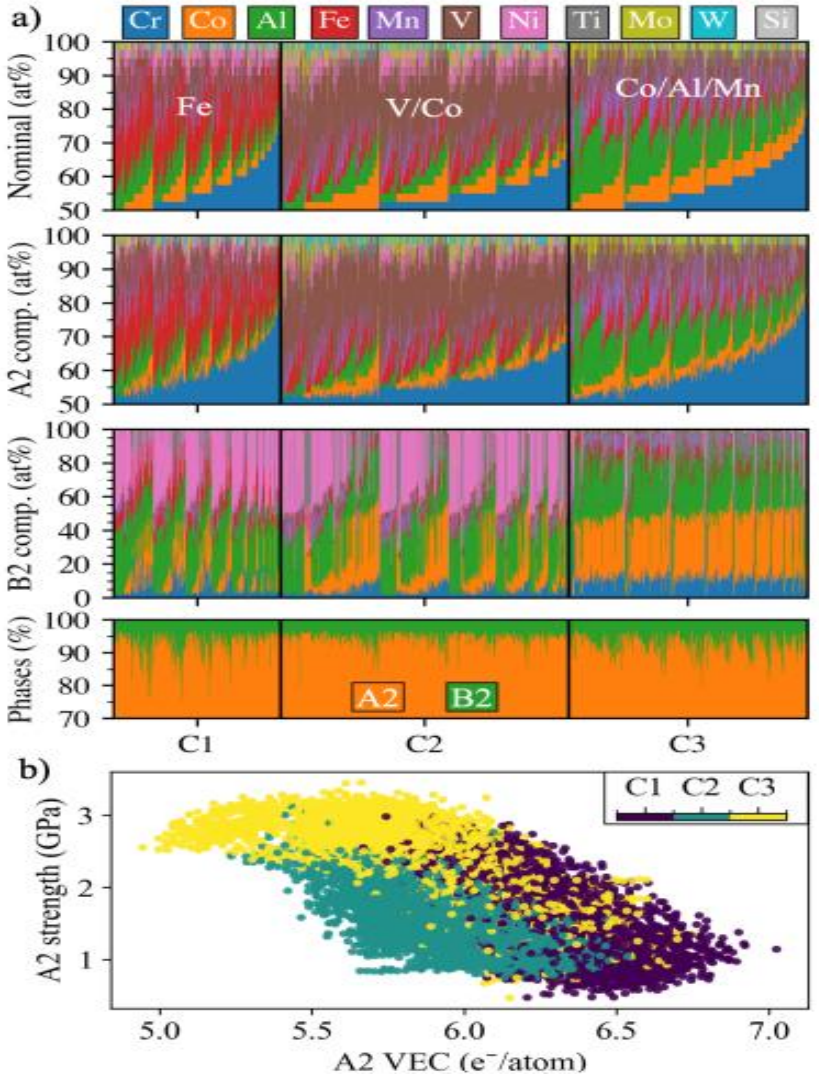
EBSD Images, Fracture Surface Analysis

Case Study: 5 (Finland)

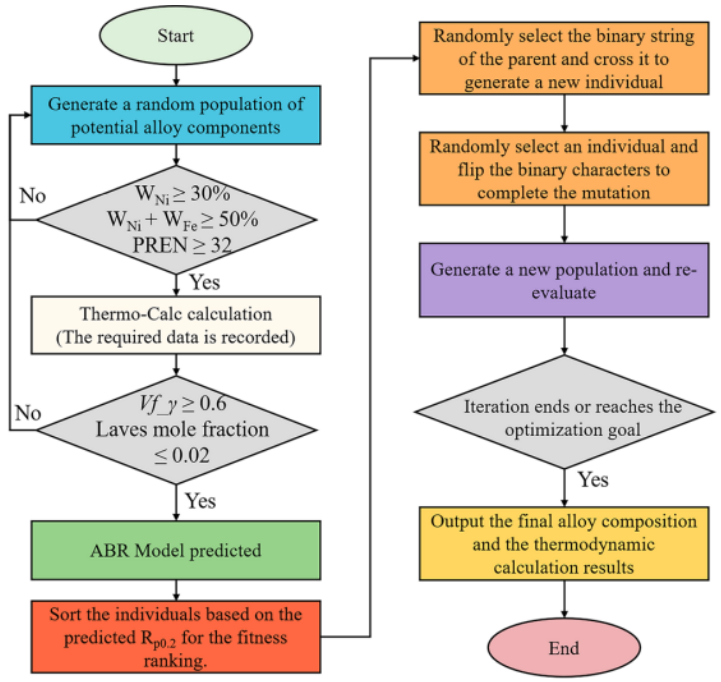
- Multi-criteria discovery workflow for Cr-based A2+B2 superalloys
- Deep-neural-network surrogate model trained on CALPHAD data to sample and screen 11-element compositional space
- Layered filtering: CALPHAD revalidation, thermodynamic screening, mechanical property checks, DFT calculations and feasibility check experiments
- ThermoCalc equilibrium data using TC-Python and the TCHEA5 and TCHEA7 databases used for training
- Link: https://www.nature.com/articles/s41524-026-02113-x?utm_campaign=85607618-Thermo-Calc%20Newsletters&utm_source=hs_email&utm_medium=email&_hsenc=p2ANqtz-_FTKxKNSPNdkIrskWqLlhgnLCAK364qIYUllweQ3o0jjK_GGIMw0JMRZvHxmiOTU0ipHcr

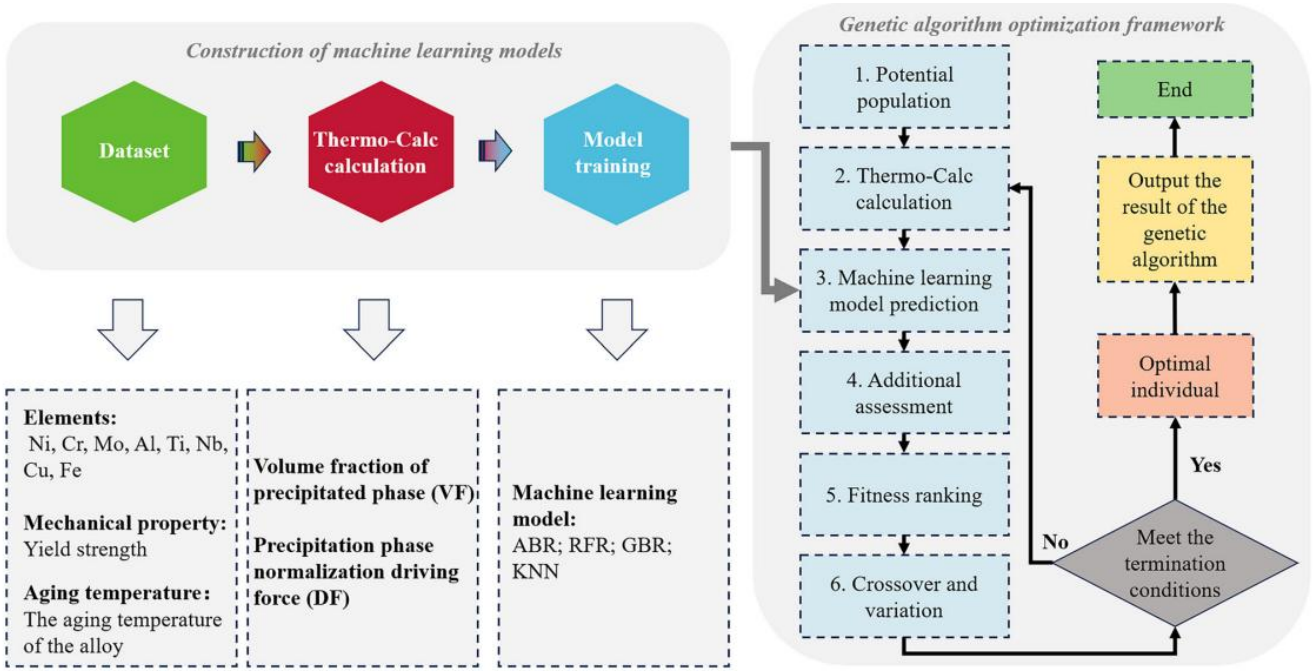



a)
Surrogate screened
Pre-clustered
CALPHAD validated
Number of compositions
10^6
10^5
10^4
b)
False positives (%)
60
40
20
0
1
2
3
4
5
Surrogate loss threshold τ_{loss} (%)


a)
Cr
Co
Al
Fe
Mn
V
Ni
Ti
Mo
W
Si
Nominal (at%)
Fe
V/Co
Co/Al/Mn
A2 comp. (at%)
B2 comp. (at%)
Phases (%)
A2
B2
C1
C2
C3
b)
C1 C2 C3
A2 strength (GPa)
A2 VEC (e⁻/atom)

Case Study: 6 (China)





- Composition–microstructure–property optimization model by integrating genetic algorithm, machine learning, and thermodynamic calculations for ultra-high-strength nickel alloy design
- Thermo-Calc (via TC-Python) calculated γ′/γ″ phase volume fractions and precipitation driving forces, used as input data for model
- Trained AdaBoost regressor embedded into the genetic algorithm as the fitness function to perform constrained composition optimization
- Optimized alloy, ML1, exhibited excellent mechanical performance under various heat treatment conditions
- Link: https://www.nature.com/articles/s41524-026-02068-z

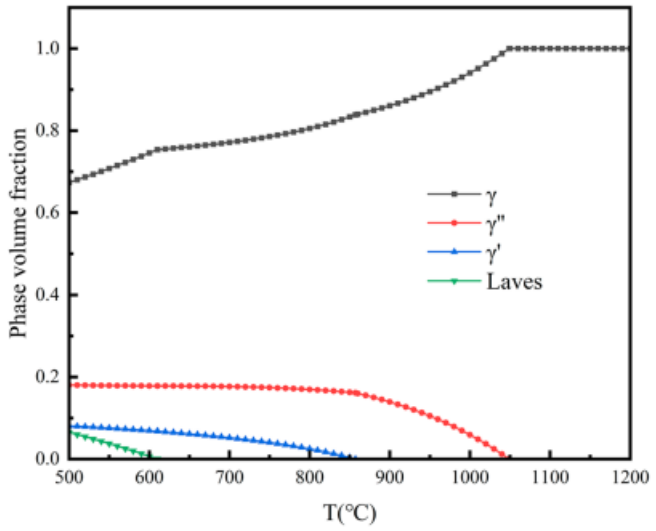


Fig. 6 | Curve of the volume fraction of the candidate alloy ML1 phase changing with temperature calculated using Thermo-Calc.

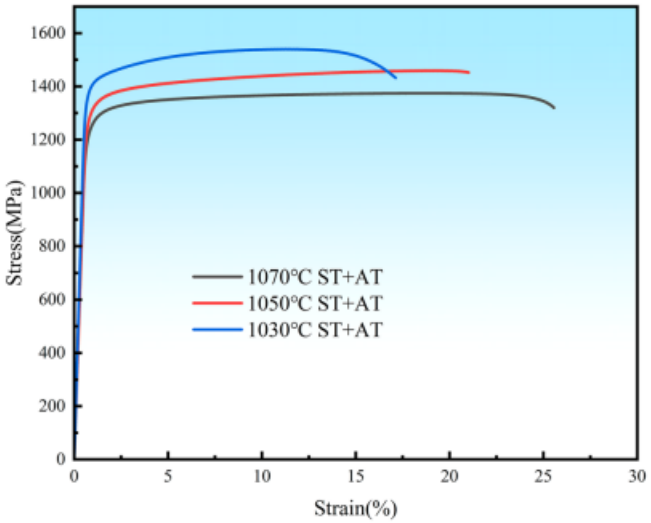


Fig. 7 | Tensile stress–strain curves of ML1 alloy after solution treated at 1050 ± 20 °C and subsequent aging treatment.

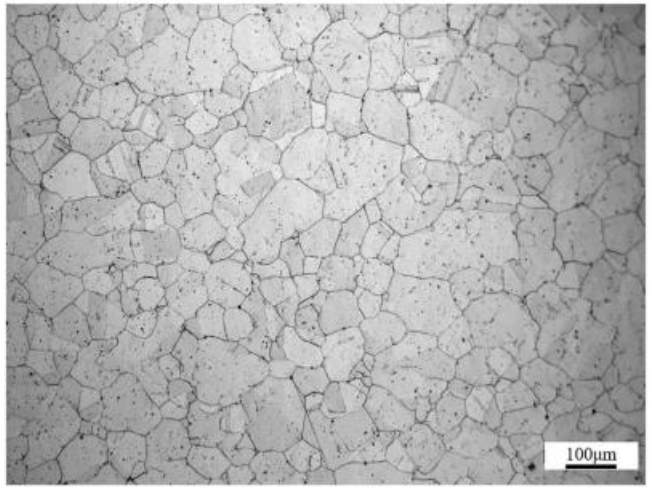


Fig. 8 | Optical microstructure of ML1 alloy after solution treatment at 1030 °C followed by aging.

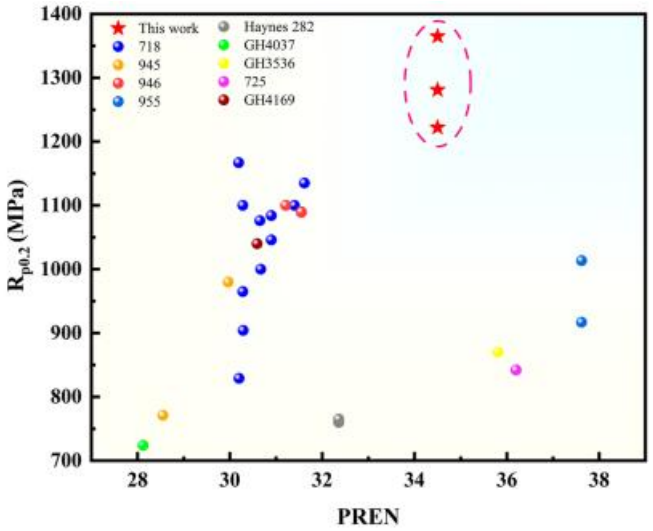


Fig. 9 | Comparison of PREN and $R_{p0.2}$ between the ML1 alloy and representative commercial age-hardened Nickel-based corrosion-resistant alloys reported in the literature.

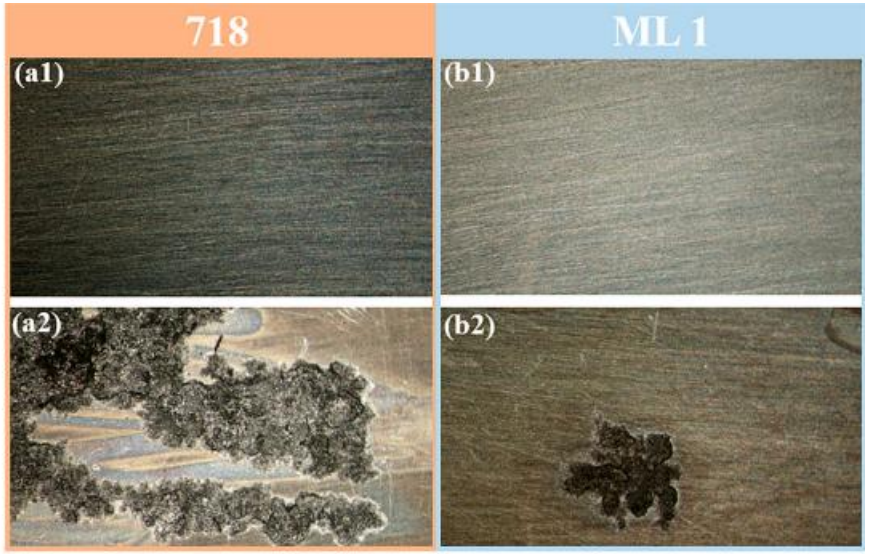


Fig. 10 | Macroscopic morphology of nickel-based alloy after 720 h of immersion in different environments. (a1-b1) is under the condition of 150 °C-6 MPa CO_2-3 MPa H_2S; (a2-b2) is under the condition of 205 °C-3.5 MPa H_2S-3.5 MPa CO_2.

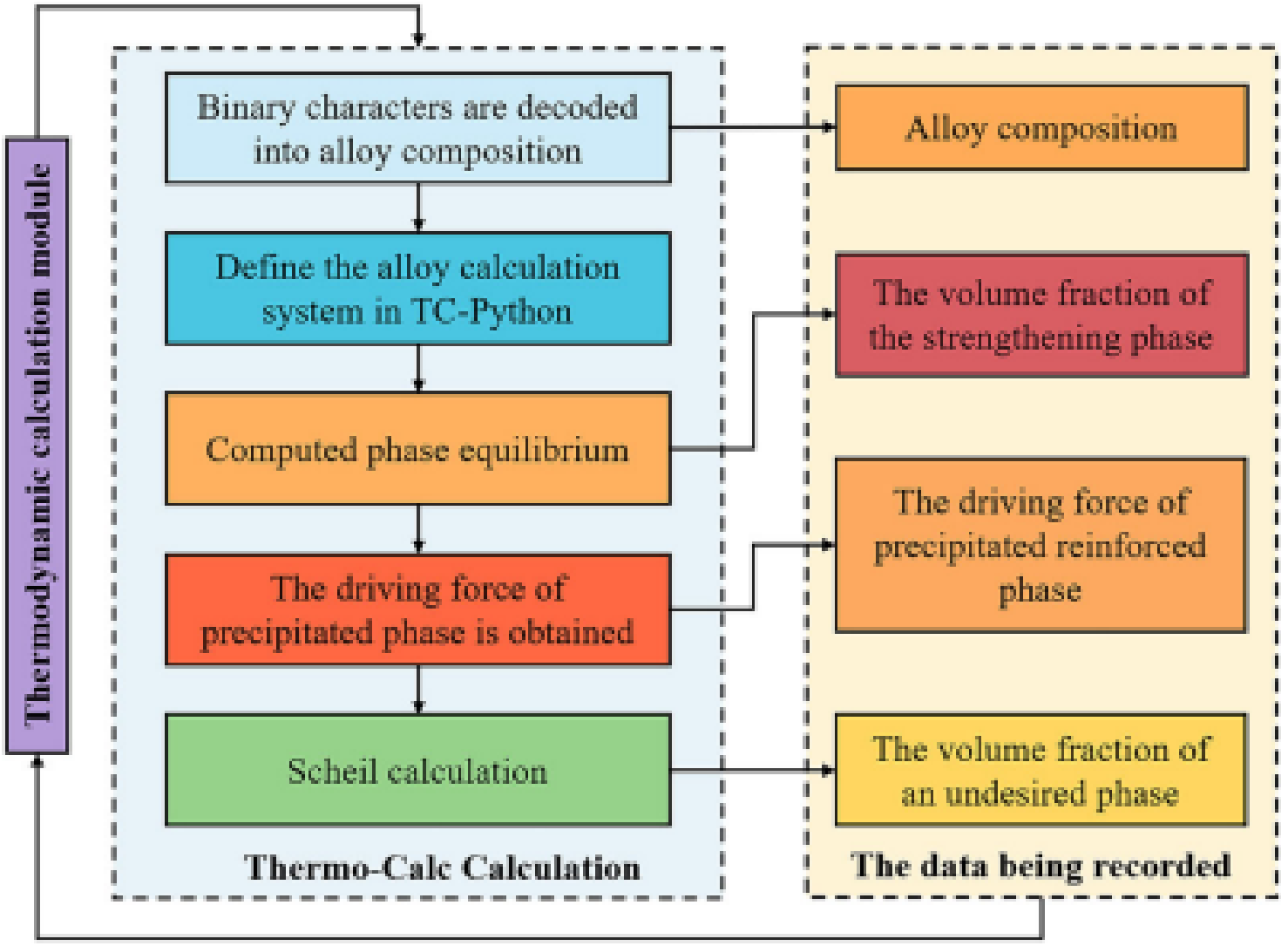


Fig. 15 | Schematic diagram of the thermodynamic calculation system.

TC-Python | A Python API for Thermo-Calc: Enabling Automted, High-Throughput Materials Design

TC-Python provides a Python-based software development kit (SDK) to access Thermo-Calc's CALPHAD calculations, enabling automated workflows, integration with the Python ecosystem, and high-throughput materials discovery.

1. What is TC-Python?

Thermo-Calc (GUI)



Manual, single calculations

TC-Python

Programmable API/SDK

Automated, high-throughput, integrated workflows

TC-Python turns Thermo-Calc from a GUI tool into a programmable materials design engine that can be controlled from Python scripts or Jupyter notebooks.

2. How it works

- Python script / Jupyter notebook
- Composition, T, P, processing conditions
- Databases (TC, TT, etc.)

→ **TC-Python API** →

Thermo-Calc CALPHAD Engine

Gibbs energy minimization, thermodynamic & kinetic calculations

Outputs

- Phases & phase fractions (equilibrium)
- Thermodynamic properties (G, H, S, Cp, ...)
- Diagrams (phase, property)
- Kinetic results (diffusion, precipitation, etc.)
- Custom property models

3. Key Features

- **Python-based API/SDK** — Control Thermo-Calc programmatically
- **High-throughput calculations** — Batch and parallel runs for large composition spaces
- **Access to major calculation types** — Equilibrium, phase diagrams, properties, Scheil, diffusion (DICTRA), precipitation (TC-PRISMA)
- **Integration with Python ecosystem** — NumPy, SciPy, pandas, matplotlib, scikit-learn, etc.
- **Custom property models** — Develop and integrate user-defined models
- **Flexible workflows** — Python scripts or Jupyter notebooks

4. Calculation Capabilities (via TC-Python)

Equilibrium calculations
- Phase stability
- Phase fractions
- Thermodynamic properties

Phase & property diagrams
- Binary, ternary, higher order
- Isopleths, isotherms
- Property diagrams

Scheil solidification
- Solidification path
- Phase evolution
- Microsegregation

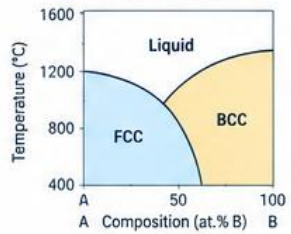
Diffusion (DICTRA)
- Diffusion profiles
- Interdiffusion coefficients
- Multiphase diffusion

Precipitation (TC-PRISMA)
- Nucleation & growth
- Precipitate size/volume fraction
- Time-temperature evolution

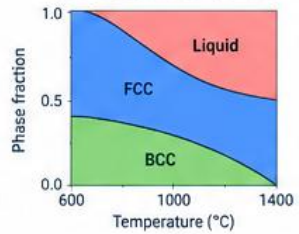
Custom property models
- User-defined models
- Mechanical, magnetic, oxidation properties, etc.

5. Example Results (generated via TC-Python)

Equilibrium phase diagram



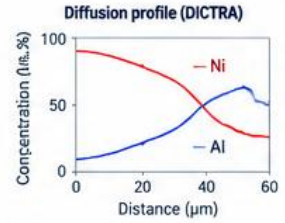
Phase fractions vs temperature



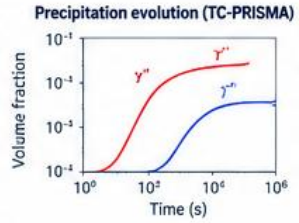
Thermodynamic properties

Property	Value
Gibbs energy	−325 kJ/mol
Enthalpy	−310 kJ/mol
Entropy	120 J/mol-K
Density	7.8 g/cm³
Liquidus T	1420 °C
Solvus T	980 °C

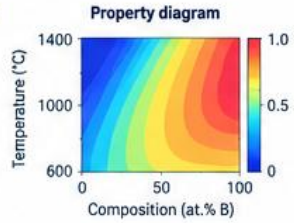
Diffusion profile (DICTRA)



Precipitation evolution (TC-PRISMA)



Property diagram



6. Python Integration

NumPy · SciPy · pandas · matplotlib · scikit-learn · TensorFlow

- Process and analyze Thermo-Calc results
- Build high-throughput workflows
- Use ML/optimization for materials discovery
- Create custom visualizations and dashboards
- Run in Python scripts or Jupyter notebooks

7. Typical Applications

- Aerospace superalloys
- Battery materials
- Coatings (TBC, protective)
- Process optimization
- Materials discovery

8. Example Workflow: High-Throughput Alloy Screening

1. **Define design space** — Select elements, composition ranges, temperature, etc.
2. **Generate compositions** — Automatically generate thousands of compositions
3. **Run CALPHAD (via TC-Python)** — Automated batch calculations
4. **Extract outputs** — Phases, phase fractions, properties
5. **Apply screening criteria** — Filter based on phase stability, temperature, density, cost, etc.
6. **Candidate alloys** — Shortlist for further analysis and experimental validation

9. Sample Python Code Snippet (conceptual)

```python
from thermocalc import *
# create a system and load database
system = System()
system.select_database("TCFE")
system.set_elements(["Al", "Cr", "Ni"])

# define conditions and run equilibrium calculation
cond = {"T": 1000, "P": 101325, "X(Ni)": 0.3, "X(Cr)": 0.2}
result = system.calculate_equilibrium(cond)

# extract phase fractions
phases = result.get_phases()
for p in phases:
    print(p.name, p.fraction)
```

```python
# import TC-Python

# initialize system
# choose thermodynamic database
# define elements

# calculation conditions
# run equilibrium calculation

# get list of phases
# print results
```

References:

- https://calphad.com/calphad_method/
- https://pmc.ncbi.nlm.nih.gov/articles/PMC4912057/pdf/nihms-784023.pdf
- https://calphad.com/wp-content/uploads/2025/07/Introduction_to_CALPHAD_Method.pdf
- https://thermocalc.com/about-us/methodology/the-calphad-methodology/

Materials Discovery: Thermodynamic Modelling

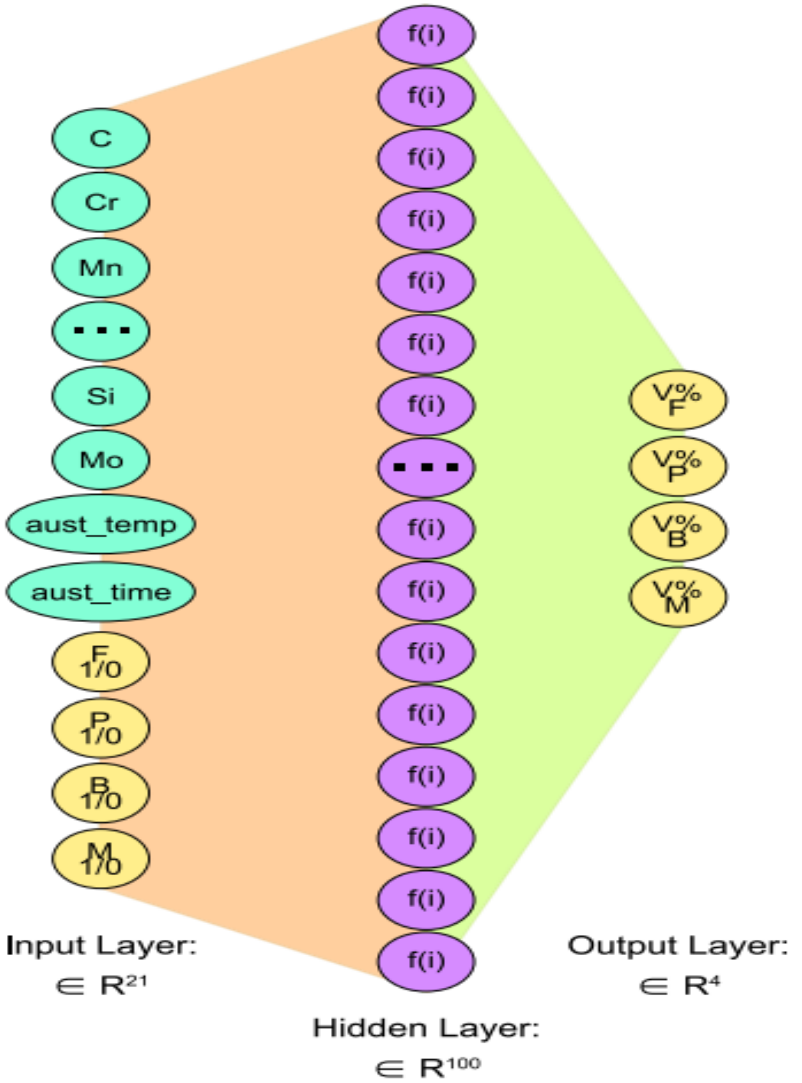
A Deep Dive into ThermoCalc, MatCalc, Pandat, etc in Materials Discovery

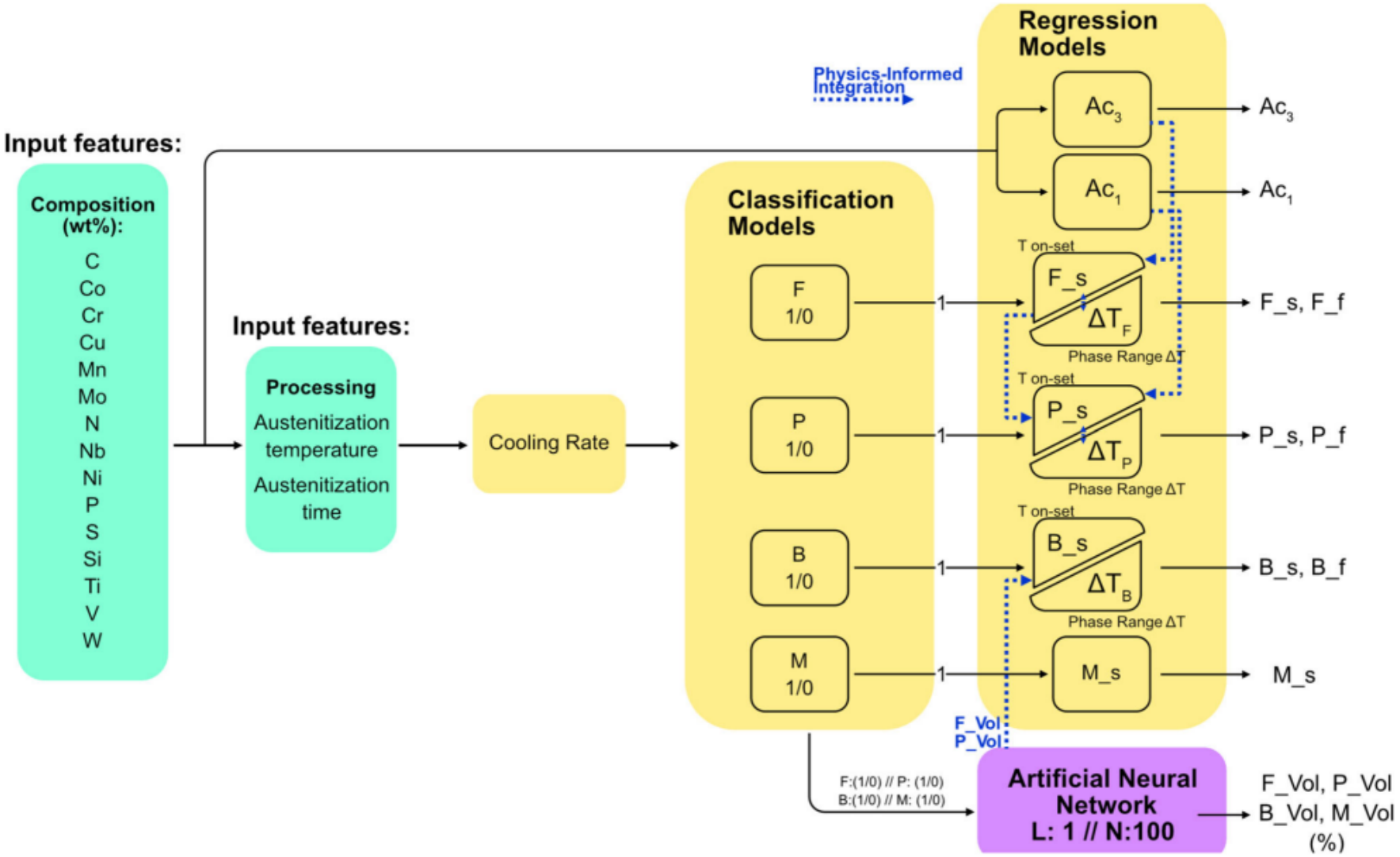
Case Study: 1 (Sweden)

- Creation of a computational framework to develop CCT diagram for steels
- Dataset: 4100 CCT diagrams and 35820 datapoints from CCT and TTT Diagrams of Steels database
- Sub-models detect phases (classification) and predict transformation temperatures (regression): Committee of 14 neural networks to reconstruct CCT diagram
- Physics Integration:

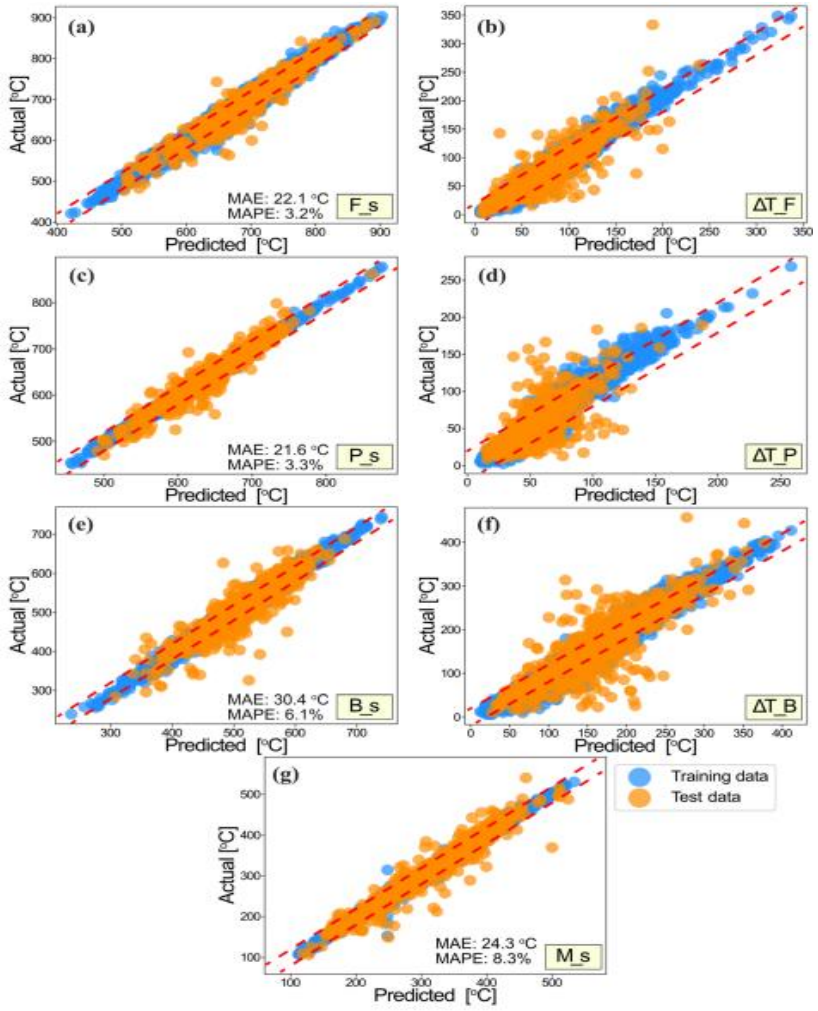
  - Synthetic data points
  - Thermodynamic boundary conditions
  - Sequential phase transformations
  - f(Bainite) = f(Pearlite, Ferrite)

- Phases F1-score: ~ 88% F1-score; TT MAEs: ~ 20°C
- Link: https://link.springer.com/article/10.1007/s11661-026-08181-4



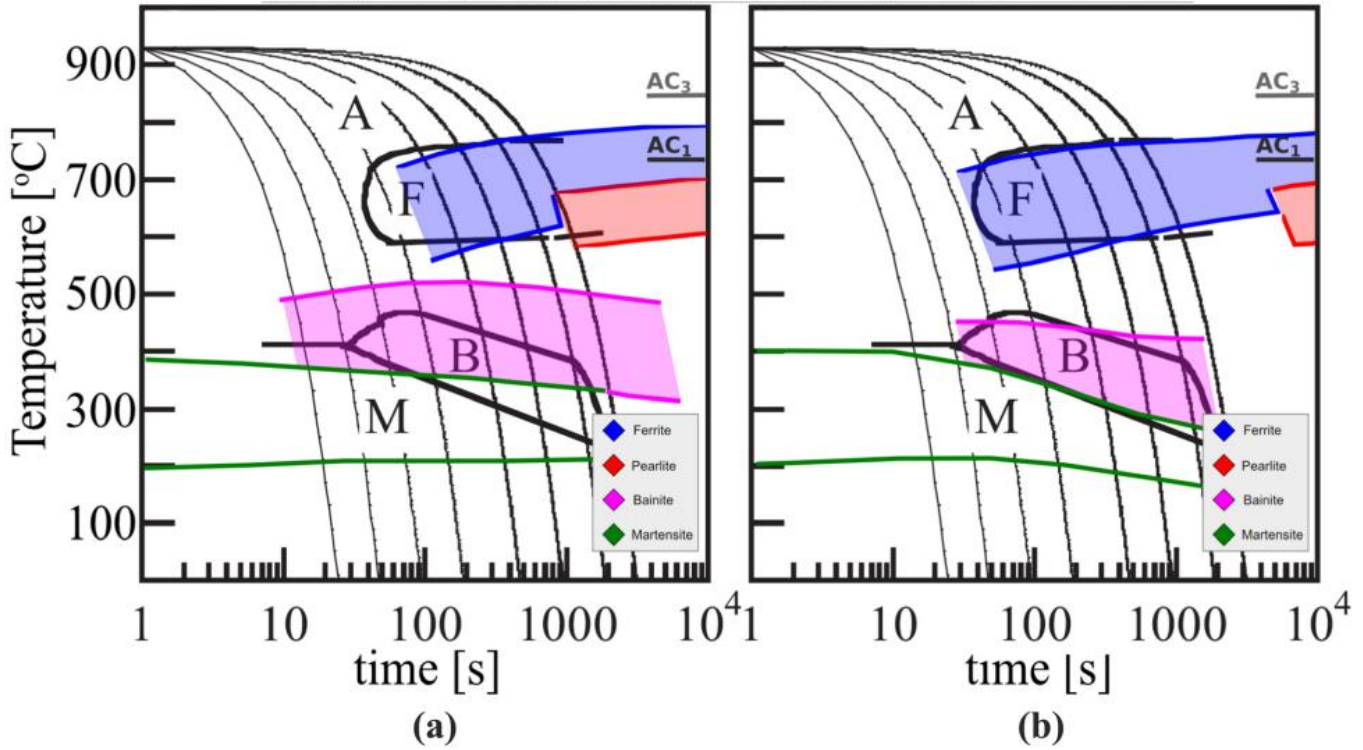


Committee architecture consisting of sub-models



Measured VS Predicted Phase Transformation Temperatures

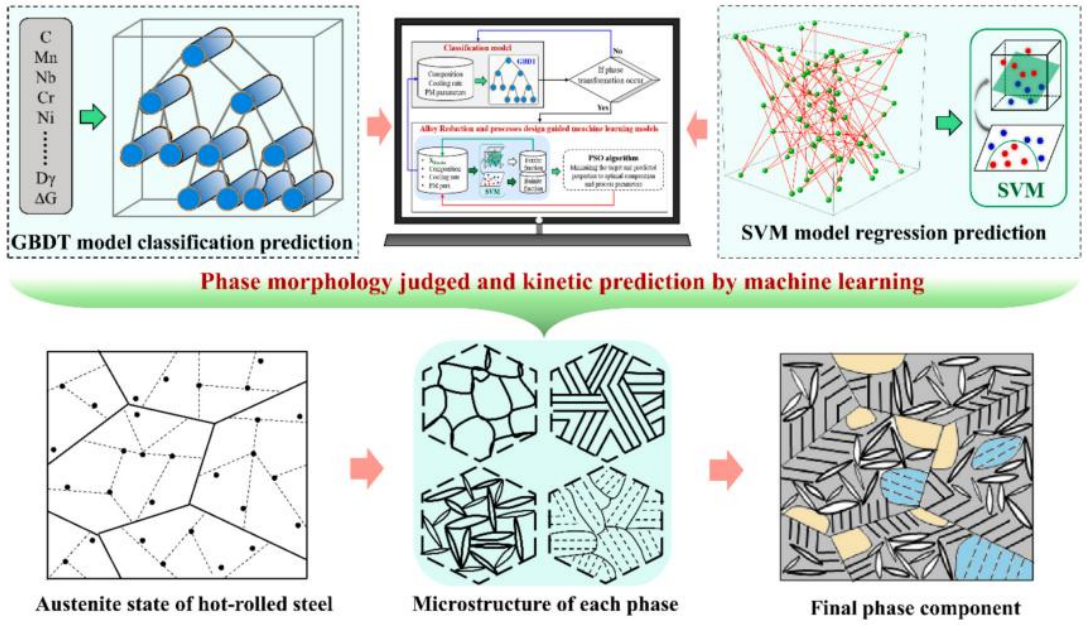
Wt.% (Fe Balance)											T_{aus} [°C]	t_{aus} [s]
C	Mn	Si	Cr	Mo	S	P	Al	Nb	N	O	910	1200
0.18	1.49	1.38	1.02	0.20	0.0029	0.043	0.043	0.05	0.001	0.0015		

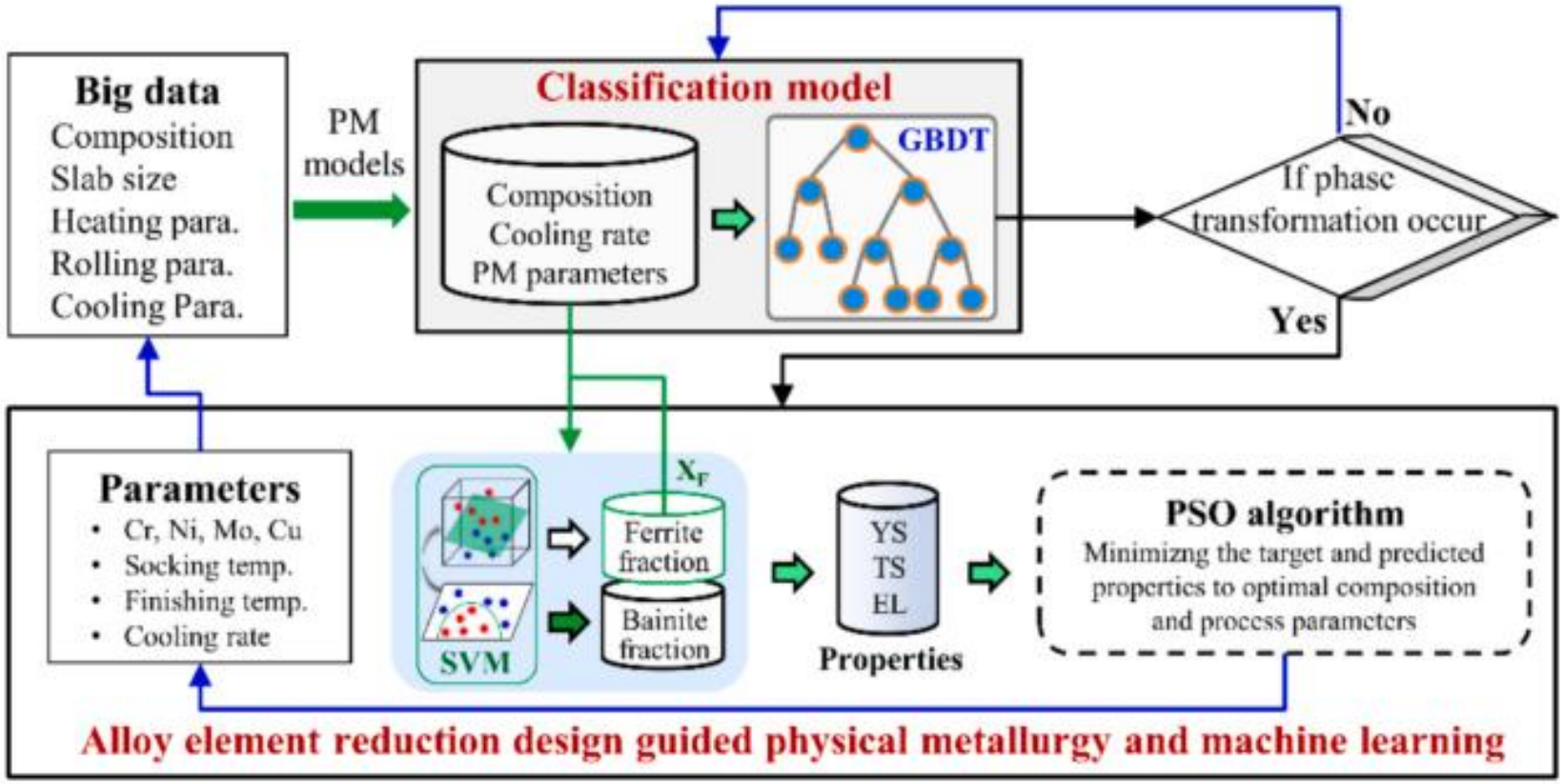


Predicted CCT data overlaid on experimental CCT data

Case Study: 2 (China)

- ML based phase fraction and morphology prediction of products after hot rolling of steel
- The reheating and deformation inputs are converted into physical quantities: Prior Austenite Grain Size, deformation energy and ferrite fraction
- Data gathered from experiments and literature
- Two-stage model used, A GBDT classifier decides which phases occur, SVM predicts each phase fraction
- Particle swarm optimisation with a weighted fitness function used to keep YS, TS and EL close to the original steel, maximises AF fraction (for toughness), and minimises yield ratio, along with constraining soaking and rolling temperature
- Validation: Optimised samples rolled into 20 mm plates, subject to mechanical testing.
- Link: https://www.sciencedirect.com/science/article/pii/S0264127525010627?ref=pdf_download&fr=RR-2&rr=a444d4deee6e4da5





Pipeline Architecture

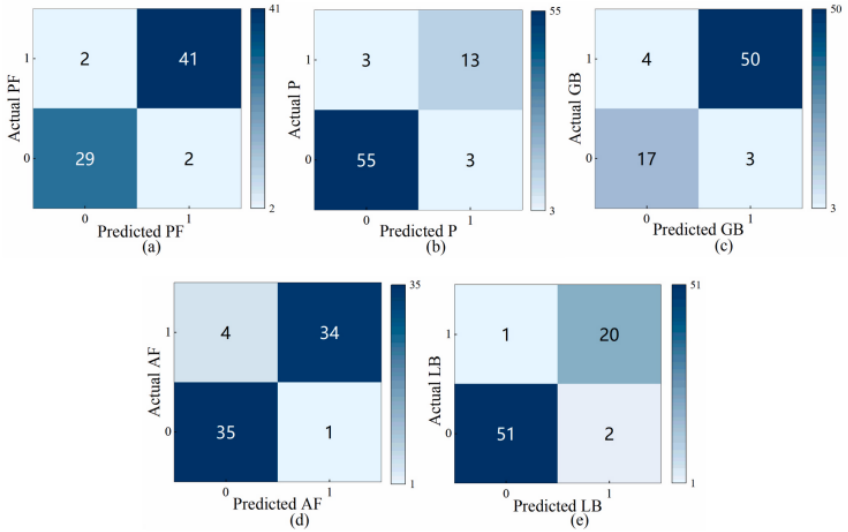
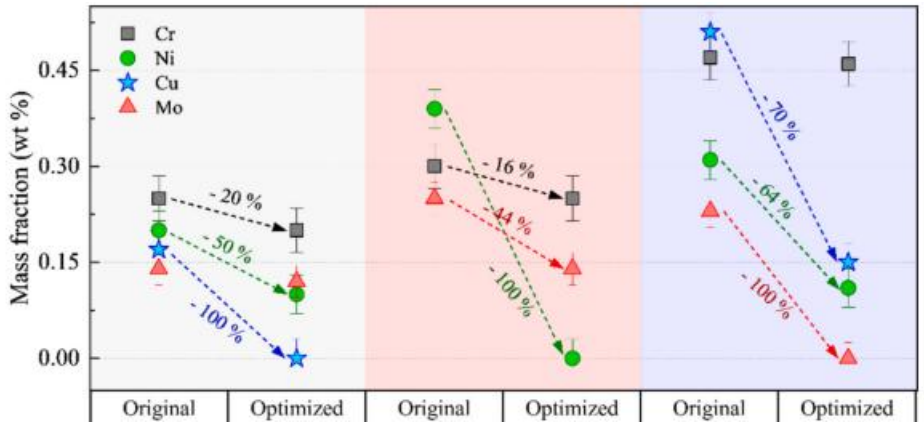


Fig. 4. Confusion matrix of GBDT model for phase structure classification.



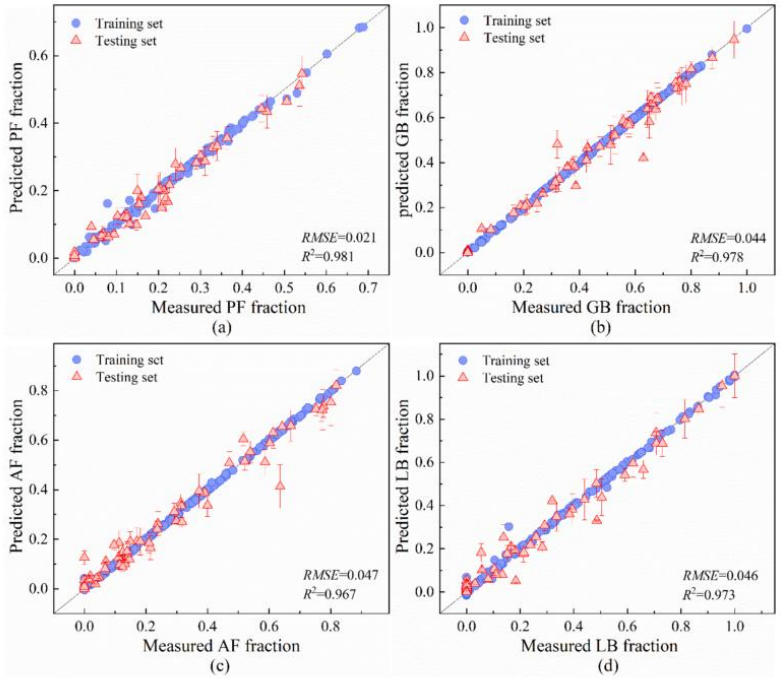


Fig. 5. Comparison between predicted and measured values in training and testing datasets (a) PF, (b) GB, (c) AF, (d) LB.

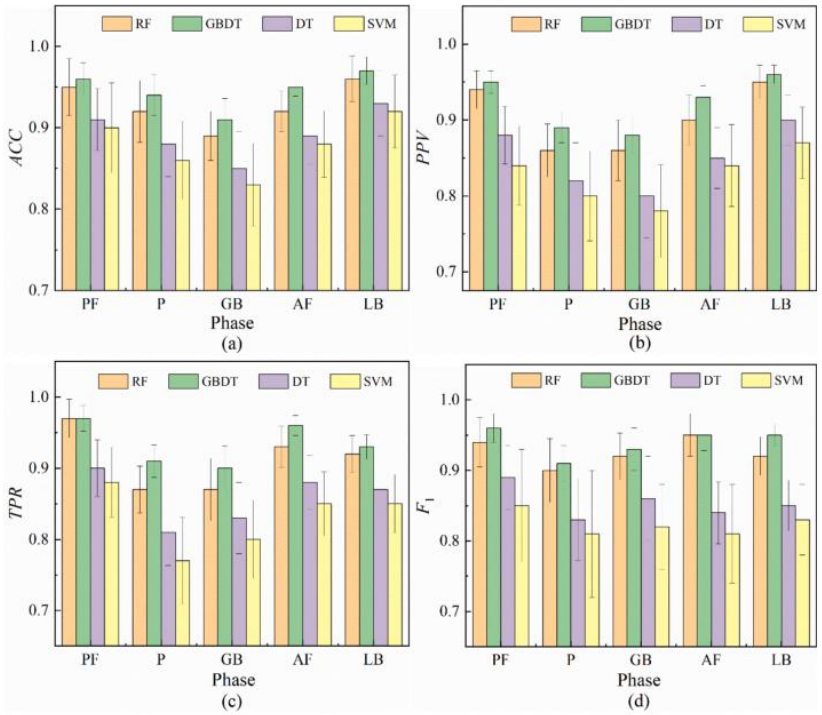
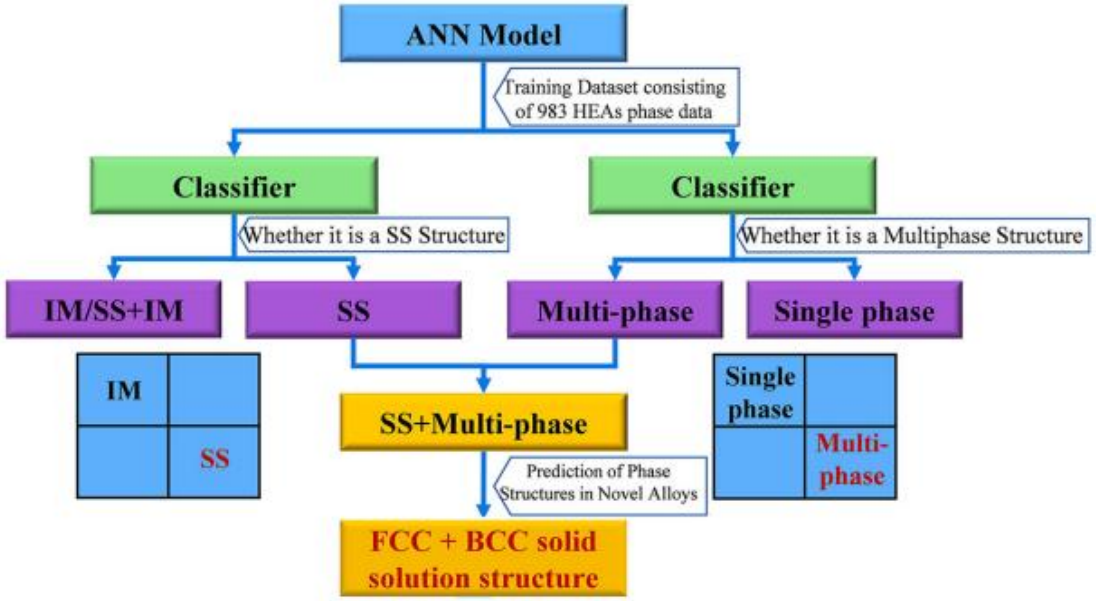


Fig. 3. Comparison of classification performance of ML models based on different classification algorithms.

Case Study: 3 (China)

- Designing BCC/FCC dual-solid-solution RHEAs using CALPHAD + machine learning + experimental approach.
- Binary phase diagrams and mixing enthalpies were used to identify suitable refractory-element combinations, Thermo-Calc and Scheil-Gulliver simulations were then used to assess equilibrium phase formation
- The dataset is 983 alloys from CALPHAD databases, and eleven Hume-Rothery descriptors were used as ML features
- Two binary neural networks were trained, whose intersection was used to identify candidate BCC/FCC dual-solid-solution alloys
- Experimental validation & key result: Arc-melted NiMoVW experimentally exhibited BCC + FCC solid-solution phases, consistent with the predictions
- Link: https://www.nature.com/articles/s41524-025-01597-3



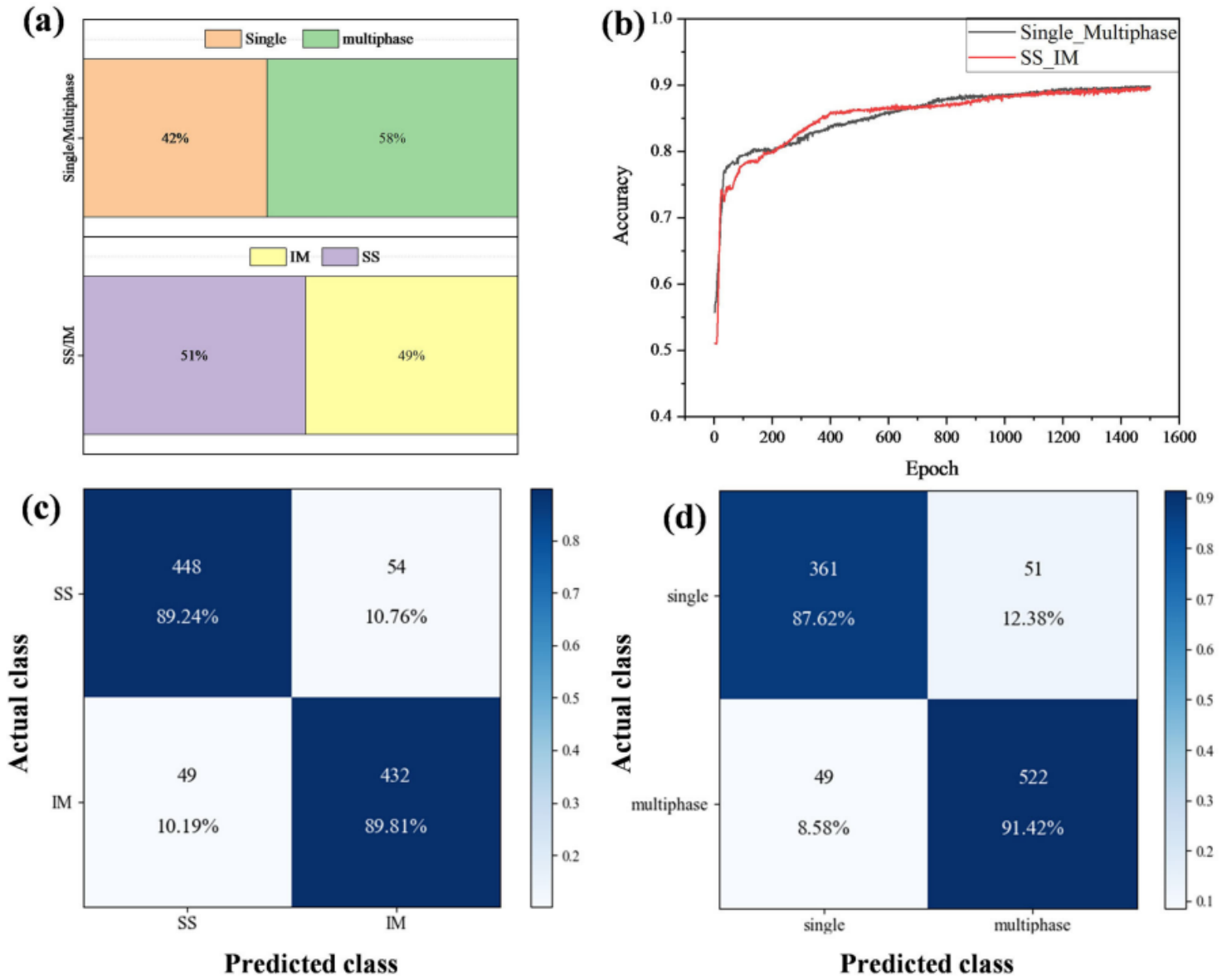


Fig. 5 | Performance evaluation of binary classification models for phase prediction in HEAs. a Distribution of the training data across the two binary classification models. **b** Comparison of accuracy for both models. **c**, **d** Confusion matrices obtained from training on the full dataset.

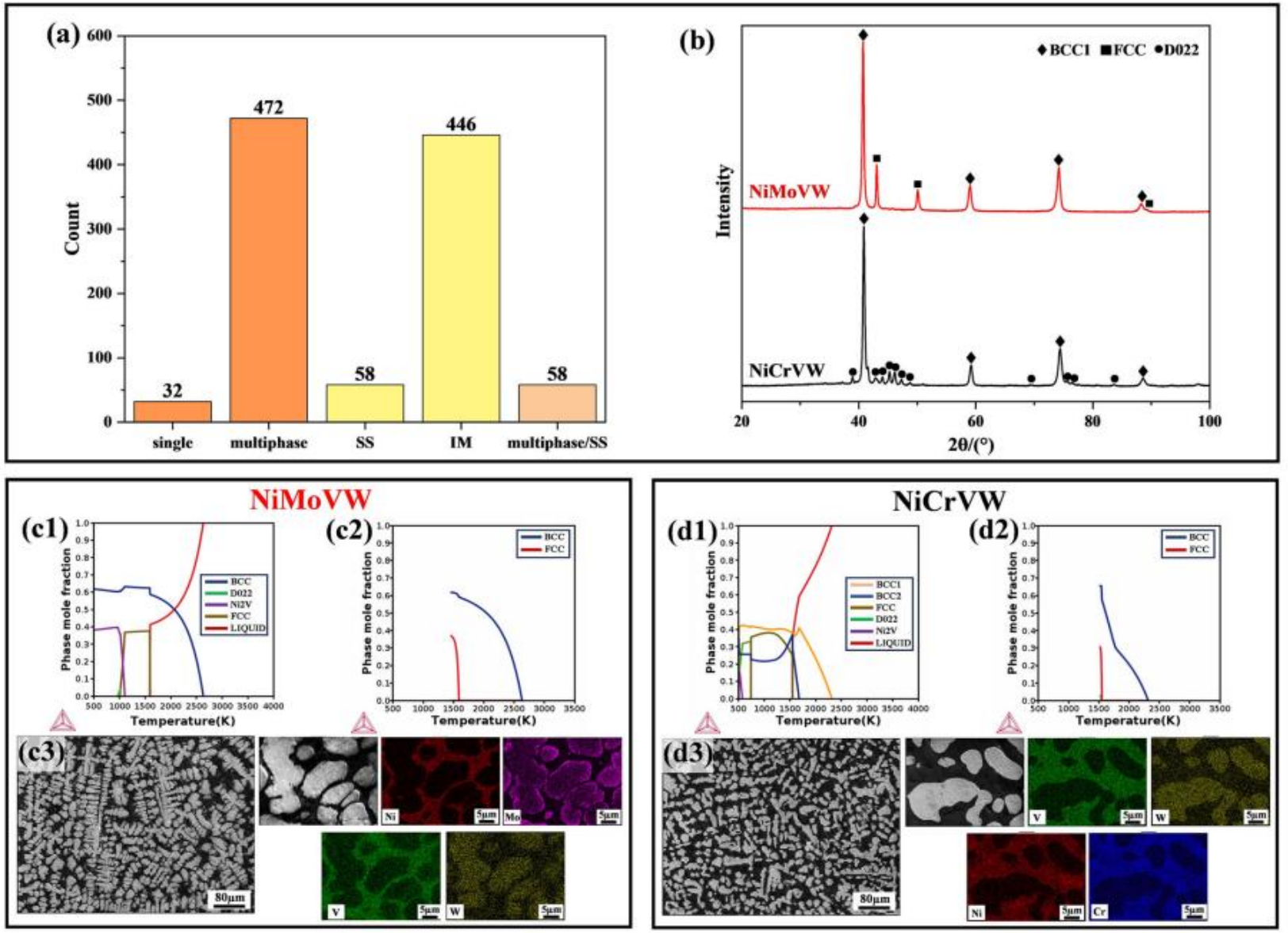
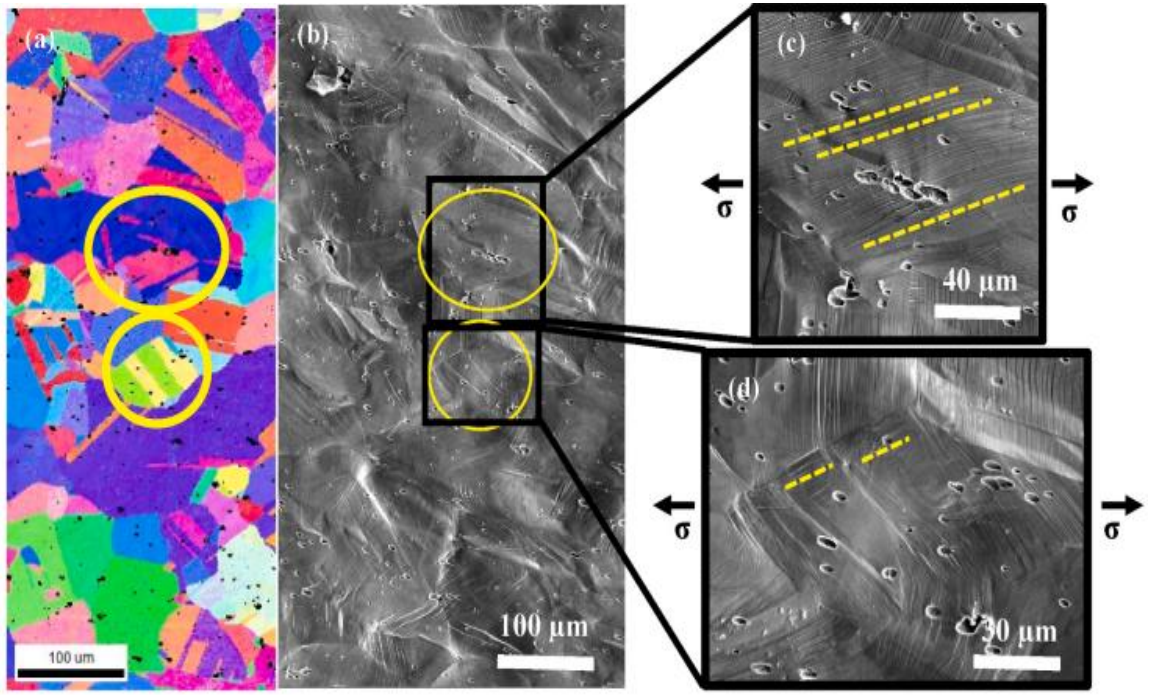


Fig. 6 | Phase prediction results of optimized models and experimental validation. a Phase prediction using optimized models. **b** XRD patterns of NiMoVW and NiCrVW alloys. Mole fraction of **c1**, **d1** equilibrium and **c2**, **d2** non-equilibrium phases at different temperatures. **c3**, **d3** BSE images and corresponding EDS mappings of NiMoVW and NiCrVW alloys.

Case Study: 4 (Sweden, Netherlands)

- High-throughput CALPHAD screening of a non-equiatomic Cantor alloy for cost reduction
- Brute-force CALPHAD screening for single phase FCC structure. 1.78 million compositions computed every 50 K from 500 to 2500 K
- Constraints applied with elemental compositions to maximize entropy of mixing. 5D FCC “island” projected into binary maps, and its centre at 1000 K selected as candidate: $Co_{10}Cr_{12}Fe_{43}Mn_{18}Ni_{17}$
- SFE and SSH property models run on candidate. Surface and strain energy terms added to stacking fault energies and Labusch’s theory modified to make it solvent free for concentrated alloys
- Arc-melted, hot-rolled and homogenised alloy gave single-phase FCC as predicted. 40% cost reduction was achieved, with comparable yield strengths
- Link: https://www.sciencedirect.com/science/article/pii/S0921509321015616?via%3Dihub



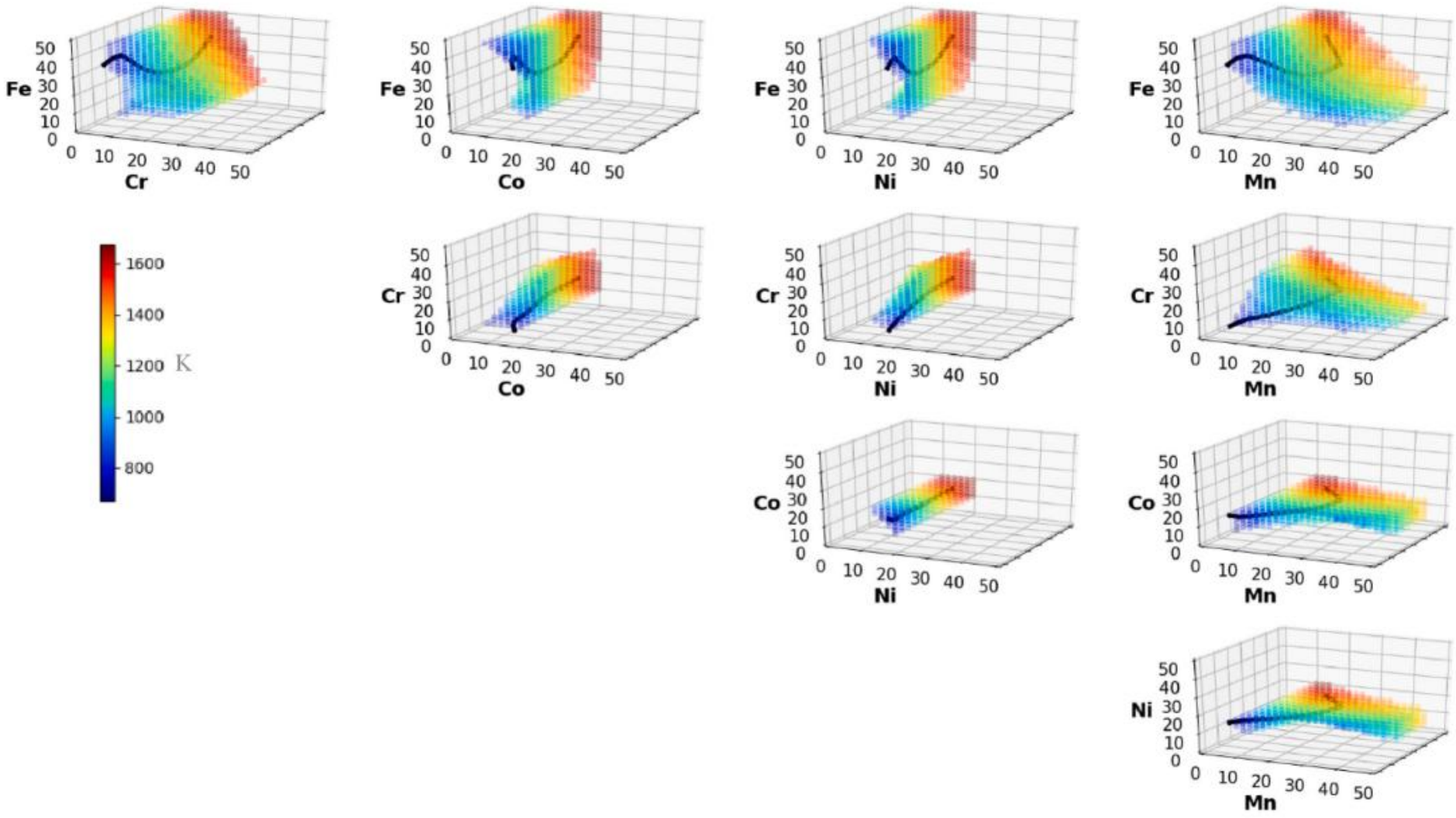


Fig. 2. Regions in binary compositional space (*at.%*) projected from the 5-element data-set that predict a single phase FCC structure within the specified constraints. Black lines indicate the centre of the compositional space at increasing temperatures.

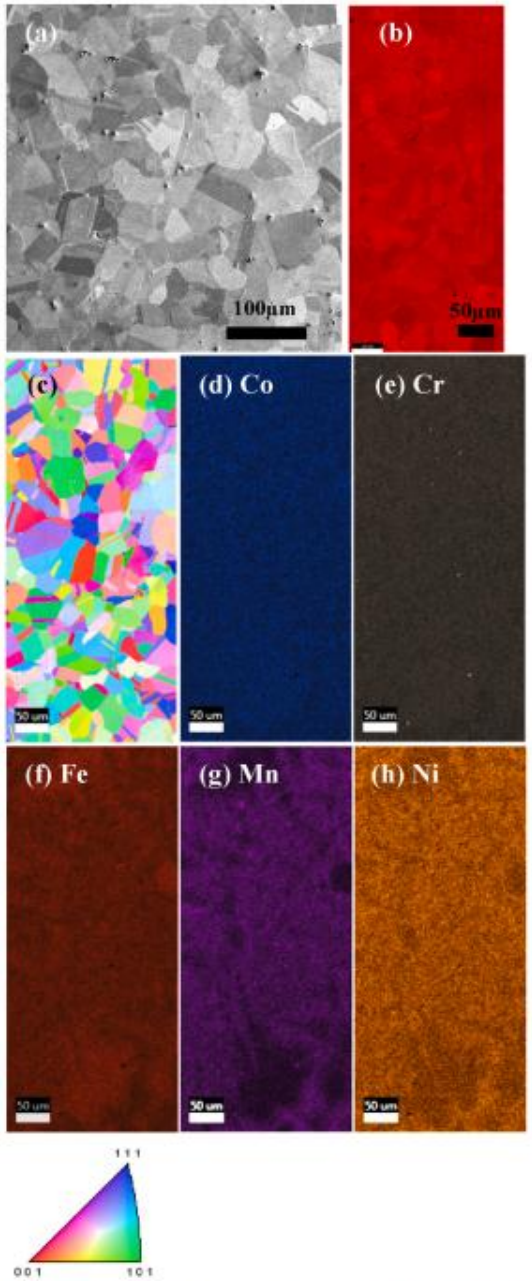


Fig. 3. (a) SEM, (b) phase map (red FCC), (c) Inverse Pole Figure (IPF), and (d–h) EDS maps of $Co_{10}Cr_{12}Fe_{43}Mn_{18}Ni_{17}$ after homogenisation. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

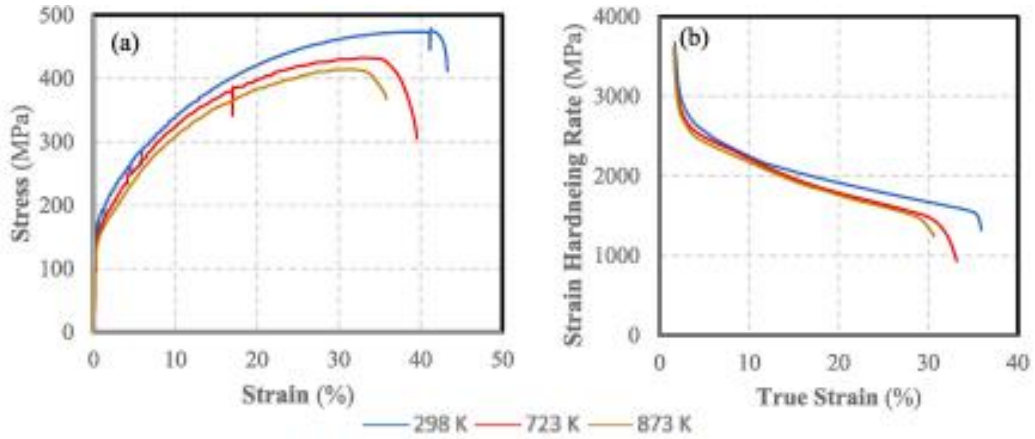


Fig. 4. (a) Engineering stress-strain (tensile) curves and (b) strain hardening rate of $Co_{10}Cr_{12}Fe_{43}Mn_{18}Ni_{17}$ at 293, 723 and 873 K during in-situ testing. The interruptions in the curves are EBSD acquisition points.

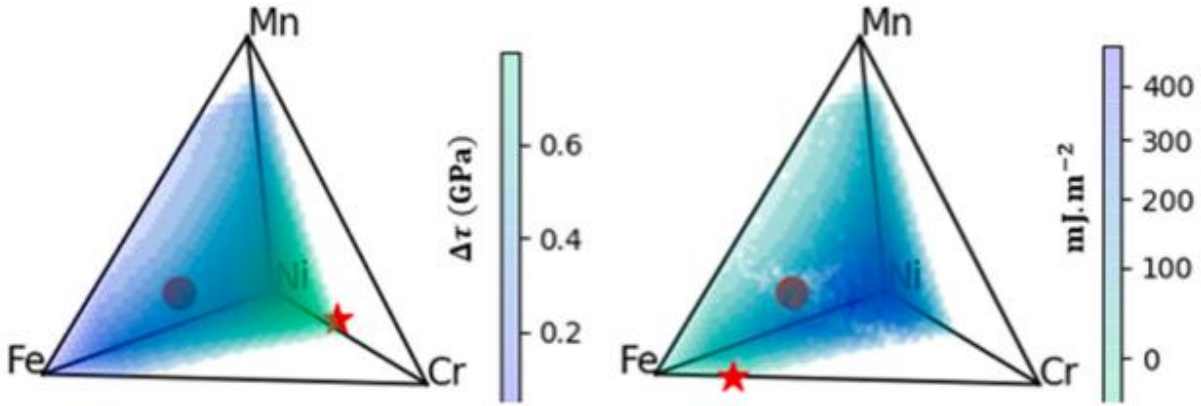


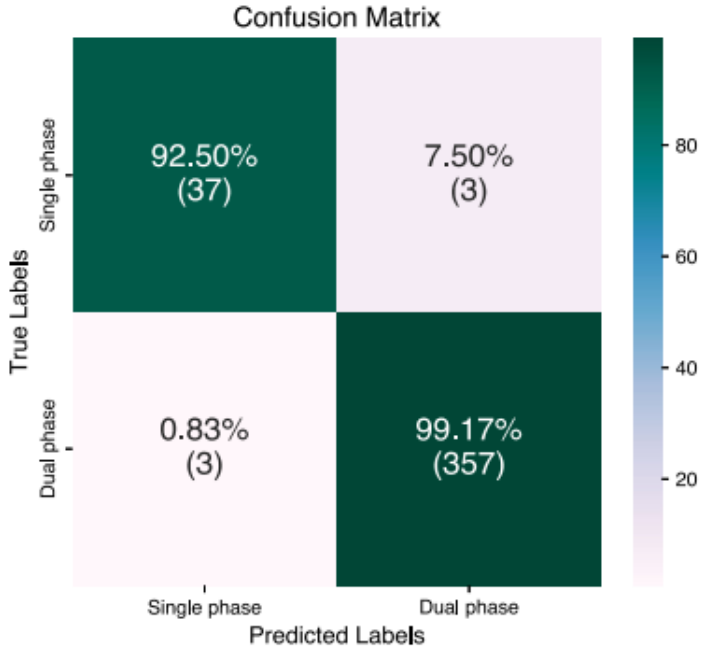
Fig. 11. Quaternary diagrams at 10 at.% Co identifying the location of the alloy studied here (circle) and the optimum composition (red star) for (a) maximising SSH and (b) minimising SFE. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Case Study: 5 (India)

- CALPHAD-trained ML framework to design a lightweight Al-Ti-V beampipe alloy with high radiation length, low density and high stiffness (combined as high Figure of Merit)
- Dataset consists of 2000 Al-Ti-V compositions chosen by Latin hypercube sampling, with TC-Python generating phases, density, X_0 and liquidus from Thermo-Calc
- Two-stage model: An XGBoost classifier decides single- vs dual-phase, and linear regression (Model A) or random forest (Model B) predicts density and X_0
- Scheil-Gulliver property model used to predict optimized non-equilibrium solidification pathways
- High accuracy metrics are seen with Thermo-Calc reproducing the predicted phase sequence. Arc-melted alloy-1 shows Al_3Ti + (Al)ss by XRD, SEM and EDS
- Link: https://journals.aps.org/prab/abstract/10.1103/PhysRevAccelBeams.28.043101

Model A:
XGBoost Classifier:
Prediction of Number of Phases
Linear Regression:
Prediction of Density and Radiation Length

Model B:
XGBoost Classifier:
Prediction of Number of Phases
Random Forest Regression: Prediction of Density and Radiation Length





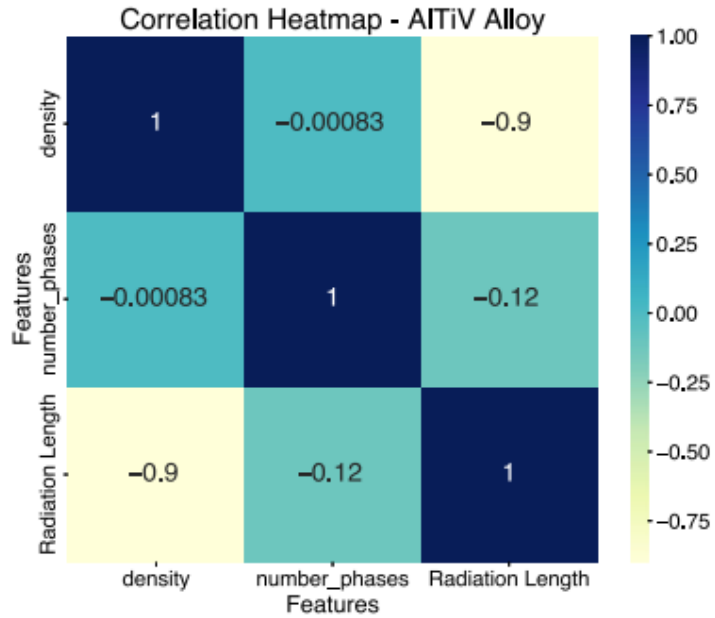


FIG. 3. Confusion matrix for the classification of the phases using XGBoost model.

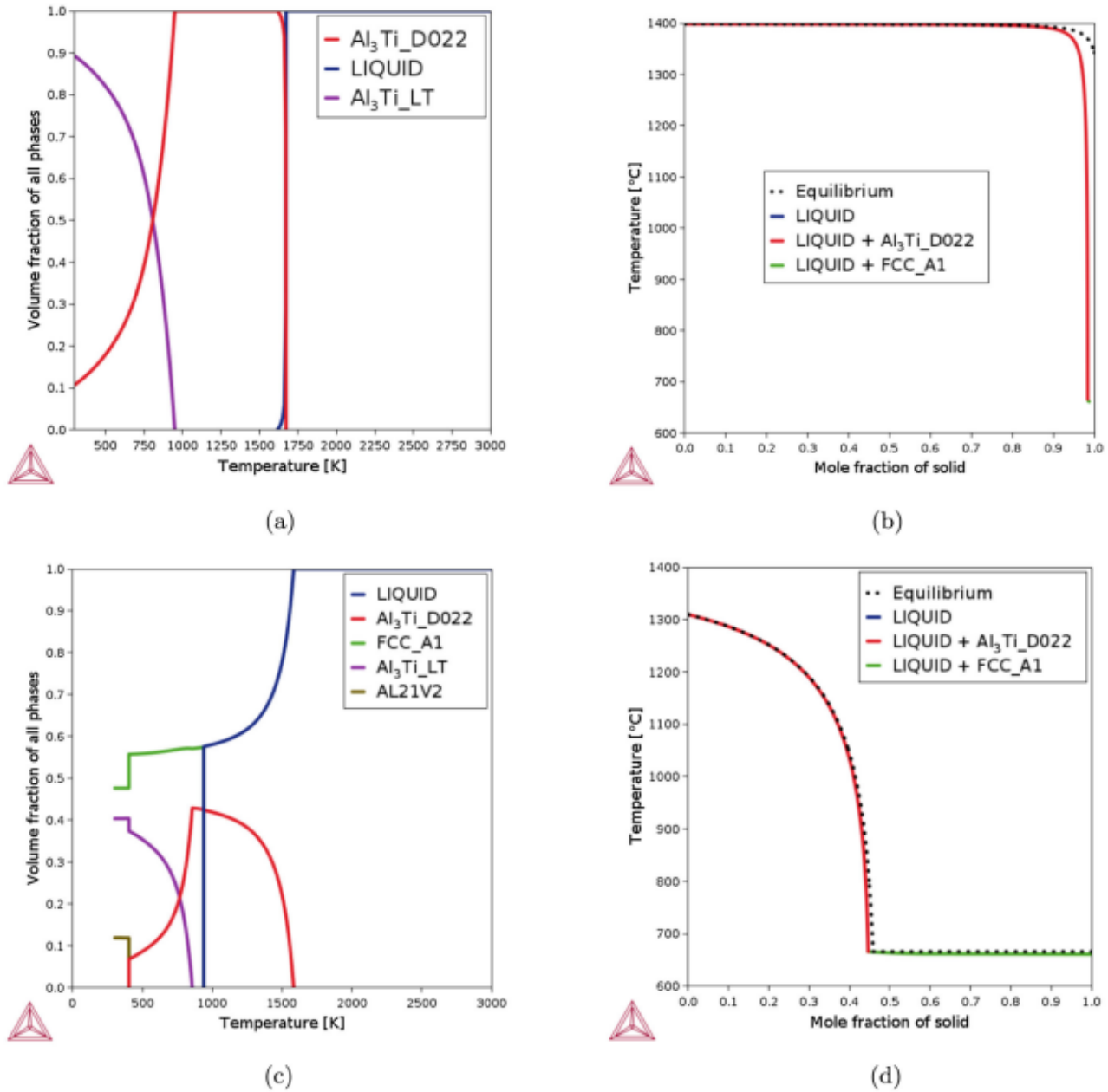
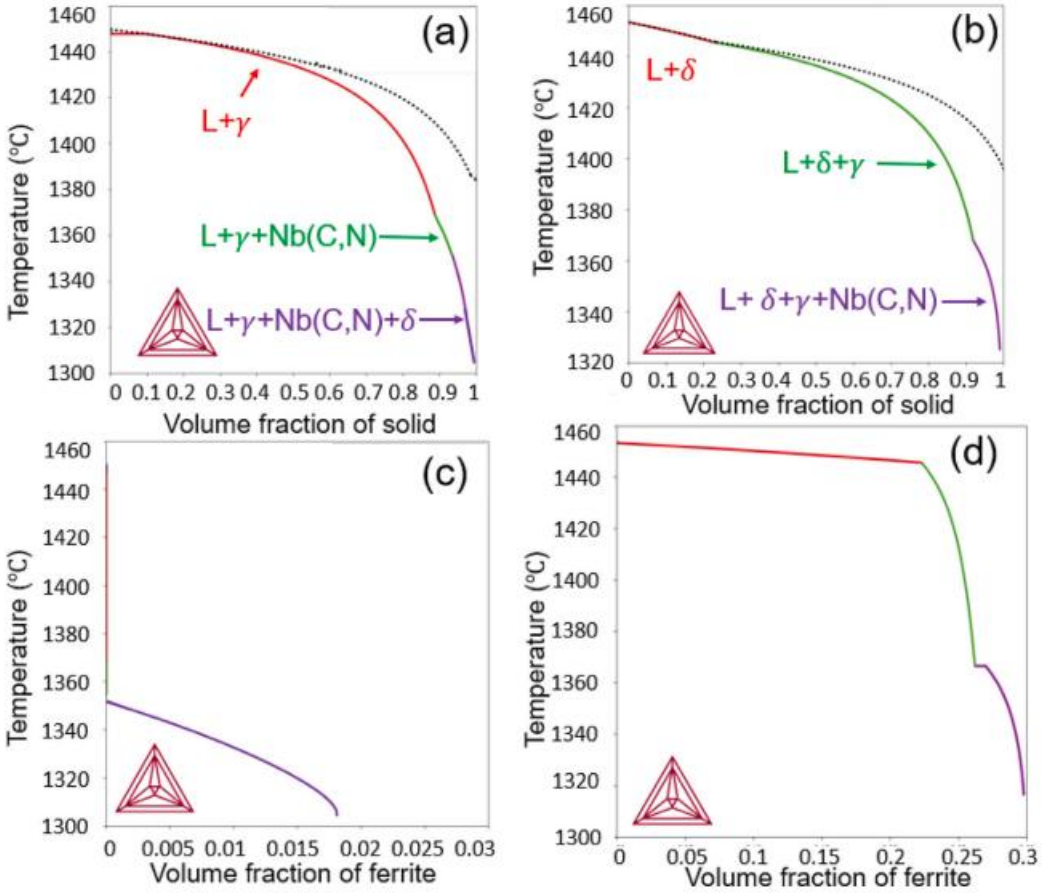


FIG. 4. Machine-learning predictions for (a) Volume fraction of solids in a single-phase alloy. (b) Mole fraction of solids in a single-phase alloy. (c) Volume fraction of solids in a dual-phase alloy. (d) Mole fraction of solids in a dual-phase alloy.

Case Study: 6 (UK, South Korea)

- Genetic-algorithm design of a precipitation hardening stainless steel with martensitic matrix that self-ages during printing
- No machine learning surrogate is used. A GA in Matlab calls Thermo-Calc (TCFE10) through TC-toolbox for every candidate with elemental compositional constraints
- Generated candidates must clear sequential physics gates with regards to crack resistance, corrosion and martensitic transformation temperature
- The fitness is the Cu-precipitate strengthening factor, volume fraction times driving force. The GA does not maximise strength directly. It searches for a printable composition window
- The GA designed alloy displayed superior mechanical properties under identical LPBF settings
- Link: https://www.sciencedirect.com/science/article/pii/S1359645424003707?ref=pdf_download&fr=RR-2&rr=a407e781adb43c4f



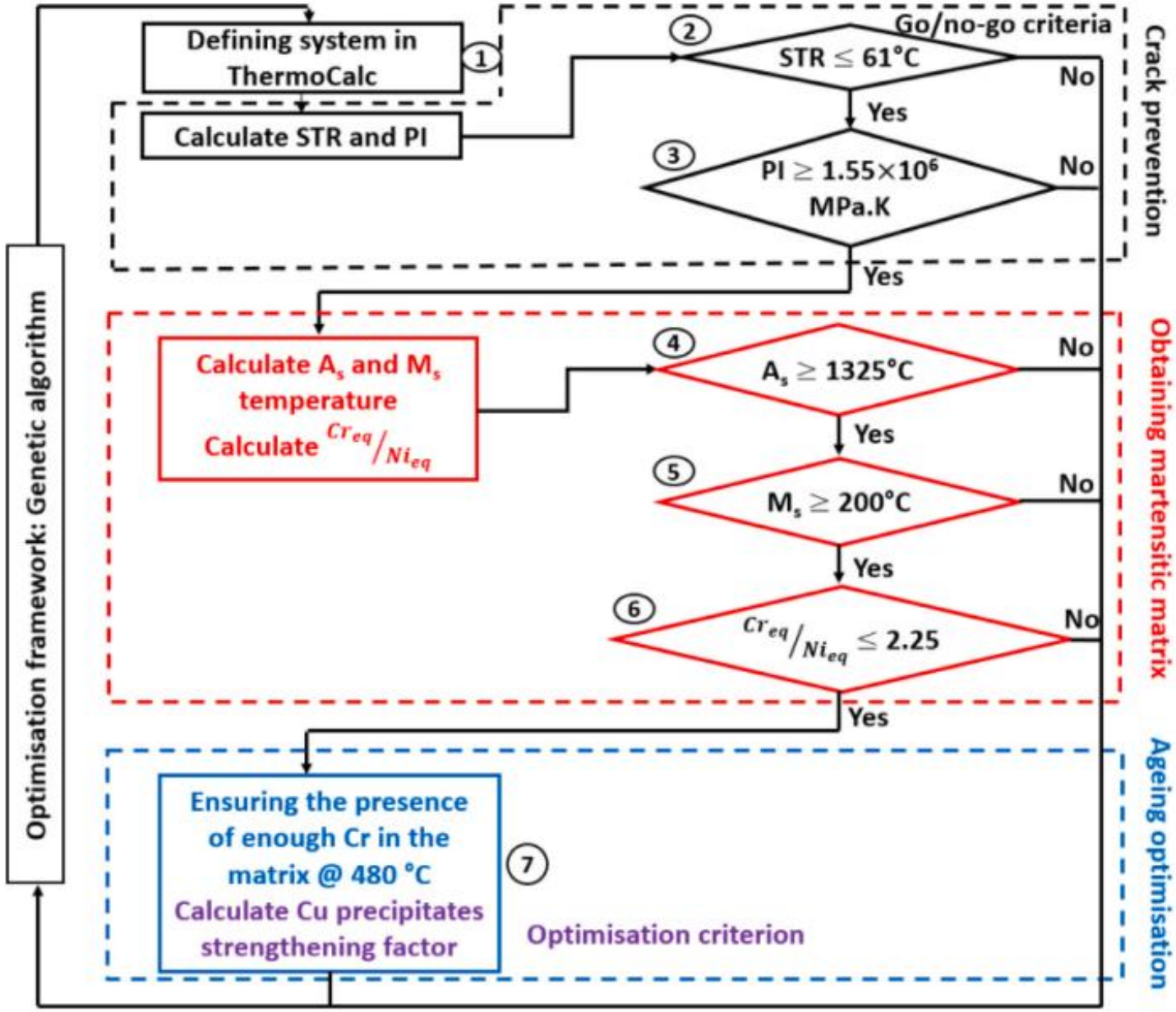


Fig. 1. Genetic algorithm combined with thermodynamic calculations flowchart used in this study to design a novel PH SS.

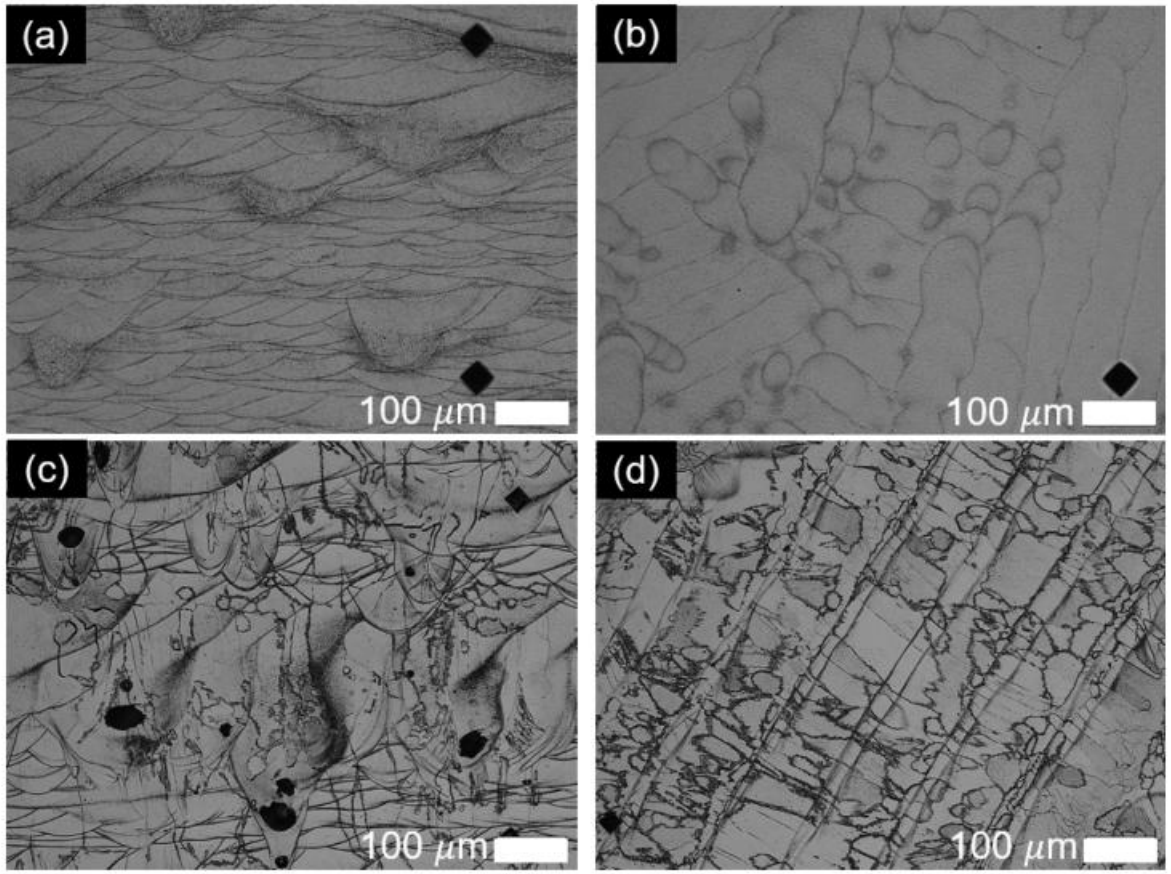


Fig. 2. Optical micrographs showing microstructure of the as-built (a) 17-4 mod in the build direction, (b) cross section of 17-4 mod, (c) 17-4 PH in the build direction, and (d) cross section of 17-4 PH.

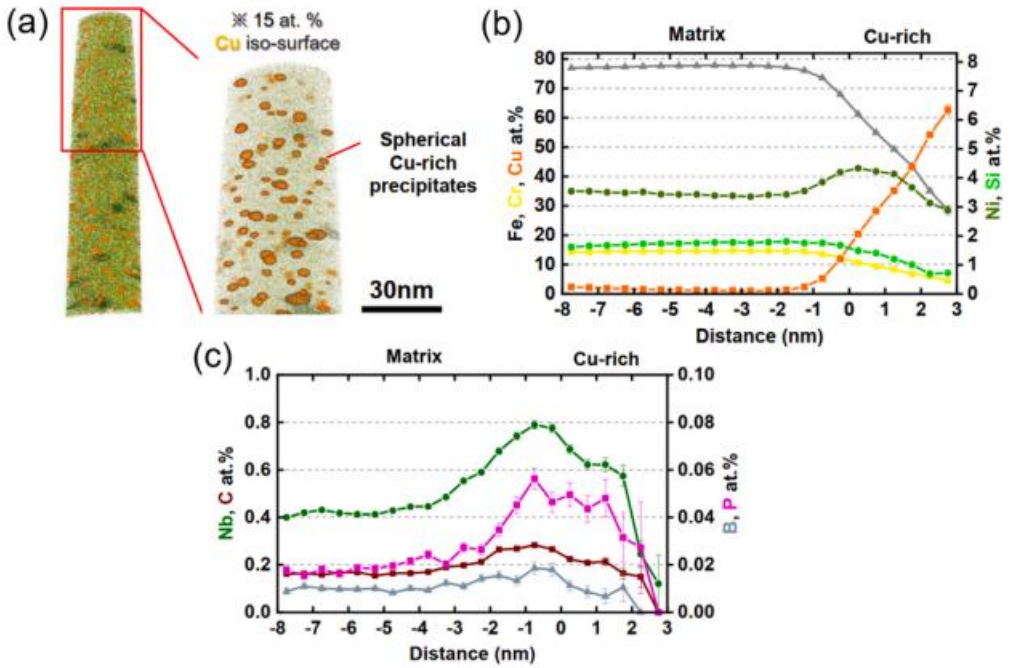


Fig. 11. (a) Cu-rich precipitates focusing on iso-concentration surfaces with 15 at.% Cu (orange). (b) and (c) Proximity histogram of Cu-rich precipitates and the matrix.

ThermoCalc Partnership Projects

ESA-GSTP:

- **Timeline:** Announced 30 Oct 2025, a nine-month effort
- **Collaborators:** Thermo-Calc Solutions AB, Skyrora Ltd, Metalysis
- **Details:**
  - The project develops and tests Tanbium, an ultra-high temperature Hf-free Ta-Nb alloy as an alternative to C-103 for 3D printed rocket components
  - Alloy powder made by a solid-state process, test specimens manufactured by directed energy deposition and run through mechanical and high-temperature trials
  - Thermo-Calc Solutions tailors the composition for the launch vehicle
- **Outcome:** Ongoing, with no final results reported



Eureka Network AM4Plasma

- **Timeline:** 1 Feb 2026 to 31 Jan 2029.
- **Collaborators:** The University of Turku coordinates the project. Finnish partners are VTT, Electro Optical Systems Finland, Top Analytica and Delva
- **Details:**
  - The project uses laser powder bed fusion and multi-material metal 3D printing to combine high-conductivity copper with refractory metals, redefining additive manufacturing
  - The target is plasma generator hardware that survives heat fluxes above 17 MW/m², with conformal cooling channels and graded interfaces
- **Outcome:** Ongoing. The intended results are better efficiency and lifetime for plasma spray systems, with applications in fusion and aerospace propulsion training



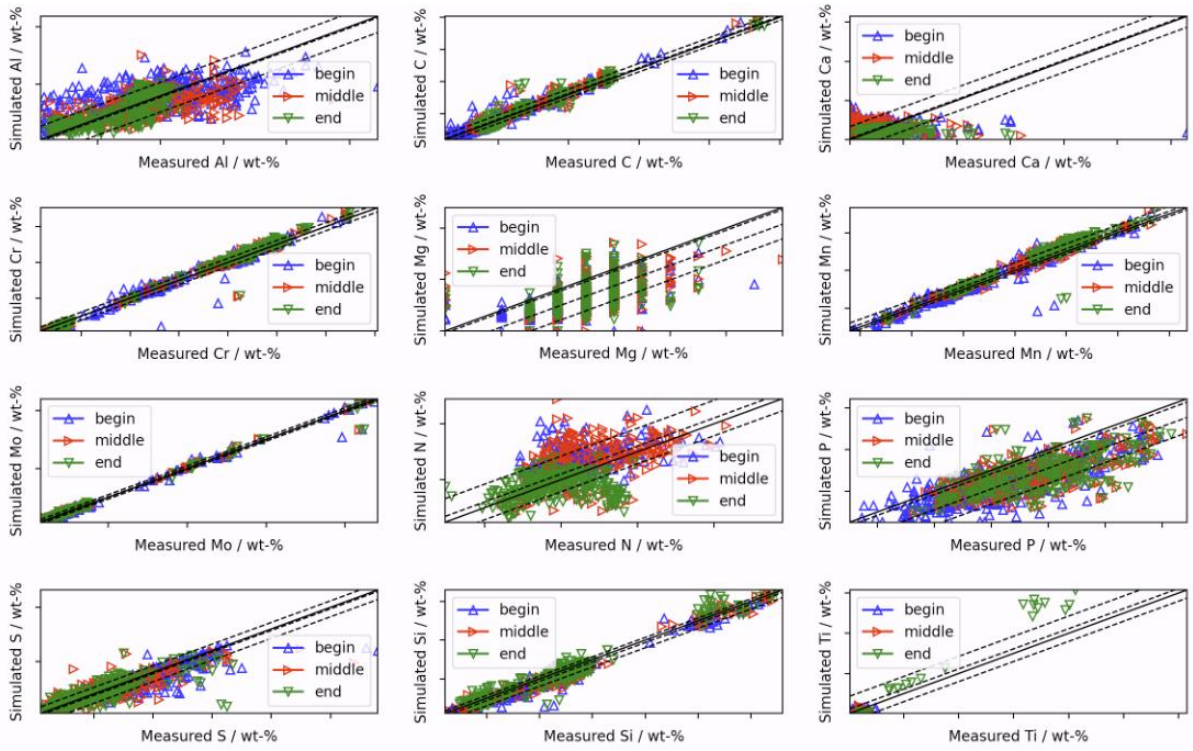
WISE ACCESS

- **Timeline:** Launched August 2025, a two-year project. Announced 27 Nov 2025
- **Collaborators:** Thermo-Calc and KTH Royal Institute of Technology
- **Details:**
  - Aims to reduce reliance on critical minerals in Li-ion batteries and to support the move to solid-state batteries.
  - It integrates CALPHAD with DFT and experimental data to build a thermodynamic and kinetic database for battery properties such as open-circuit voltage and phase transformation sequences
- **Outcome:** Ongoing. The first version of the database is expected in early to mid-2027.



GMH Collaboration

- **Timeline:** Started over four years before the announcement. No exact start date mentioned
- **Collaborators:** Thermo-Calc Software and the German steel producer GMH
- **Details:**
  - Aiming to develop physical model of sustainable steelmaking metallurgy (Green Steel)
  - It uses an effective equilibrium reaction zone (EERZ) approach built on Thermo-Calc's thermodynamic engine, the TCOX database and TC-Python
  - Simulates the EAF tapping, ladle furnace, vacuum degassing and final treatment steps
- **Outcome:** Model in routine use at the GMH plant, fed directly by production data. First fully automated model of its kind in the industry. The vacuum degassing and final treatment calibration is still being fine-tuned.



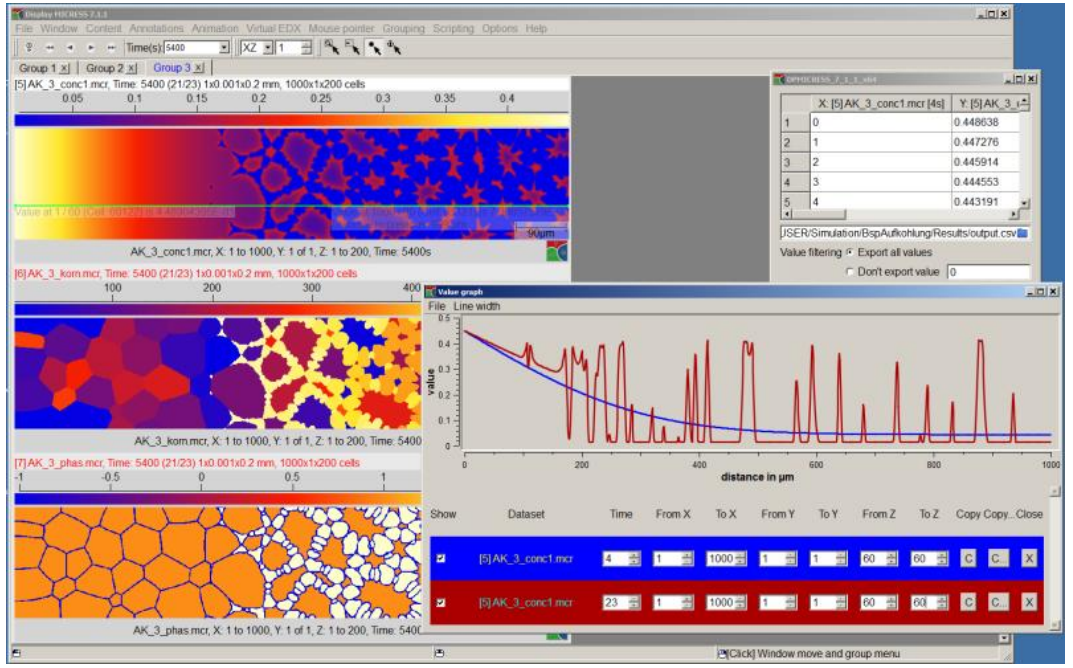
ADVANCE

- **Timeline:** Project has concluded
- **Collaborators:** Thermo-Calc, Max-Planck-Institut für Eisenforschung, Helmholtz-Zentrum and Montanuniversität Leoben
- **Details:**
  - An experimental programme to build an accurate CALPHAD database for lightweight, high-temperature titanium aluminide alloys used in aero-engine blades. - - - 84 ternary and quaternary Ti-Al-X alloys and more than 360 samples were made, heat-treated to equilibrium, and analysed
- **Outcome:** The results improved the Titanium and TiAl-based Alloys database (TCTI) and went into TCTI5. Replacing Ni-base engine parts with TiAl could save about 50% in weight and potentially halve the time to introduce new alloys



MICRESS_TQ

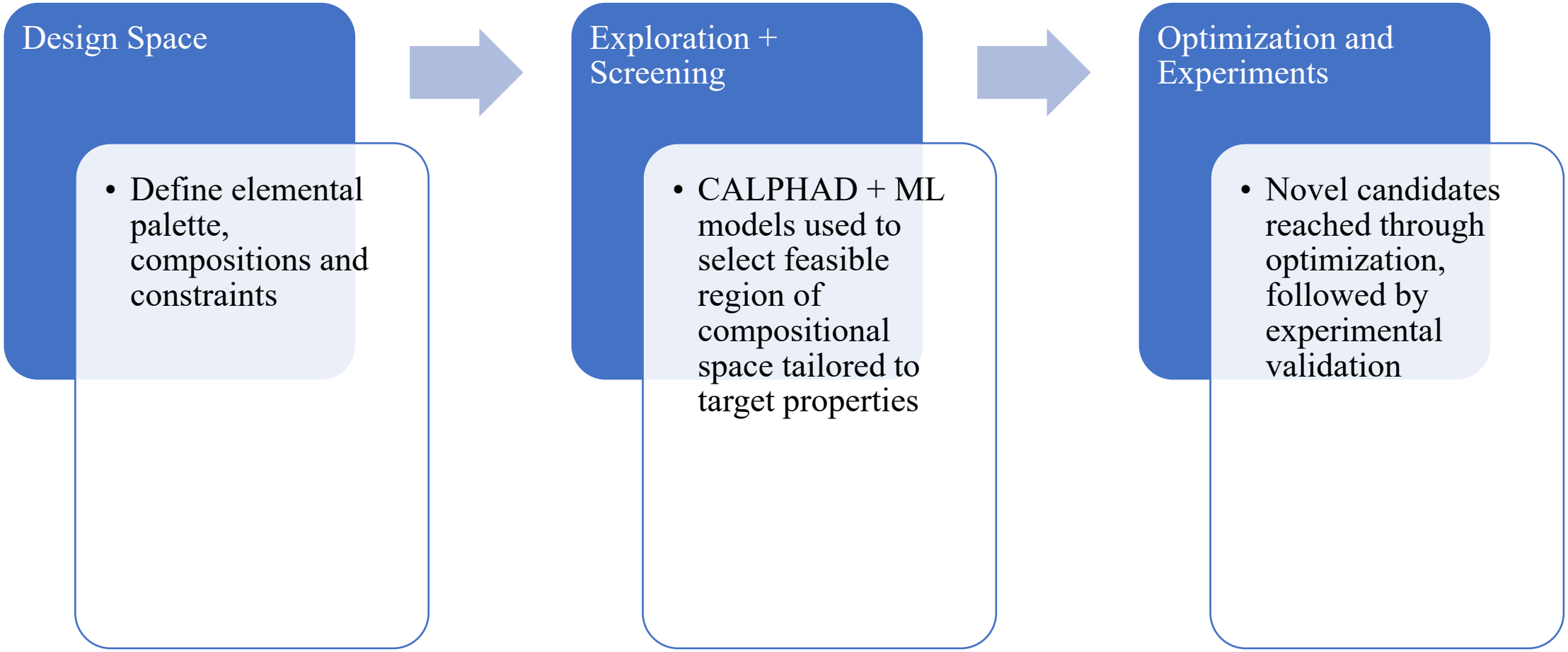
- **Timeline:** The underlying collaboration goes back to the EU VESPISM project (2000–2004)
- **Collaborators:** Thermo-Calc Software and ACCESS e.V
- **Details:**
  - MICRESS_TQ is a runtime coupling between MICRESS phase-field simulations and Thermo-Calc thermodynamic and mobility databases, through the TQ-Interface
  - It handles full multicomponent diffusion matrices and composition- and temperature-dependent molar volumes.
  - It covers solidification, solid-state transformations and heat treatment in casting, welding, soldering and additive manufacturing.
- **Outcome:** This is a released commercial tool, not a time-boxed project. It needs a MICRESS_TQ licence, a Thermo-Calc licence and the relevant databases



Key Focus: TC-Python

- Python based SDK to couple and automate Thermo-Calc calculations across a range of composition
- Crucial for high throughput computational screening
- Easy integration with NumPy, SciPy, TensorFlow for ICME
- Development of customized property prediction and optimization model
- Documentation: https://thermocalc.com/wp-content/uploads/Documentation/Current_Static/tc-python-api-programmer-guide.pdf

Integrated Materials Discovery Pipeline



CALPHAD Applications:

- High Throughput Screening: Tool covering thousands of compositions satisfying desired checks like presence and stability of phases, processing window requirements, solidification and precipitation windows, etc.
- Thermodynamic Descriptors: Generating thermodynamic descriptors to make the model physics-informed
- ML Data Generation: The aforementioned data is used to train the ML models
- Validation Layer: Final calculation check on finalized candidates before experimentation

ML Applications:

- Structure Classification and Property Prediction: A rapid method of classification and prediction of structure, phases and properties
- Surrogate for Expensive CALPHAD: Models trained on CALPHAD data give a workaround to performing expensive calculations every time
- Screening and Exploration: Often used as preliminary screening before CALPHAD
- Fitness Function for Optimization

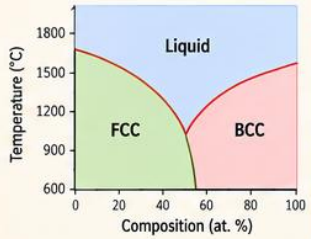
How CALPHAD and Machine Learning are Combined in Materials Discovery

Multiple integration strategies illustrated with workflows and real examples from recent studies

CALPHAD

Physics-based thermodynamic modelling

Thermo-Calc (via TC-Python)



- Phase stability
- Phase fractions
- Thermodynamic properties (G, H, S, Cp, etc.)
- Processing/solidification
- Precipitation/transformations
- Property models (SFE, driving forces, etc.)

Integrated Approach

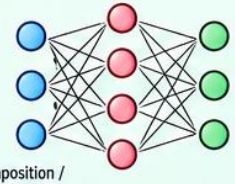
(via TC-Python)

TC-Python
(Automation & Integration)

- Automate high-throughput calculations
- Connect with ML, optimization and data processing

Machine Learning

Data-driven prediction and surrogate modelling



Composition / CALPHAD descriptors

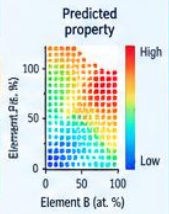
- Phase/structure classification
- Property prediction (mechanical, thermal, corrosion, etc.)
- Surrogate models for CALPHAD
- Uncertainty quantification (e.g. Gaussian process)
- Active learning
- Fitness function for optimization

Different Ways CALPHAD and ML are Combined in Materials Discovery

A — CALPHAD → ML Surrogate → High-Throughput Screening

Use CALPHAD to generate a large dataset, train an ML surrogate model, and rapidly screen huge composition spaces.

Large composition space (thousands - millions)
↓
High-throughput CALPHAD (via TC-Python)
↓
Dataset: composition → phases/properties
↓
Train ML surrogate
↓
Rapid screening (millions of candidates)
↓
CALPHAD revalidation (top candidates)
↓
Experiment



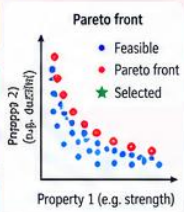
Example from review:

- 11-element HEA design using DNN surrogate trained on CALPHAD data (TCHEAS/TCHEA7).
- Used for large-scale screening with multi-stage validation (CALPHAD → DFT → experiment).

B — CALPHAD → ML → Multi-objective Optimization

Use CALPHAD data to train ML models that predict multiple properties, and couple them with optimization algorithms.

CALPHAD calculations (SFR, GRF, HCS, PSC, etc.)
↓
Train ML surrogate models
↓
Multi-objective optimization (e.g. NSGA-3)
↓
Pareto-optimal candidates
↓
CALPHAD validation
↓
Experiment



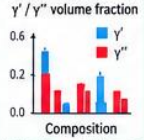
Example from review:

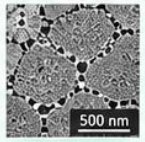
- AM metallurgy-guided ML alloy design.
- CALPHAD-generated SFR, GRF, HCS, PSC used to train ML surrogates.
- NSGA-3 multi-objective optimization.
- Experimental validation via laser-directed energy deposition.

C — CALPHAD Descriptors → ML → Optimization

Use CALPHAD-derived thermodynamic/phase descriptors as input features for ML models, and optimize the target property.

Composition
↓
CALPHAD (via TC-Python)
↓
Thermodynamic descriptors (e.g. phase fractions, driving forces, liquidus, SFE, etc.)
↓
Train ML model (e.g. AdaBoost, ANN)
↓
Optimization (e.g. Genetic Algorithm)
↓
Optimal composition
↓
Experiment





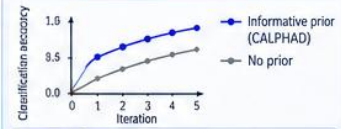
Example from review:

- Ni-based superalloy design.
- CALPHAD-derived γ'/γ'' volume fractions and precipitation driving forces used as ML inputs.
- AdaBoost model as GA fitness function.
- Identified high-performance compositions.

D — CALPHAD as Physics Prior → Physics-informed ML

Incorporate CALPHAD predictions as a prior or constraint in ML models to improve accuracy, especially with limited data.

CALPHAD predictions (physics prior) + Experimental data (limited)
↓
Physics-informed ML (e.g. Gaussian Process)
↓
Uncertainty-guided candidate selection
↓
Experiment / CALPHAD
↺ Iterate (active learning)



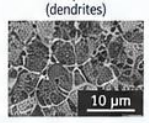
Example from review:

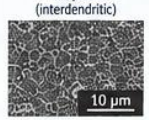
- Physics-informed Guassian process classification.
- CALPHAD used as prior mean function.
- Active learning improved model performance.
- Demonstrated effectiveness of physics-informed ML.

E — CALPHAD-generated Examples → Improved ML Training (Phase Discovery)

Use CALPHAD (e.g. binary phase diagrams) to enrich the ML training dataset with underrepresented phase examples.

Binary phase diagrams + CALPHAD screening
↓
Add relevant examples (e.g. BCC/FCC alloys)
↓
Expanded training dataset
↓
Train ML phase classifiers (e.g. ANN)
↓
Screen new compositions
↓
Experiment





Example from review:

- Refractory HEA phase discovery (BCC/FCC dual solid-solution).
- Binary phase diagrams and CALPHAD used to add missing BCC/FCC examples to dataset.
- Two-stage ANN classifiers (SS vs IM, single vs multiphase).
- Identified 52 candidate alloys, experimentally validated NiMoVW.

Overall Integrated Workflow: High-Throughput, Closed-Loop Discovery

1. Define composition space (and objectives)
→ 2. CALPHAD (high-throughput calculations via TC-Python)
→ 3. Train / use ML (surrogate, classification or property prediction)
→ 4. Screening (apply constraints)
→ 5. Optimization (GA, Bayesian, NSGA-III, etc.)
→ 6. Validation (CALPHAD / DFT)
→ 7. Experiment (synthesize & test)

↺ New data (experimental or computational) → Update ML models → Refine search space (Active Learning)